



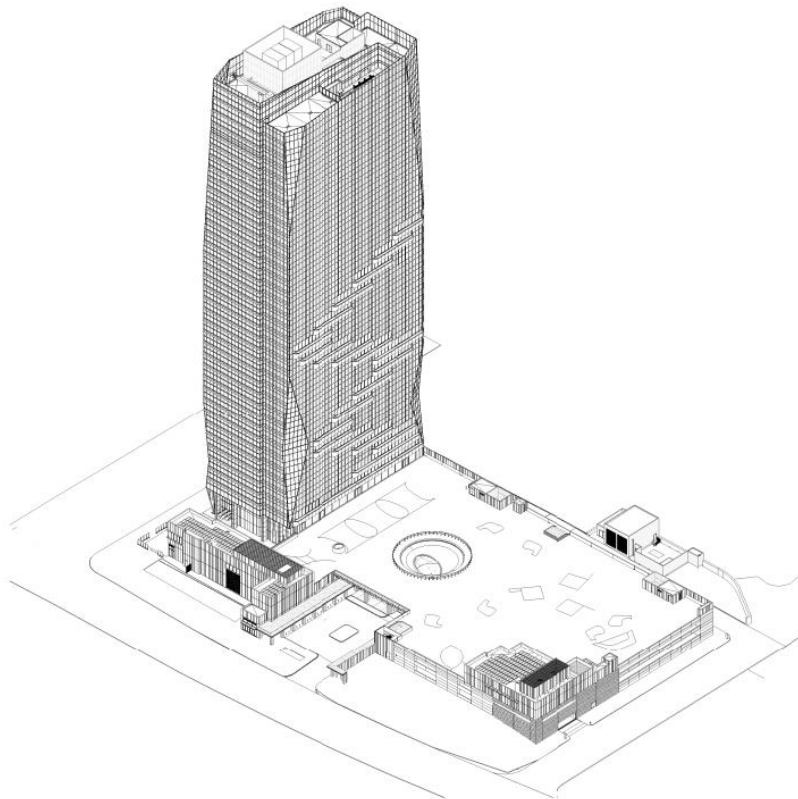
SMOKE CONTROL RATIONAL ANALYSIS

First Submittal

Creative Artists Agency Tenant Space
Los Angeles, California

18 February 2026

SGH Project 231440



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EXECUTIVE SUMMARY

The purpose of this smoke control rational analysis (SCRA) is to summarize analysis methods, design assumptions, conceptual exhaust and supply arrangement, simulation results pertaining to the smoke control system, equipment requirements, and smoke control sequencing requirements proposed for the Creative Artists Agency (CAA) tenant improvement at 1950 Avenue of the Stars in Los Angeles, California.

This SCRA details the approach taken for the CAA tenant space, which spans from Level 4 to Level 21. The base building was designed as a core and shell space, with an original SCRA prepared by CCI, dated 23 March 2020, with revisions through Revision Bulletin #2.2 dated 20 May 2024, hereto referred to as the "Base Building SCRA." The smoke control systems were not sized or designed in this analysis. The existing supply and exhaust quantities outlined in the Base Building SCRA were utilized as maximum values in this analysis. The smoke control systems in the building discussed herein are the stair pressurization system(s) and the floor depressurization systems. These systems extend into and through the tenant space; as such, the full building must be analyzed to confirm the appropriate performance objectives are met.

The Base Building SCRA approach is adopted and slightly modified herein. The modifications relate to passive subzones. Specifically, the configuration of selected air transfer openings and the classification of certain spaces as passive subzones will be adjusted. The following aspects of the smoke control design established in the SCRA remain unchanged:

- The overall smoke control strategy (pressurized stairs and floor depressurization for the high-rise office spaces)
- The design performance criteria (pressure differentials, door opening forces, tenability objectives, and duration of operation) as defined in the SCRA
- The acceptance testing and special inspection approach required by LABC 91.909.3 and the applicable Los Angeles Department of Building and Safety procedures

The changes described in this document are consistent with, and in several respects increase the robustness of, the design basis in California Building Code (CBC) Section 909 and the SCRA:

1. Stair 2 stair vestibule openings remain in place with 90 min. fire dampers, maintaining the intended pressure relationships between stair, vestibule, and floor that support the smokeproof enclosure concept.
2. Lounge and convenience stair zones are treated as passive subzones bounded by two-hour fire resistance rated construction and side-acting fire shutters, with air transfer openings to the main office zone removed. This improves compartmentation, reduces potential smoke migration paths between zones, and supports the floor-to-floor smoke control objectives established in the rational analysis.
3. Small utility and non-rated vestibule spaces are treated as passive subzones to the main floor. Eliminating their dedicated air transfer openings reduces the actively depressurized volume and the number of dampers, improving system reliability and increasing the margin for meeting the pressure and door-force criteria under CBC Section 909.

No changes are proposed to the design objectives, input assumptions, or acceptance criteria established in the SCRA under CBC Section 909.4. Any supporting calculations and modeling associated with the redistribution of leakage (due to removal of the subject air transfer openings) are updated within the SCRA to reflect these revisions.

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1. INTRODUCTION

This report provides details of the smoke control rational analysis (SCRA) as required by Los Angeles Building Code (LABC) 404 and LABC 909 for the Creative Artists Agency (CAA) tenant space in the Culver City Center (CCC) at 1950 Avenue of the Stars in Los Angeles, California.

1.1 Project Background

The full CCC building description and system description can be found in the Base Building SCRA; however, a brief description is provided herein. The CCC building consists of a 37-story high-rise office tower with two levels of above-grade parking, four levels of subterranean parking (Levels P4 through P1), and an attached two-story mixed-use component. The parking garage and two-story mixed-use component are considered a part of the high-rise building.

At the street and lower levels, the building features several specialized assembly and transition spaces. Level 1 serves as the primary entrance, featuring a two-story entry lobby and retail volume that atmospherically connects to Level 3, noting that Level 1 is a two-story high retail volume with no occupied floor at Level 2. Level 3 also contains a portion of the screening room lobby and social lounge area.

The main use of the building is an office tower, with the CAA tenant space located on Levels 4 through 21, and other office spaces located on Levels 22 through 37.

The CAA tenant space consists mainly of open office space. This tenant space also includes a series of communicating convenience stairs that cascade throughout all levels of the CAA tenant space. These convenience stairs are separated such that each portion of the convenience stairs connect a maximum of two stories. The convenience stairs themselves are separated using two-hour side-acting fire shutters. Each of the convenience stairs also has a lobby surrounding the convenience stairs. This lobby is also separated by two-hour rated fire barriers with 90 min. rated doors. Specific details related to the convenience stairs and lobby can be found in the approved Requests for Modification (RFMs).

2. OVERVIEW OF SMOKE CONTROL SYSTEMS

The project is required to have smoke control in accordance with LABC 403 because the building is a high rise. The goal of the smoke control system is to maintain a tenable environment for the evacuation and relocation of occupants in the event of a fire. A combined zoning approach for passive and active smoke control will be implemented such that smoke movement is directed or restricted, thereby limiting the spread of smoke between smoke zones. This CAA SCRA will specifically discuss criteria and systems located in the tenant space. For further information about systems outside of this scope of work, refer to the Base Building SCRA.

2.1 Passive Smoke Control

The design team will implement passive smoke control to control smoke spread between smoke zones.

- All below-grade parking areas, including storage and mechanical, electrical, and plumbing spaces, on the parking levels will be passive as permitted by LABC 909.2 and in accordance with permissible leakage identified in LABC 909.5. These areas are outside of the CAA scope of work and, for the purposes of this report, will not be discussed further.
- Levels below the CAA scope of work (P4 through Level 3) include all passive subzones, such as retail areas, mechanical and utility spaces, common use areas, and amenity spaces as permitted by LABC 909.2 and in accordance with permissible leakage identified in LABC 909.5. These areas are outside of the CAA scope of work and, for the purposes of this report, will not be discussed further.
- The CAA scope of work levels (Levels 4 through 21) include passive subzones identified in Section 3.
- Office floors above the CAA scope of work (Levels 22 to 37) include passive subzones similar to the CAA scope of work. For the purposes of this report, these will not be discussed further.
- Roof areas, inclusive of mechanical penthouse and occupiable floor zones (if any), will be passive as permitted by LABC 909.2 and in accordance with permissible leakage identified in LABC 909.5.

Following is a detailed description of passive subzones and their approach and the types of fire resistance rated assemblies that will be implemented to achieve passive smoke control.

2.1.1 Smoke Barriers

Smoke barriers will principally be implemented between smoke zones (horizontal and vertical). When used, smoke barriers are required to comply with LABC Section 709 and must be constructed and sealed to limit leakage areas exclusive of protected openings. The maximum allowable leakage area is the aggregate area calculated using the leakage area ratios per LABC 909.5 (Table 1). Smoke barriers are used throughout in various locations.

Table 1 – Maximum Allowable Leakage Area for Smoke Barrier Construction

Smoke Barrier Construction	Maximum A/A_w or A/A_f
Walls (interior only)	0.00100
Interior exit stairway and ramps, and exit passageway	0.00035
Enclosed exit access stairways and ramps, and all other shafts	0.00150
Floors and roofs	0.00050

where:

A = Total leakage area, square feet

A_f = Unit floor or roof area of barrier, square feet

A_w = Unit wall area of barrier, square feet

2.1.2 Smoke Partitions

Smoke partitions may be used within smoke zones where it is desired to control enclosure leakage for the purposes of door fan testing or for any other reason that is consistent with the intent of the rational analysis in support of limiting or directing smoke movement within the building.

2.1.3 Fire Partitions

Fire partitions may be used in the office in lieu of smoke partitions to compartment a smoke zone into discrete areas when it is desirable to limit smoke movement. One-hour fire partitions are being used at all elevator lobbies, excluding fire service access elevator (FSAE) lobbies. These partitions will be equipped with 45 min. smoke control and draft doors in accordance with LABC Sections 3006.3 and 716.5.3.

2.2 Active Smoke Control

Active smoke control is implemented via mechanical pressurization and depressurization to achieve required pressure differences across fire barriers at smokeproof enclosures in accordance with LABC 909.6 and 909.20. Open office spaces on Levels 4 through 21 will be depressurized to achieve required pressure differentials across horizontal smoke barriers (floor/ceiling assemblies) from the open office within the smoke zone of origin to the open office in smoke zones above and/or below.

The primary active smoke control systems are as follows:

- Stairs 1, 2, and 3, serving tower levels, inclusive of the CAA scope of work will be designed using pressurization systems per LABC 909.20.
- Stairs 5, 6, and 7, serving parking levels only, will be designed using pressurization systems per LABC 909.20. Note that vestibules were omitted in accordance with the approved RFM in the Base Building SCRA.
- All open office spaces on Levels 4 through Level 21 are designed using a depressurization system.

2.2.1 Minimum Pressure Difference

In accordance with LABC 909.6.1, the minimum pressure difference across a smoke barrier shall be 0.05 in. water gage (0.0124 kPa) in fully sprinklered buildings. It is the intent of this rational analysis that this provision applies to actively depressurized open office levels to the adjacent levels' smoke zone(s) above and/or below the level of alarm. A minimum pressure difference of 0.05 in. water gage (0.0124 kPa) will be maintained between the actively depressurized open office spaces in the smoke zone of origin on Levels 4 to Level 21 and adjacent smoke zones; a 0.05 in. water gage (0.0124 kPa) pressure differential will not necessarily be achieved across smoke barriers within the same smoke zone. All other floors smoke control approaches are further discussed in the Base Building SCRA.

At smokeproof enclosures, the minimum pressure differences within the vestibule with the doors closed will be 0.05 in. water gage (0.0124 kPa) positive pressure relative to the fire floor and 0.05 in. water gage (0.0124 kPa) negative pressure relative to exit enclosure, where they exist. No pressure difference is required relative to a non-fire floor. Where stairs communicate with

exterior spaces and vestibules are not present, it is the intent of this rational analysis that the minimum pressure difference across the stair enclosure shall be 0.10 in. water gage. Stairs 5, 6, and 7 have omitted vestibules and will achieve a pressure differential of 0.10 in. water gage.

2.2.2 Maximum Pressure Difference

In accordance with LABC 1010.1.3, the force for pushing or pulling open interior swinging egress doors, other than fire doors, may not exceed 5 lbs (22 N). For other swinging doors, as well as sliding and folding doors, the door latch must release when subjected to a 15 lb (67 N) force. The door is required to swing to a full-open position when subjected to a 15 lb (67 N) force.

Doors serving the pressurized stairs will meet the requirements for a smoke and draft control door assembly tested in accordance with UL 1784 (LABC 710.5.2.2). The air leakage rate of the door assembly shall not exceed 3.0 cu ft per minute per square foot of door opening at 0.10 in. of water for both the ambient temperature and the elevated temperature exposure test. Installation of smoke doors shall be in accordance with NFPA 105.

To address door forces and the associated requirements for automatic door operators (ADOs), Eqn 9-1 of LABC 909.6.2 (Eqn 6.32 from *Principles of Smoke Management*) is applied, assuming that the door closer device is set to 5 lbs (force), and that the average door is 36 in. wide and 84 in. (7 ft) high:

$$\Delta p = \frac{2(W - d)(F - F_R)}{K_d W A}$$

Where:

- Δp = Pressure differential across door (in. w.g.)
- F = Total door opening force (lbf)
- F_R = Force to overcome door closer (5 lbf)
- K_d = Coefficient (5.20)
- W = Door width (36 in.)
- A = Door leaf area (21 to 24 sq ft)
- d = Distance from door handle to edge of door (0.3 ft)

For exit doors subject to a maximum door opening force of 15 lbf (excluding the force required to retract latch bolts or disengage other devices that hold the door or gate in a closed position) per California Building Code (CBC) 11B-404.2.9, the corresponding maximum pressure differential is approximately 0.17 in. water gage for 7 ft tall doors. Where the pressure differential on exit doors exceeds 0.17 in. water gage due to stack effect, provisions for the installation of ADOs may be required. Per CBC 909.20.6.2, the pressure difference across doors serving pressurized stair enclosures shall not exceed the 30 lbf (133 N) maximum force to begin opening the door, corresponding to approximately 0.34 in. water gage.

ADOs will be installed on all doors that have a measured door opening force in excess of 15 lbf (excluding the force required to retract latch bolts or disengage other devices that hold the door or gate in a closed position per CBC 11B-404.2.9), including all doors serving the smokeproof enclosures that access the exterior environment. Accessible control switches susceptible to tampering will not be permitted at ADOs.

ADOs shall meet the following requirements:

- Electronic motion sensor features should not be included on ADOs.
- Doors using ADOs must close and latch.
- ADOs must function after initial transfer to emergency power in addition to normal power conditions; as such, battery backup or emergency power will be required.
- Dedicated emergency circuits with breaker locks are required for the ADOs.

Final decisions on which doors need to be equipped with ADOs shall be based on the air balancing test during which door opening forces will be physically measured. It has been frequently observed that the actual door opening forces measured in the field are less than 15 lbs where the associated pressure differentials are more than 0.17 in. water gage; this is because the pressure across doors can be balanced through adjusting the door sweep, which deviates the effective flow area of such doors simulated in CONTAM models. Additionally, the door closers themselves can be adjusted to accommodate the maximum door opening force. Programming the variable frequency drive of a stair pressurization fan to maintain different speeds for different levels can also help to adjust the pressures and door opening forces.

The doors requiring ADOs, based on modeling results, are provided in Appendix B. Final ADO requirements will be determined upon completion of smoke control system commissioning, subject to specific as-built conditions.

3. PASSIVE SUBZONES IN CAA TENANT SPACE

Per LABC Section 91.909, which adopts CBC Section 909 by reference, the smoke control system for this building is supported by a rational analysis prepared by a registered design professional. That analysis is documented in the Base Building SCRA prepared by CCI, dated 23 March 2020, with revisions through Revision Bulletin #2.2 dated 20 May 2024 and this SCRA.

This SCRA modifies the Base Building SCRA, the configuration of selected air transfer openings, and the classification of certain spaces as passive subzones. The following aspects of the smoke control design established in the SCRA remain unchanged:

- The overall smoke control strategy (pressurized stairs and floor depressurization for the high-rise office spaces)
- The design performance criteria (pressure differentials, door opening forces, tenability objectives, and duration of operation) as defined in the Base Building SCRA
- The acceptance testing and special inspection approach required by LABC 91.909.3 and the applicable Los Angeles Department of Building and Safety (LADBS) procedures

The changes described are consistent with, and in several respects increase the robustness of, the design basis in CBC Section 909 and the SCRA:

1. Stair 2 stair vestibule openings remain in place with 90 min. fire dampers, maintaining the intended pressure relationships between stair, vestibule, and floor that support the smokeproof enclosure concept.
2. Lounge and convenience stair zones are treated as passive subzones bounded by two-hour fire resistance rated construction and side-acting fire shutters, with air transfer openings to the main office zone removed. This improves compartmentation, reduces potential smoke migration paths between zones, and supports the floor-to-floor smoke control objectives established in the rational analysis.
3. Small utility and non-rated vestibule spaces are treated as passive subzones to the main floor. Eliminating their dedicated air transfer openings reduces the actively depressurized volume and the number of dampers, improving system reliability and increasing the margin for meeting the pressure and door-force criteria under CBC Section 909.

No changes are proposed to the design objectives, input assumptions, or acceptance criteria established in the Base Building SCRA under CBC Section 909.4. Any supporting calculations and

modelling associated with the redistribution of leakage (due to removal of the subject air transfer openings) are updated within this SCRA to reflect these revisions.

3.1 Passive Subzone Discussion

The proposed use of passive subzones on the high-rise office floors is consistent with the intent and technical requirements of CBC Section 909 (as adopted by LABC Section 91.909), National Fire Protection Association (NFPA) 92, *Standard for Smoke Control Systems*, and accepted smoke control engineering practice. CBC Section 909.1 permits smoke control systems to be mechanical, passive, or a combination of both, and the International Code Council (ICC) *International Building Code Commentary* to Section 909 discusses zoned smoke control as a performance-based strategy in which containment is achieved at smoke zone boundaries rather than by providing active smoke control in every individual room. NFPA 92 (Chapters 4 and 5, together with the Annex A discussion of zoned smoke control) similarly frames the design objective as limiting smoke movement out of the zone of origin through the use of pressure differences *and/or* compartmentation across smoke control zone boundaries. Neither CBC/International Building Code Section 909 nor NFPA 92 requires every enclosed space within a smoke control floor to be mechanically pressurized or depressurized, nor do they mandate that every subcompartment achieve the nominal 0.05 in. water column pressure differential; that criterion is applied across smoke control zone boundaries, such as between the event floor and adjacent floors and smokeproof enclosures.

Within this framework, treating the lounges, convenience stair compartments, and small core utility and vestibule spaces as passive subzones of the actively depressurized office floor is consistent with NFPA 92 and CBC Section 909 and is typical of high-rise smoke control designs documented in the Society of Fire Protection Engineers (SFPE) *Handbook of Fire Protection Engineering* and the SFPE *Engineering Guide: Design of Smoke Management Systems*. In this project, in the CAA scope of work, the lounge and convenience stair zones are bounded by two-hour fire resistance rated construction and side-acting fire shutters while the small core utility and vestibule spaces are modest in size and accessory in nature. These spaces are therefore treated as part of the same smoke control zone as the open office floor for purposes of floor depressurization and are not provided with dedicated smoke control air transfer

openings. Under the NFPA 92 zoned smoke control concepts, these rooms are simply internal subcompartments within the event zone model, and their volume and leakage characteristics are incorporated into the rational analysis rather than being treated as independent zones requiring separate pressure criteria.

This approach is aligned with the rational analysis requirements of CBC Section 909.4 and the NFPA 92 design methodology, which require the registered design professional to demonstrate that the overall system maintains the required pressure differentials and tenability at the smoke zone boundaries under design fire and door-opening conditions—not to apply the 0.05 in. water column criterion between every interior room and the adjoining office area. For this building, eliminating the dedicated air transfer openings between the office floor and the lounge, convenience stair zones, and small core utility vestibule spaces reduces the active depressurized volume, decreases the number of dampers and control points, and consolidates leakage paths into more robustly detailed boundaries. This configuration improves the reliability and operating margin of the floor depressurization system while maintaining or improving compartmentation between stories and between the floor and the smokeproof enclosures. As documented in the Base Building SCRA and this SCRA, the resulting system continues to satisfy the design objectives, pressure differential criteria, and acceptance testing framework established under CBC Section 909 and NFPA 92 while using a configuration that is consistent with both code commentary and widely accepted smoke control engineering practice for high-rise buildings.

3.2 Elimination of Air Transfer Openings in Tenant Improvement Areas

The air transfer openings currently prescribed by the Base Building SCRA are in various locations throughout the building. These air transfer openings were originally provided in an attempt to assist with the differential pressures required between smoke zones. These locations are described in the table on p. 6 of the Base Building SCRA (Figure 1); see the Base Building SCRA for more details.

The air transfer openings fall into the following categories:

1. Stair-to-vestibule air transfer openings for Stair 2, located on Levels 4 through 37

2. Air transfer openings from the lounge to office space, located on Levels 4 through 21
3. Air transfer openings between small utility spaces (mechanical, electrical, janitor, storage) and the office, located on Levels 4 through 37; and passive vestibules to adjacent spaces located on Levels 19 through 36

Air Transfer Openings		
Level	Location	Net Free Area
<u>4-1820</u>	<u>Lounge to Office Space</u>	<u>2.00 SF</u>
<u>4-37</u>	<u>Mechanical Room to Office Space</u>	<u>0.50 SF</u>
<u>4-37</u>	<u>Electrical Room to Office Space</u>	<u>0.50 SF</u>
<u>4-37</u>	<u>Emergency Electrical Room to Office Space</u>	<u>0.25 SF</u>
<u>5-8</u>	<u>Janitor/Storage Closet to Office Space</u>	<u>0.25 SF</u>
<u>10-13</u>		
<u>15-18</u>		
<u>27-28</u>		
<u>19</u>	<u>IT Closet to Vestibule</u>	<u>0.50 SF</u>
	<u>Vestibule to Office Space</u>	<u>0.25 SF</u>
<u>20-25</u>	<u>Janitor/Storage/IT Closet to Vestibule</u>	<u>0.50 SF</u>
	<u>Vestibule to Office Space</u>	<u>0.50 SF</u>
<u>26</u>	<u>Janitor Closet to Vestibule</u>	<u>0.50 SF</u>
<u>31-33, 36</u>	<u>Janitor Closet to Vestibule</u>	<u>0.25 SF</u>

Figure 1 – Air Transfer Openings Table from Base Building SCRA

3.2.1 Types of Air Transfer Openings

There are three principal types of air transfer openings located in this project as currently defined. They consist of rated and non-rated penetrations. Each type of air transfer opening introduced above will be further described herein.

3.2.1.1 Stair to Vestibule Opening

These openings are provided between the stair and vestibule in Stair 2 on Levels 4 through 37 (Figure 2). These openings are equipped with a 90 min. fire damper. These openings are utilized to allow for the differential pressures required between (1) the stair and vestibule and (2) the vestibule and the open office levels. It is proposed to leave these openings unchanged. The differential pressures measured will remain between the stair and vestibule, and the vestibule and floor, as measured utilizing a standard manometer.

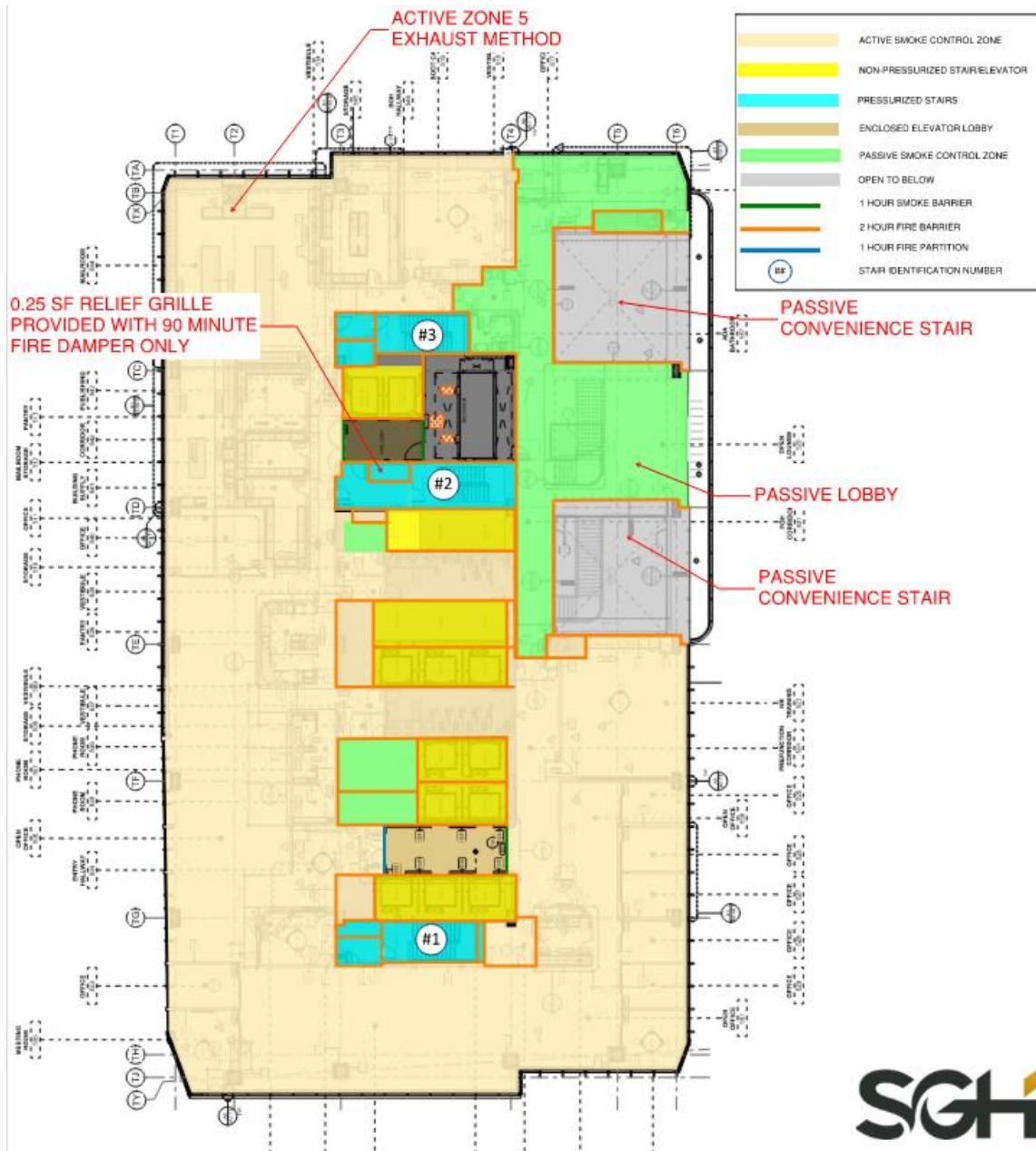


Figure 2 – Air Transfer Openings Exemplar Level 5

3.2.1.2 Air Transfer Openings from the Lounge to Office Space

These openings are currently prescribed by the Base Building SCRA to be provided in the two-hour fire resistance rated fire barriers between the lounge space and the remainder of the open office. These openings are currently prescribed to be provided on Levels 4 to 21, in the CAA scope of work. These openings were originally intended to assist with obtaining a pressure differential between the lounge and the overall office level. See Base Building SCRA for further details.

The lounge is an area that is separated from the remainder of the floor by two-hour fire barrier construction. The lounge also contains a rated separation that separates the floor opening for the convenience stairs serving two levels. These convenience stair openings serve two total levels and are separated at the top level by two-hour fire resistance rated side-acting fire shutters. These side-acting fire shutters make it such that only one occupied level is connected to a non-occupiable smoke reservoir above. This separation alone limits the interconnection between no more than two stories. The additional rated lobby surrounding these convenience stairs and fire shutters serves as an additional level of safety to mitigate the risk from a single point of failure allowing more than two stories to be interconnected. Further details on this compartmentation can be found in the RFM dated 3 April 2023, located in Appendix A to the Base Building SCRA.

As a function of this SCRA, the lounge and convenience stair spaces are classified as passive subzones to the active depressurized office floor levels. See Figure 2 for the proposed change.

The use of floor depressurization is useful for mitigating the risk of floor-to-floor smoke spread between levels in high-rise buildings. Floor depressurization is also useful in assisting to relieve some of the built-up pressure created by smokeproof enclosure pressurization systems. With buildings being constructed of tighter construction, the typical avenues for leakage through exterior facades and other pathways are less available than in the past. As such, without a means of relieving this built-up pressure, or allowing for a location for this stair pressurization air to flow, differential pressures at the smokeproof enclosures cannot often be achieved.

As such, due to the double containment on Levels 4 through 21, this SCRA proposes that the lounge and convenience stair zones be classified as passive subzones to the floor depressurization zone. The air transfer openings will be removed from the two-hour fire resistance rated fire barrier on each level. This air transfer opening removal will result in better compartmentation, as it will be a permanently closed two-hour fire barrier, as opposed to an opening provided with a fire smoke damper.

3.2.1.3 Air Transfer Openings from Small Utility Spaces

The remaining air transfer openings on each floor are limited to openings between small utility-type spaces in the core—for example, mechanical, electrical, janitor, and storage rooms, and passive vestibules adjacent to the open office space—and the office areas. These spaces comprise small zones. In many other projects, similar spaces are treated as passive subzones of the main floor. It is proposed that these spaces be treated as passive subzones and no longer require air transfer openings.

The small, accessory nature of these spaces minimizes the risk associated with not depressurizing them directly. Removing the air transfer openings will also reduce the impact on the occupied open office areas if a fire occurs in one of these small zones. In addition, eliminating air transfer openings serving these small accessory spaces will improve the performance and operating window of the main floor depressurization system by decreasing the overall volume that must be depressurized.

4. SMOKE ZONES AND APPROACHES

The following is a summary of the smoke zones in the CAA scope of work. The remaining smoke zones in this SCRA analysis assume the same smoke zones set forth in the Base Building SCRA.

- **CAA Open Office Smoke Zone.** These smoke zones each constitute a single level which contain open office spaces, associated accessory spaces, amenities and passive subzones. Each level has one depressurized open office area, with the passive subzones discussed in Section 3. These zones are located on Levels 4 through 21.

Table 2 – Smoke Zone Description

Smoke Zone ID	Level(s)	Approach
CAA Open Office Level XX Smoke Zone	Level 4 to Level 21	• Enclosed stairs will be smokeproof enclosures (pressurized per LABC 909.20) using variable-speed drive fans. • Open office zones will be depressurized by variable-speed drive fans. • Passive elevator lobbies (LABC 403.4.7 and 3006.3). • Passive convenience stair and lobby areas (LABC 403.4.7). • All other enclosed spaces within each smoke zone are passive subzones (LABC 403.4.7).
Outside of CAA Scope of Work Smoke Zone	Level P4 – Level 3; Level 22 – Roof	• All other smoke zones are described in the Base Building SCRA.

5. SYSTEM ACTIVATION AND SEQUENCE OF OPERATIONS

In accordance with LABC 909.17, the smoke control system will be automatically activated by an appropriately zoned automatic sprinkler system complying with Section 903.3.1.1, smoke detectors required by engineering analysis, and manual controls that are readily accessible to the fire department in the fire command center. Manual activation of the system will result in the same sequence of operations as automatic activation. Pursuant to LABC 909.12.3.1 and 909.12.4, mechanical smoke control systems using the pressurization method at smokeproof enclosures are required to have completely automatic control.

5.1 Smoke Control Initiating Devices

The following details pertain to initiating devices of the smoke control systems:

1. Waterflow switches will initiate smoke control systems as described in the Smoke Control Functional Matrix (Figure 3). At a minimum, separate waterflow switches will be provided for each smoke zone and as required by NFPA 13 based on maximum allowable area per sprinkler zone.
 - a) In accordance with LABC 403.3, a single sprinkler waterflow alarm-initiating device and a control valve with a supervisory signal-initiating device will be provided at the lateral connection to the riser for each floor to initiate the smoke control system.
 - b) Waterflow switches will be coordinated with smoke zones such that no single waterflow switch activates more than one smoke zone.
 - c) In future two-story or high-bay ceiling areas, sprinklers are to be supplied from waterflow devices on the level that the sprinklers protect and not on the level above the ceiling.
2. Smoke detectors will initiate smoke control as described in the Smoke Control Functional Matrix. Smoke detection will be provided per LABC 907.2.13.1.1 (area detection in each elevator machine room and in elevator lobbies) and per LABC 907.2.13.1.2 (duct smoke detection).
 - a) Duct and corridor smoke detectors (if provided) will initiate the smoke control system, alarm sequence, and shut down HVAC.
 - b) Smoke detectors in the elevator lobbies will initiate elevator recall, release elevator lobby doors on hold-opens, and cause all stair pressurization fans to start. The elevator lobby is part of the active floor zone.

3. A single manual pull station will be provided at a location approved by the Los Angeles Fire Department (LAFD) per LABC 907.2 will initiate smoke control as described in the Smoke Control Functional Matrix. The manual pull station will result in a general alarm condition (evacuation signal throughout).
4. When elevator hoistway or machine room detectors are activated, elevator lobby doors throughout will close and the elevator will recall to the primary recall level.
5. Other doors on hold-opens (if provided) on all floors will also release upon detection of smoke.
6. Manual controls at the smoke control panel will have the highest control priority.

	Shut Down HVAC in Related Smoke Zones per Sequence of Operations	Shut Down Local (Related) HVAC Equipment	Close Smoke/Fire Dampers in Smoke Barriers per Sequence of Operations	Close Automatic Fire Doors, fire shutter, and Release Doors on Hold-Opens per Sequence of Operations	Activate alternative elevator hoistway protection where present per Sequence of Operations (e.g. close elevator lobby roll down shutter)	Initiate Stairwell Fan Pressurization (All Stairwell Fans)	Initiate Open Office Depressurization on Smoke Zone of Origin	Initiate Parking CO Ventilation (Smoke Purge)	Recall Elevators to Primary or Alternate Recall Level per Sequence of Operations	Initiate Evacuation Signal in Related Smoke Zone(s) per Sequence of Operations
INITIATING DEVICE (LABC 907.4)										
Waterflow										
Waterflow in P4 through P1 Smoke Zone	X		X	X	X	X		X	X	X
Waterflow in Level 1 through Level 3 Smoke Zone	X		X	X	X	X			X	X
Waterflow in Level 4 through Level 21 Smoke Zone (CAA Scope of Work)	X		X	X	X	X	X		X	X
Waterflow in Level 22 through 37 Smoke Zone	X		X	X	X	X	X		X	X
Waterflow in Roof Smoke Zone	X		X	X	X	X			X	X
Smoke Detection										
Duct Detectors at Supply Fan in any Smoke Zone	X	X	X	X	X	X			X	X
Duct Detector at Return Fan (non-smoke control) in any Smoke Zone	X	X	X	X	X	X			X	X
Smoke Detection in Elevator Lobby				X	X	X			X	X
Smoke Detection in Elevator Hoistway or Elevator Machine Room				X	X	X			X	X
Smoke Detection in P4 through P1 Smoke Zone	X	X	X	X	X	X		X		X
Smoke Detection in Level 1 through Level 3 Smoke Zone	X	X	X	X	X	X				X
Smoke Detection in Level 4 through Level 21 Smoke Zone (CAA Scope of Work)	X	X	X	X	X	X	X			X
Smoke Detection in Level 22 through 37 Smoke Zone	X	X	X	X	X	X	X			X
Smoke Detection in Roof Smoke Zone	X	X	X	X	X	X				X
Manual Controls										
Manual Pull Station						X				X

Figure 3 – Smoke Control Functional Matrix

5.2 Sequence of Operation

The smoke control system will be automatically activated by an appropriately zoned automatic sprinkler system complying with LABC Section 903.3.1.1, smoke detectors required by LABC 907, and manual controls that are readily accessible to the fire department (LABC 909.12.3). Table 3 provides the sequence of operations for the CAA smoke zone based on the location of the first initiating device.

Table 3 – Sequence of Operations

Smoke Zone ID	Location of First Initiating Device	Sequence of Operations
CAA Scope of Work Smoke Zone	Level 4 through Level 21	1. Initiate evacuation sequence and associated alarms per NFPA 72 and local requirements, i.e., evacuation signals to operate in the smoke zone of origin, the smoke zone above, and the smoke zone below. a. When the smoke zone of origin is CAA Scope of Work Smoke Zone, the evacuation sequence will sound on the Fire Floor, the Level above, and the Level below. 2. Start all stairwell pressurization fans serving smokeproof enclosures at Stairs 1, 2, 3, 5, 6, and 7. 3. Shut down all shared HVAC systems serving the smoke zone of origin and the immediately adjacent smoke zones. a. Supply and return systems with a capacity of greater than 2,000 cfm and provided with duct detectors in accordance with LABC 907.2.13.1.2 will shut down when the associated supply or return duct detector activates. b. From the Firefighter's Smoke Control Station diagram on sheet M7.0.15 fans that shut down should be Toilet Exhaust Fan EF-39-03; Tenant Exhaust Fan EF-39-02, and the Floor AHUs for the floor of alarm, the floor above, and the floor below. 4. Close combination fire/smoke dampers (CFSDs) serving common supply and exhaust shafts throughout. Specifically, close all toilet exhaust dampers. close tenant normal exhaust dampers on the fire floor, the level above, and the level below. 5. Open CFSD serving the smoke exhaust shaft on the fire floor only. 6. Start smoke exhaust fan serving the common smoke exhaust shaft, Smoke Exhaust SEF-39-01. 7. Release doors on hold-opens or initiate closure of automatic-closing fire assemblies at elevator lobbies, in the smoke zone of origin, and the immediately adjacent smoke zones.* a. When the smoke zone of origin is CAA Scope of Work Smoke Zone, close elevator lobby doors on the level above, the level below, and the fire floor.

Smoke Zone ID	Location of First Initiating Device	Sequence of Operations
		8. Deploy horizontal fire shutter for convenience stair zone and associated lobby on the floor of alarm. The two level convenience stair will close fire shutters on both the bottom and top of the convenience stair zone. 9. Recall elevators as described in the Smoke Control Functional Matrix. Ground Level is the primary recall level, and Level 2 is the secondary recall level (see elevator stop information in the Base Building SCRA for specific recall conditions).

* Door hold-open devices shall be connected to emergency power and/or provided with a listed power-interruption ride-through (battery/capacitor) so that momentary power loss or transfer does not cause nuisance release. Door hold-open devices shall release upon the applicable fire alarm/smoke control activation. Note that the ground floor may not be provided with elevator lobby doors.

5.3 Activation of Subsequent Devices

Upon receipt of subsequent automatic signals from an initiating device from any smoke zone anywhere in the building, no change to the automatic system configuration will occur. Only a single active smoke zone, in addition to pressurized smokeproof enclosures throughout the project, is intended to automatically activate. This rational analysis does not intend that multiple smoke zones within the tower be simultaneously operational.

5.4 Manual Control

Under normal power, manual controls will allow for independent or simultaneous operation of any or all smokeproof enclosures serving the project and any single smoke zone. Under standby power, a single active smoke zone and all smokeproof enclosures will be capable of operating simultaneously via manual controls. Manual activation of smoke control zones at the firefighter's control panel will result in the same sequence of operations as automatic activation. It is not the intent of this document that multiple floor smoke zones be capable of operating simultaneously under standby power conditions.

6. SMOKE CONTROL EQUIPMENT

Equipment including, but not limited to, fans, ducts, automatic dampers, and balance dampers are required to be suitable for their intended use, suitable for the probable exposure temperatures that the rational analysis indicates, and as approved by the fire code official (LABC 909.10).

6.1 Control Systems

- Control systems for mechanical smoke control systems shall include provisions for verification. Verification shall include positive confirmation of actuation, testing, manual override, the presence of power downstream of all disconnects, and through a preprogrammed weekly test sequence, reporting abnormal conditions audibly, visually, and by printed report.
- Detection and control systems must be clearly marked at all junctions, accesses, and terminations.
- Control systems shall be equipped with a control unit complying with UL 864 and listed as smoke control equipment.
- Variable speed fan drive controls serving smoke control equipment shall have a lock-out capability so that unauthorized personnel cannot reset or change settings (e.g., Hertz settings).

6.2 Control Air and Differential Pressure Verification Tubing (LABC 909.13)

Control air and differential pressure verification tubing will be provided at variable speed drive fans used to pressurize smokeproof enclosures.

- Control air tubing is required to be of sufficient size to meet the required response times.
- When used, tubing must be flushed clean and dry prior to final connections and is required to be adequately supported and protected from damage. Tubing passing through concrete or masonry must be sleeved and protected from abrasion and electrolytic action.
- Control air tubing is required to be hard-drawn copper, Type L, ACR¹ in accordance with ASTM B42, ASTM B43, ASTM B68, ASTM B88, ASTM B251, and ASTM B280. Fittings are required to be wrought copper or brass, solder type in accordance with ASME B16.18 or ASME B16.22. Changes in direction are required to be made with

¹ Air conditioning and refrigeration

appropriate tool bends. Brass compression-type fittings must be used at final connection to devices; other joints are required to be brazed using a BCuP5 brazing alloy with solidus above 1,100°F (593°C) and liquids below 1,500°F (816°C). Brazing flux may be used on copper-to-brass joints only.

6.3 Combination Fire Smoke Dampers (LABC 909.10.4 and 909.12)

CFSDs will be provided at shafts serving common supply and exhaust shafts in the building.

- Automatic dampers, regardless of the purpose for which they are installed within the smoke control system, shall be listed and conform to the requirements of approved, recognized standards.
- The status of dampers shall be determined using limit or proximity switches installed at the damper or incorporated into the damper actuator. Where multiple dampers are grouped together in an assembly requiring one or more actuators, each damper shall be independently controlled by a separate actuator and provided with an individual limit or proximity switch, or the dampers shall be linked together by a reliable and durable mechanical or otherwise permanent means into one or more groups, with each group provided with a common limit or proximity switch.
- Damper status is required for CFSDs in shafts serving common fresh air systems. Dampers are required to be monitored for power and position; dampers that fail safe upon loss of power will not be required to be monitored for power.
- Smoke damper leakage ratings are required to be no less than Class II. Elevated temperature ratings may not be less than 250°F (121°C).
- As possible, dampers will utilize external actuators and linkages.
- Dampers will be monitored for position, regardless of the final position (open or closed) of such dampers.

6.4 Doors

Automatic closing doors on hold-opens or self-closing doors will be used throughout passive and active areas of the project.

- Door assemblies in fire-resistive assemblies will comply with LABC 716, which references NFPA 80. A 3/4 in. gap will be allowed at the bottom of the door at the time of installation.
- In accordance with LABC 1010.1.3, the force for pushing or pulling open interior swinging egress doors, other than fire doors, may not exceed 5 lbs (22 N). For other swinging doors, as well as sliding and folding doors, the door latch must release when

subjected to a 15 lb (67 N) force. The door is required to swing to a full-open position when subjected to a 15 lb (67 N) force.

- The maximum pressure differential allowed at stairs for the swing force (excluding the force to set the door in motion, i.e., break the seal) is 0.17 in. water gage, unless field adjustments permit.
- Where the pressure differential on exit doors exceeds 0.17 in. water gage due to stack effect and the swing door force exceeds 15 lbs, provisions for the installation of ADOs (motorized door openers) will be made.
- ADOs will also be installed at all doors that connect the smokeproof enclosures to the exterior (e.g., at the exit discharge door and on the amenity deck). ADOs will be provided with dedicated emergency circuits and breaker locks and will meet applicable requirements of LABC Chapter 10 and NFPA 80.
- Heavy duty adjustable door closers capable of operating under dynamic pressure conditions (e.g., pressure differentials in excess of 0.1 in.) will be installed at automatic closing elevator lobby doors.
- Where magnetic door hold-open devices are used, provide a listed power-interruption ride-through (battery/capacitor) or emergency power so that momentary power loss or transfer does not cause nuisance release. Door hold-open devices shall release upon the applicable fire alarm/smoke control activation.
- If used, accordion door assemblies shall be capable of resisting an air pressure differential to 0.1 in. water column with the addition of anti-sway brackets placed interior to the door and lead post outriggers.
- Fixed door sweeps will be provided in lieu of adjustable door sweeps at all doors across which a pressure gradient is being measured.

6.5 Ducts

Ducts used as part of the stair pressurization will meet the requirements of LABC 909.

- Duct materials and joints are required to be capable of withstanding the probable temperatures and pressures to which they are exposed as determined in accordance with Section 909.10.2.
- Ducts are required to be constructed and supported in accordance with the Los Angeles Municipal Code. Positively pressurized ducts that pass through multiple smoke zones are required to be leak tested to 1.5 times the maximum design pressure in accordance with nationally accepted practices. Measured leakage may not exceed 5% of design flow. Results of such testing will be a part of the documentation procedure.

- Ducts are required to be supported directly from fire resistance rated structural elements of the building by substantial, noncombustible supports.

6.6 Power Systems (LABC 909.11)

All fans, dampers, and controls (including initiating devices and notification appliances) in connection with the smoke control system will meet the provisions of LABC 909.11.

- The smoke control system will be supplied with two sources of power. Primary power will be from the normal building power systems. Secondary power will be from an approved standby source complying with LABC Chapter 27.
- The standby power source and its transfer switches are required to be in a room separate from the normal power transformers and switch gear and ventilated directly to and from the exterior. The room will be enclosed with not less than one-hour fire barriers constructed in accordance with LABC Section 707 or horizontal assemblies constructed in accordance with LABC Section 712, or both.
- The transfer to full standby power is required to be automatic and within 60 sec of failure of the primary power.
- Elements of the smoke management system relying on volatile memories or the like must be supplied with uninterruptible power sources of sufficient duration to span a 15-minute primary power interruption. Door hold-open devices on magnetic door hold-opens should also have battery backup or similar uninterruptible power supplies.
- Elements of the smoke management system susceptible to power surges are required to be suitably protected by conditioners, suppressors, or other approved means.
- An approved standby power-generating system shall be provided. The system shall be equipped with a suitable means for automatically starting the generator system upon failure of the normal electrical service and for automatic transfer and operation of all of the required electrical functions at full power within 60 sec of such service failure.

6.7 Detection and Wiring (LABC 909.12 and 909.14)

- Fire detection systems providing control input or output signals to mechanical smoke control systems or elements thereof shall comply with the requirements of LABC 907.
- In addition to meeting requirements of the California Electrical Code, all wiring, regardless of voltage, shall be fully enclosed within continuous raceways (LABC 909.12).

6.8 Firefighter's Smoke Control Panel (LABC 909.16)

A single firefighter's smoke control panel will be provided for the project. The panel will be located in a fire control room located on the ground level (Level 1) and will comply with

Section 911. The firefighter's smoke control panel will comply with Sections 909.16.1 to 909.16.3 and any specific LAFD requirements pertaining to the use of lights and switches on the panel.

- The firefighter's smoke control panel is required to comply with all the requirements of LABC 909.16.
- The firefighter's smoke control panel must monitor and be capable of overriding the Mechanical Test and Inspect Panel.

6.8.1 Status Indication

Status indicators are required to be provided for all smoke control equipment, annunciated by fan and zone and by approved indicators as follows per LABC 909.16.1:

- Fans, dampers, and other operating equipment in their OFF or CLOSED status—RED
- Fans, dampers, and other operating equipment in their ON or OPEN status—GREEN
- Fans, dampers, and other operating equipment in a FAULT status—YELLOW/AMBER

The firefighter's smoke control panel is required to provide control capability over the complete smoke control system equipment within the building as follows per LABC 909.16.2:

- ON-AUTO-OFF control over each individual piece of operating smoke control equipment that can also be controlled from other sources within the building. This includes stairway pressurization fans; smoke exhaust fans; supply, return and exhaust fans; elevator shaft fans; and other operating equipment used or intended for smoke control purposes.
- OPEN-AUTO-CLOSE control over individual dampers relating to smoke control and that are also controlled from other sources within the building.
- ON-OFF or OPEN-CLOSE control over smoke control and other critical equipment associated with a fire or smoke emergency and that can only be controlled from the firefighter's control panel.

Controls and indicators may be combined on a per zone basis within the firefighter's smoke control panel as approved by the LAFD. The firefighter's smoke control panel will be specified and designed by others.

6.8.2 Control Actions and Priorities

The following are key requirements pertaining to control action and priorities within the smoke control system (LABC 909.16.3):

- ON-OFF and OPEN-CLOSE control actions will have the highest priority of any control point within the building. Once issued from the firefighter's smoke control panel, no automatic or manual control from any other control point within the building may contradict the control action.
- Manual activation of the smoke control system at the firefighter's smoke control panel is required to result in the same sequence of operations as automatic activation.
- Where automatic means are provided to interrupt normal, nonemergency equipment operation or produce a specific result to safeguard the building or equipment (e.g., duct freezestats, duct smoke detectors, high-temperature cutouts, temperature-actuated linkage and similar devices), such means are required to be capable of being overridden by the firefighter's control panel.
- The last control action as indicated by each firefighter's smoke control panel switch position will prevail.
- In no case will control actions require the smoke control system to assume more than one configuration at any one time.
- Only the AUTO position of each three-position firefighter's smoke control panel switch will allow automatic or manual control action from other control points within the building. The AUTO position will be the NORMAL, nonemergency, building control position.
- Where a firefighter's smoke control panel is in the AUTO position, the actual status of the device (on, off, open, closed) will continue to be indicated by the status indicator described above.
- When directed by an automatic signal to assume an emergency condition, the NORMAL position will become the emergency condition for that device or group of devices within the zone. In no case will control actions require the smoke control system to assume more than one configuration at any one time.

6.9 System Response Time (LABC 909.17)

- Smoke control system activation shall be initiated immediately after receipt of an appropriate automatic or manual activation command. Smoke control systems shall activate individual components (such as dampers and fans) in the sequence necessary to prevent physical damage to the fans, dampers, ducts, and other equipment.

- For purposes of smoke control, the firefighter's control panel response time shall be the same for automatic or manual smoke control action initiated from any other building control point.
- The total response time, including that necessary for detection, shutdown of operating equipment and smoke control system startup, shall allow for full operational mode to be achieved before the conditions in the space exceed the design smoke condition. Upon receipt of an alarm condition at the fire alarm control panel, fans, dampers, and automatic doors shall have achieved their proper operating state and the final status shall be indicated at the smoke control panel within 90 sec.
- The system response time for each component and their sequential relationships shall be detailed in the required rational analysis and verification of their installed condition reported in the required final report.

6.9.1 Weekly Testing

A preprogrammed weekly test sequence is required to report abnormal conditions audibly, visually, and by printed report. The preprogrammed weekly test must operate all devices, equipment, and components used for smoke control.

7. SPECIAL INSPECTION AND COMMISSIONING

The smoke control systems are subject to the provisions of Section 909 of the LABC and are required to undergo special inspections and tests sufficient to verify the proper commissioning of the smoke control design in its final installed condition. The design submission accompanying the construction documents must clearly detail procedures and methods to be used, as well as the items subject to the inspections and tests. The commissioning will be in accordance with generally accepted engineering practice and, where possible, based on published standards for the particular testing involved. The special inspections and tests required by this section will be in compliance with Section 909.3 of the LABC.

Refer to Appendix F for details of compliance pertaining to special inspections.

8. LIMITATIONS

At the request of the Client, Simpson Gumpertz & Heger Inc. (SGH) prepared this rational analysis. References to key applicable standards and specifications are listed in this report. The requirements and strategies listed here are consistent with the drawings available as of February 2026.

Implementation of recommendations presented herein will require the development of architectural/engineering design and specifications by others. Final determination of code compliance will be made by the Authority Having Jurisdiction, whose interpretation of code provisions may differ from that of SGH. Evaluation or review of any facilities system(s) feature(s) or component(s) not specifically included in this report are beyond the scope of this narrative.

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Appendix A

Rationality

The following rational analysis supports the types of smoke control systems to be employed, their methods of operation, the systems supporting them, and the methods of construction to be utilized.

Stack Effect – LABC 909.4.1

Stack effect is most readily observed in shafts, including stair enclosures, hoistways, and ventilation shafts. Stack effect is driven by the temperature differential between the outside air and the air within the building due to the effect of temperature on the relative buoyancy of the air in those areas and the corresponding pressure differentials that cause air to rise or fall.

Climatic data considered in this analysis are based on the Los Angeles area and are consistent with the Base Building SCRA (e.g., ASHRAE weather station data from Santa Monica Municipal Airport). A 1% dry bulb winter temperature of 44.4°F and a 1% dry bulb summer temperature of 83.7°F were used as the basis of evaluation of normal and reverse stack effect. The stack effect has been modeled using the network pressurization model CONTAM. Stack effect is described in Principles of Smoke Management.

For the building in question, the greatest temperature gradient is expected when the inside temperature of the building is 68°F² and the corresponding 1% dry bulb winter temperature is 44.4°F. The 23.6°F gradient will result in greater development of normal stack than the corresponding 10°F to 30°F reverse stack condition (depending on the degree to which interior air conditioning impacts inside temperatures during the periods of higher outside temperatures).

Modeling Inputs

Airflows within the modeled building are impacted by various input parameters.

- **Building Geometry:** The geometry input in the CONTAM model will be based on the base building architectural drawings dated 20 December 2024. Many additional drawings were utilized for the analysis including CAA tenant mechanical and architectural drawings. Refer to the plans for all dimensions used as inputs into the model.

² Max Indoor Design Comfort Temp, ASHRAE Std 55, Thermal Environmental Conditions for Human Occupancy.

- **Smoke Zones:** Stairwells and stair vestibules, elevators, corridors, and open office areas are explicitly modeled. Other rooms, including but not limited to, electrical rooms, service rooms, restrooms, and the like are implicitly modeled by including compartments with equivalent area, volume, and leakage to the represented room. Refer to Appendix E for screenshots of the CONTAM model.
- **Supply Air Quantity:** Section 4 of the report provides the supply flow rates for each smoke zone.
- **Leakage:** The leakage areas for building components provided in Table A-1 are used in the CONTAM model based on the data provided in *Principles of Smoke Management*³ and *Handbook of Smoke Control Engineering*.⁴ As applicable, the leakage areas used in the CONTAM model comply with the maximum leakage areas required for smoke barriers per LABC 909.5. The leakage area around a door depends upon the fit of the door to the door frame. Typical 36 in. wide doors contain leakage values ranging from 0.090 to 0.414 sq ft.⁵ While the doors will most likely be tight fitting, a leakage area of 0.385 sq ft per door is used in the model for stairway doors, and leakage of 0.53 sq ft is used for hoistway doors. The leakage values used for elevator doors, walls, and floors represent average to loose construction.⁶ All leakages and openings are implemented in the model using a mass flow power-law formulation for one-way flow.⁷

Note: Door leakage assumptions used in the CONTAM model should be reconciled with the specified door assembly leakage ratings (e.g., UL 1784 where applicable) and field-installed gasketing/sweeps. If installed leakage is lower than assumed, fan controls may require differential-pressure feedback (e.g., VFD modulation) to limit maximum ΔP and door opening forces.

- **Ambient Temperature:** The ambient temperature of all spaces within the building was assumed to be 68°F.⁸ Extreme exterior temperature conditions were used to verify the performance of the smoke control system in the range of expected temperatures. A winter low temperature of 44.4°F and a summer temperature of 83.7°F were used in the model.⁹

³ Klote, J.H. and J. Milke, *Principles of Smoke Management*, ASHRAE and SFPE, 2002, Table 6.2: 98-99.

⁴ Klote, J.H., J. Milke, Turnbull, Kashef, and Ferreira, *Handbook of Smoke Control Engineering*, 2012, ASHRAE and SFPE, Table 11.3, 251

⁵ Klote, J.H. and J. Milke, *Principles of Smoke Management*, ASHRAE and SFPE, 2002.

⁶ Klote, J.H. and J. Milke, *Principles of Smoke Management*, ASHRAE and SFPE, 2002.

⁷ Walton, G.N., and W.S. Dols, *CONTAM User Guide and Program Documentation*, NIST 7251, NIST, 2013.

⁸ Max Indoor Design Comfort Temp, ASHRAE Std 55, Thermal Environmental Conditions for Human Occupancy.

⁹ ASHRAE, *ASHRAE Handbook – Fundamentals*, American Society of Heating, Refrigerating and Air Conditioning Engineers, Inc., Atlanta, GA, 2001.

Table A-1 – Building Component Leakage Data

Building Component	Leakage Area	Flow Exponent (n)
Door (assumed 3 ft wide)	0.385 sq ft per door	0.65
Elevator door (assumed 3.5 ft wide)	0.53 sq ft per door	0.65
Interior wall	0.0018 sq ft per 1 sq ft of wall area	0.65
Exterior wall	0.00017 sq ft per 1 sq ft of wall area	0.65
Corridor wall (fire partition)	0.00035 sq ft per 1 sq ft of wall area	0.65
Floor assembly	0.00017 sq ft per 1 sq ft of floor area	0.65
Exit stairwell wall	0.00011 sq ft per 1 sq ft of wall area	0.65
Elevator shaft wall	0.00084 sq ft per 1 sq ft of wall area	0.65

Modeling Outputs

A summary of pressurization capacities and key differential pressure results obtained from the CONTAM model is to be provided in Appendix D (CONTAM Results). The CONTAM analysis is intended to demonstrate that a minimum pressure difference of 0.05 in. water gage at specific building elements per LABC 909.6.1 is achieved with the calculated supply and exhaust quantities for all scenarios evaluated, while maintaining maximum pressure gradients as described herein.

Temperature Effect of Fire – LABC 909.4.2

Temperature effects associated with a fire can compromise pressurization systems designed to contain and limit smoke movement when the gas temperatures become too high to effectively counter with airflows across openings. Specifically, a fully involved fire in a non-sprinklered space can produce pressures in the range of 0.10 in. water gage (assumes temperatures of 1,600°F in a room with a 10 ft ceiling). Whereas this is twice the pressure required by LABC 909, smoke migration would be expected from a non-sprinklered fire even if the minimum pressure gradients required by LABC 909.6 were maintained.

For a sprinkler-controlled fire, pressures in the range of 0.01 to 0.02 in. water gage can be expected to occur as a result of buoyancy and expansion.¹⁰ These effects will generally be most readily observed in the fire compartment of origin, though some effects may be observed remote from the fire if alternative measures, such as compartmentation or mechanical smoke

¹⁰ Klote, J.H. and J. Milke, *Principles of Smoke Management*, ASHRAE and SFPE, 2002: 73-74.

removal, are not implemented. Figure A-1 depicts pressures as a function of buoyancy as a function of the fire compartment temperature. Therefore, for the purposes of pressurization modeling, a sprinkler-controlled fire that maintains average room temperatures at or below 100°F has been assumed as the basis for smoke control systems analysis.

Temperature effects of fire can also impact the movement of heated gases in large spaces such as atriums. Because the project in question does not contain large atriums, this is not a significant concern at the project.

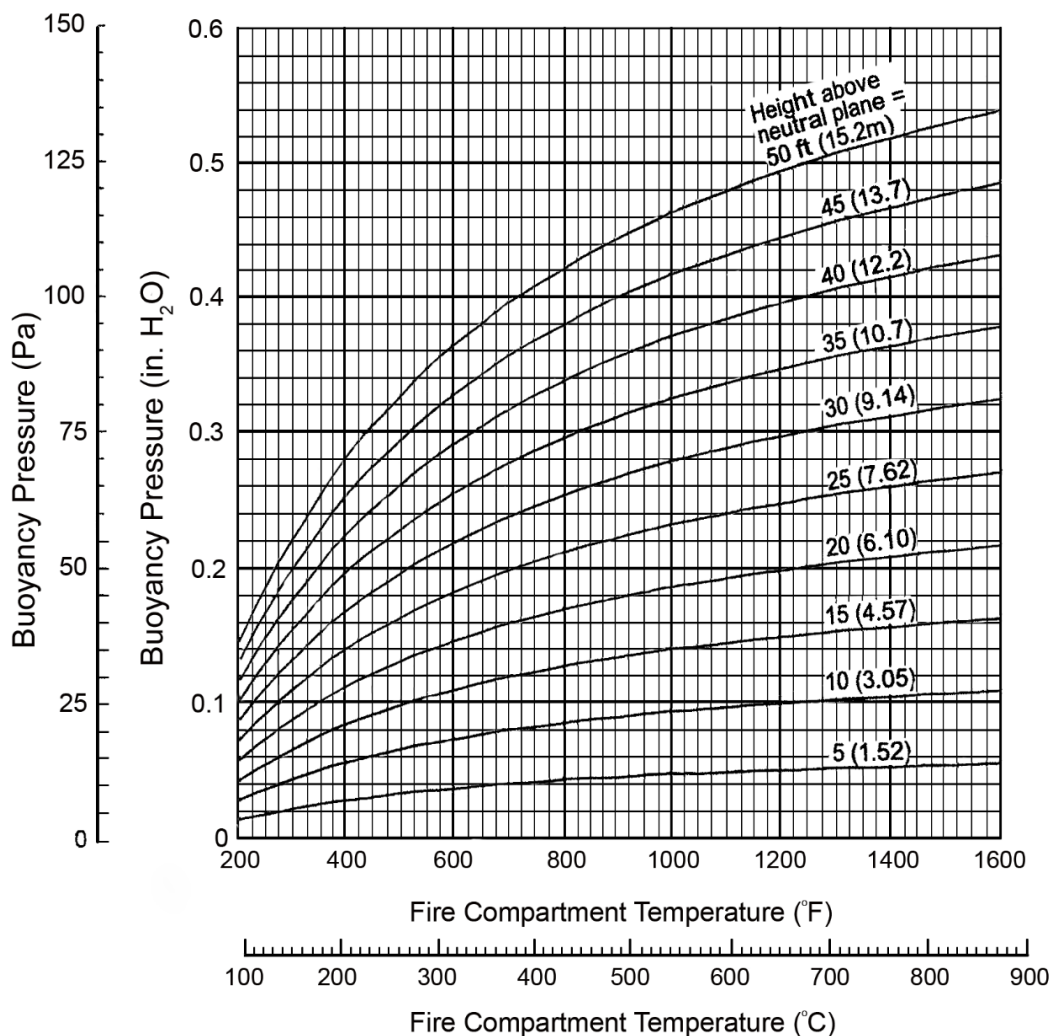


Figure A-1 – Pressures as a Function of Buoyancy¹¹

¹¹ Klotz, J.H. and J. Milke, *Principles of Smoke Management*, ASHRAE and SFPE, 2002, Table 5.1.

Design Fire Characterization

To design a smoke control system, a reasonably expected fire size must be determined (either quantitatively or qualitatively) to ensure that (1) the heat release rates (HRRs) of expected design fires can be accommodated by active and passive smoke control systems, and (2) the assumptions pertaining to average room and/or smoke zone temperatures are valid and within the bounds of what can be accommodated by the smoke control system. In the context of a pressurization analysis, it is notable that the fire protection methods of the applicable codes and standards assume that multiple simultaneous fire incidents will not occur. No design fires that would change the assumptions made regarding appropriate pressurization requirements were identified.

Maximum Heat Release Rate – Sprinkler-Controlled Design Fires

LABC 909.9.4 and its reference standard, NFPA 92, allow an engineering analysis to assume that fire growth is halted at the time of sprinkler activation. For each typical active smoke zone, ceiling height and sprinkler spacing are correlated with the fire growth rate to calculate corresponding HRR. As such, the HRR curve may be represented as an exponentially growing fire that becomes a steady state at the time of sprinkler activation.

In accordance with LABC 909.9, additional factors considered in the quantification of the design fire include the characteristics of the fuel, fuel load, (including spacing and configuration of fuel items), effects included by the fire, and whether the fire is likely to be steady or unsteady.

Available data from NFPA 92 and other published peer-reviewed sources have been used as the basis of the fire scenario selection. Sprinkler effectiveness assumptions are based on the heat detection algorithm of Heskestad and Delichatsios as documented in the Society of Fire Protection Engineers (SFPE) Handbook¹² and the Fire Dynamics Simulator (FDS) sprinkler activation algorithm.¹³ Figure A-2 depicts HRR as a function of the ceiling height. Whereas

¹² Schilfiliti, R.P., B.J. Meacham, and R.L.P. Custer, "Design of Detection Systems," *The SFPE Handbook of Fire Protection Engineering*, 3rd Edition, Society of Fire Protection Engineers, 2002: 4-10.

¹³ McGrattan, K., S. Hostikka, J. Floyd, H. Blum and R. Rehm, *NIST Special Publication 1018-5, Fire Dynamics Simulator (Version 5) Technical Reference Guide*, 1 Oct. 2007.

ceiling heights within the project are generally between 10 ft and 20 ft, maximum design fire HRRs would be expected to be less than 2,000 BTU/s.

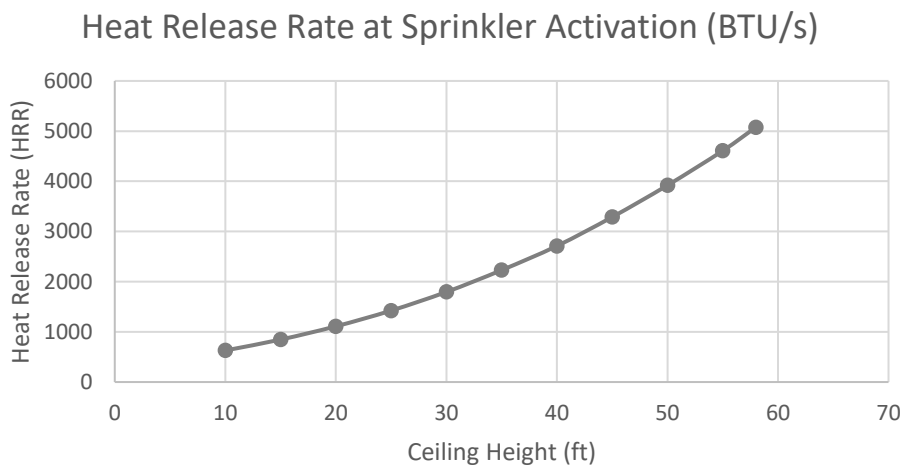


Figure A-2 – Heat Release Rate at Sprinkler Activation as a Function of Ceiling Height

Fire Growth Rate

The HRR of a fire scenario is correlated to fuel types. Design fires are typically characterized by “t-squared” growth rates, as shown in Figure A-3 (from Annex B to the 2018 Edition of NFPA 92). The growth of an unsteady design fire is based on fire test data in accordance with Section 5.2 of NFPA 92. For this analysis, we have approximated a “fast” growth rate for both fires based on the anticipated combustibles within the space and because such fires would be expected to pose the greatest challenge with respect to fire exposure. For potential fuel packages that are controlled by sprinklers, we expect the HRR to peak and then decay, as shown in Figure A-3. For larger fuel packages where shielding of the fire source may be present, it is sometimes appropriate to evaluate a steady state HRR following activation of the fire sprinklers.

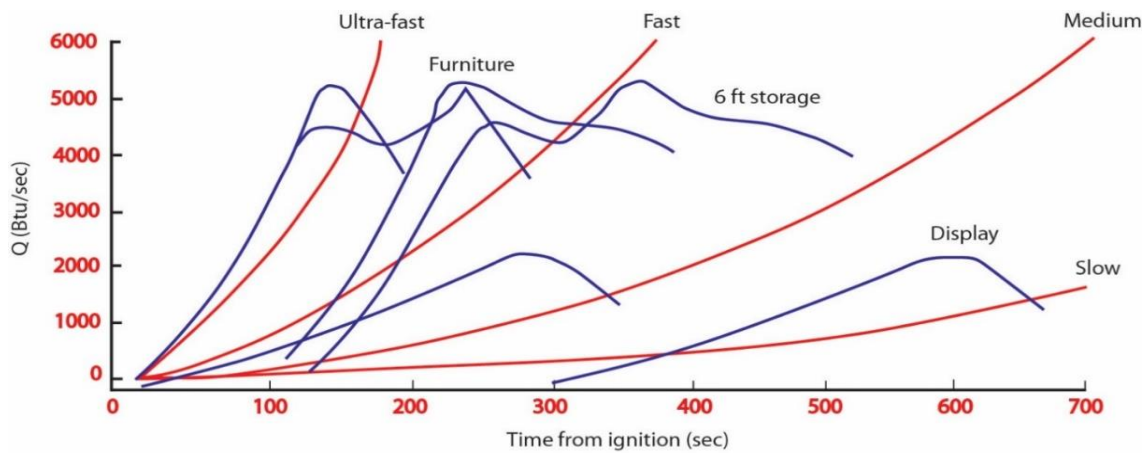


Figure A-3 – Rates of Energy Release in a t-Squared Fire

Soot Yield

Soot yield specification of typical office occupancy design fires is based on a mixture of polyurethane foam and synthetics with cellulosic material, providing an effective soot yield of 5.0% (0.050g/g) for typical combustible fuel packages, such as those used in Design Fires 1-4.¹⁴ Soot yield of an advertisement panel is based on polycarbonate material, which has a soot yield of 11.2% (0.112 g/g) and is used for Design Fires 5-6.¹⁵ A soot yield of greater than 0.05 g/g is considered a conservative soot yield for the application in question.

Heat of Combustion

The heat of combustion in connection with the above-referenced fuel packages is assumed to be 13,100 kJ/kg. This value represents a fuel load comprising cellulosic materials and expended polyurethane foam. It is notable that in the context of pressurization modeling, heat of combustion does not play a direct role in the analysis of smoke containment, except that it must be within the bounds of combustible heat loads that could reasonably be expected to be controlled by an automatic fire suppression system.

¹⁴ Tewarson, A, "Generation of Heat and Chemical Compounds in Fires," *The SFPE Handbook of Fire Protection Engineering*, 3rd Edition, Society of Fire Protection Engineers, 2002, Table 3-4.14.

¹⁵ The SFPE Handbook of Fire Protection Engineering, 5th Edition, Society of Fire Protection Engineers, Table A.39.

Wind Effect – LABC 909.4.3

Wind effects may be observed in a number of ways for this building type:

- Stairwell relief may be impacted negatively if the relief vents are located such that they are facing the prevailing wind direction.
- Wind entering the building lobby on the ground floor can pressurize the lobby, potentially leading to greater airflow through elevators.

Wind issues are generally addressed via self- or automatic-closing assemblies via passive smoke control methods (i.e., intervening barriers) or by way of exterior facades that limit the effects of the pressures generated from wind on the internal building systems.¹⁶ In particular, the adjustment capability and force provided by hydraulic or spring-closing assemblies is critical and must be specified by the architect to allow for a high degree of flexibility (and adjustability) in the field.

Climatic data considered in this analysis are based on a nearby ASHRAE weather station (Santa Monica Municipal Airport) and are consistent with the Base Building SCRA. The climatic data recorded in the 2012 Edition Handbook of Smoke Control, Chapter 2, Table 2.1 indicate that the fastest wind speed (99th percentile) is 14.5 mph, which is used as the basis for wind pressurization modeling. The winter and summer design temperatures (dry bulb) are listed as 43.3°F for 99.6% annual cumulative frequency of occurrence and 85.7°F for 0.4% annual cumulative frequency of occurrence, respectively. The terrain category is 1, corresponding to a city center with 50% of the buildings having a height greater than 70 ft. The prevailing wind direction is generally from the southwest toward the northeast (SW to NE), as shown in Figure A-4. A wind coefficient of 0.25 (positive or negative depending on whether it is applied to the windward or leeward side of the building) and outside air density of 0.075 lb/ft³ are assumed.

Principles of Smoke Management, Eqns 5.34 and 5.39, generally describes the effect of wind on the generation of pressure across the face of the building as a function of height.¹⁷ For high-rise buildings having an open office configuration and/or sealed facade, the effect of wind is

¹⁶ Klote, J.H., *Design Of Smoke Control Systems for Elevator Fire Evacuation Including Wind Effects*, 2nd Symposium on Elevators, Fire and Accessibility, Baltimore, MD, April 19-21, 1995.

¹⁷ Klote, J.H. and J. Milke, *Principles of Smoke Management*, ASHRAE and SFPE, 2002.

generally expected to be minimal. Wind pressurization on the windward side of the building is generally balanced out by depressurization on the leeward side of the building for open floor plans. Infiltration and exfiltration are expected to be moderate due to the anticipated tightness of the facade.

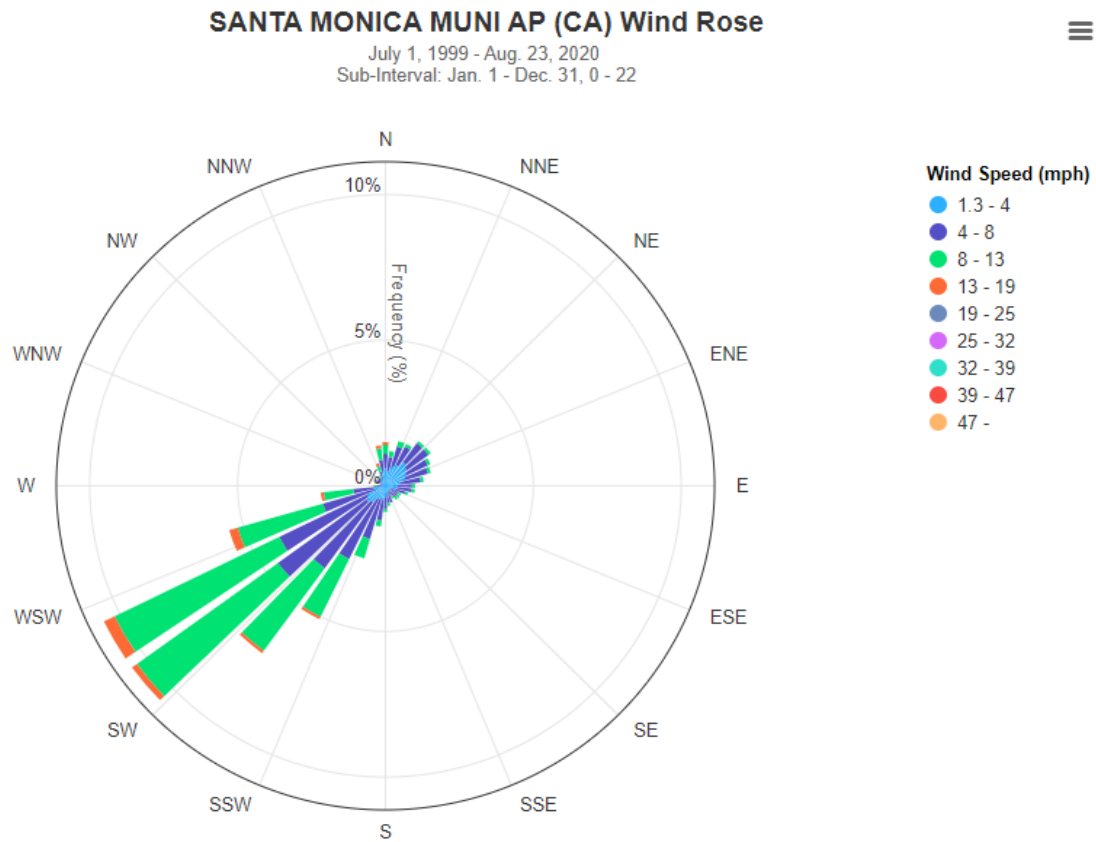


Figure A-4 – Santa Monica Municipal Airport Prevailing Wind Direction (SW to NE)

HVAC Systems – LABC 909.4.4

Mechanical heating, ventilation, and air conditioning (HVAC) systems should be shut down in the smoke zone of origin and adjacent smoke zones to minimize unwanted smoke movement in accordance with LABC 909.4.4. Specific actions are identified in the sequence of operations provided herein. It is notable that if duct detection is used for HVAC units, such detection is required to be provided on both supply and return sides of each HVAC unit.

Elevator Piston Effect

Whereas elevator piston effect would be expected to be an issue where elevators are normally operating during a fire event, elevators at the subject building will recall to the primary or alternate recall level and are not, as such, expected to generate significant pressure differentials during the initial evacuation time period. Additionally, the use of elevator lobbies with multi-car hoistways (that permit air movement within the shaft when an elevator car is moving) will significantly reduce the impacts of piston effect.¹⁸ Note that the project is not provided with any elevator hoistway relief vents as permitted by LABC Chapter 30. If relief vents are provided, such vents will need to be operable and will be assumed to be in the closed position during smoke control operations.

Climate – LABC 909.4.5

The design has been based on the effects of low and high temperatures on systems, property, and occupants. Air inlets and exhausts have been located by the Mechanical Engineer of Record so as to minimize the likelihood of reintroduction of smoke into the supply fans from exhaust fans as a function of wind, which is generally expected to be from the west, as noted in this appendix.

Duration of Operation – LABC 909.4.6

All portions of active or passive smoke control systems shall be capable of continued operation after detection of the fire event for a period of not less than either 20 min. or 1.5 times the calculated egress time, whichever is greater. For the project, a timed evacuation study has not been performed, but it is expected that occupants will be able to evacuate the smoke zone of origin within 20 min. in keeping with the intent of the prescriptive codes (LABC 909.4.6). The system is intended to operate continuously for at least eight hours under normal and emergency power conditions.

Evacuation times for high-rise buildings can vary significantly depending on numerous factors, such as the building's design, occupancy, egress configuration, alarm system, and evacuation

¹⁸ Klotz, J.H. and J. Milke, *Principles of Smoke Management*, ASHRAE and SFPE, 2002:66, 79, 129, 160.

strategies in place. Generally, the LABC and NFPA 101 (Life Safety Code) provide guidelines and standards to ensure that evacuation is safely executed, but they may not specify exact evacuation times.

- **Stairwell Capacity:** The width and design of the stairwells can influence the flow rate. NFPA 101 (Life Safety Code) provides guidelines on minimum stairwell width based on occupancy type. For an office building, this might typically allow for a flow rate of around one person per second per 22 in. of stairwell width.
- **Occupant Load and Behavior:** Times can vary depending on the number of occupants and their familiarity with the evacuation procedures.
- **Egress Pathways:** The design and number of stairwells, corridors, and other egress components can significantly influence evacuation times. Accessibility and clear signage also play a role.
- **Alarm Systems and Evacuation Strategies:** Advanced alarm systems and well-rehearsed evacuation plans can reduce times.

A study conducted by the National Institute of Standards and Technology (NIST) entitled "Evacuation of High-Rise Buildings" (NIST Technical Note 1624) notes that the average evacuation rate might be around 0.2 floors per minute per stairwell, but this is highly dependent on the factors mentioned above. For an eighteen-story building, assuming all other conditions are ideal, an estimated evacuation time might be around 30 to 60 min., but this can vary widely. The eight-hour duration is sufficiently longer than any estimate of total evacuation time, however.

Appendix B
Smoke Control Calculation Results

Stair Pressurization

Estimation Report — ASHRAE HSCE / NFPA 92 / IBC 909

Generated: February 20, 2026 06:04 AM
SGH (Simpson Gumpertz & Heger)

Section A — Input Summary

Min ΔP (closed): **12.50 Pa (0.0502 in. w.g.)** | Max ΔP (closed): **87.00 Pa (0.3493 in. w.g.)** | Max door force: **133.0 N (29.9 lbf)** | Floor exhaust ΔP: **25.00 Pa (0.1004 in. w.g.)**

Building & Conditions

Floors above grade (N)	39
Floors below grade (N_b)	4
Floor-to-floor height (h_f)	4.27 m
Building perimeter (P_bldg)	185 m
Floor area (A_floor)	2140 m²
Wall construction	average
Outdoor winter temp (T_o,w)	6.9°C
Outdoor summer temp (T_o,s)	35°C
Indoor temp (T_i)	22°C
Fire floor temp (T_f)	300°C
Wind speed (V_w)	0 m/s
Wind direction	40°
Atmospheric pressure	101325 Pa
Open doors (n_open)	1
Sprinklered	Yes
Design fire HRR	2000 kW
Fire floor	21
Stairwell temp assumption	outdoor

Elevator Shafts

Number of shafts	14
Shaft area	22.6 m²
Door type	center-opening
Vent area	0 m²

Leakage Tightness

Component	Rating
Exterior wall	average
Interior wall	average
Floor/ceiling	average
Stair door	average
Elevator door	average

Stairwell Geometry

Stair	Area	Door (W×H)	Gap	Doors/Floor	Serves	Ext Walls
OpenStair	81.8 m²	1.1 × 2.1 m	3 mm	1	8–22	0
Stair1	22.4 m²	1.1 × 2.1 m	3 mm	1	1–42	0
Stair2	19.3 m²	1.1 × 2.1 m	3 mm	1	1–42	1
Stair3	20.6 m²	1.1 × 2.1 m	3 mm	1	1–41	1
Stair5	34.8 m²	1.1 × 2.1 m	3 mm	1	1–7	1

Stair	Area	Door (W×H)	Gap	Doors/Floor	Serves	Ext Walls
Stair6	35.8 m ²	1.1 × 2.1 m	3 mm	1	1–7	1
Stair7	31.3 m ²	1.1 × 2.1 m	3 mm	1	1–7	1
Stair8	28.9 m ²	1.1 × 2.1 m	3 mm	1	5–7	1
Stair9	26.8 m ²	1.1 × 2.1 m	3 mm	1	5–7	1

Calculation Methodology — ASHRAE HSCE

This section presents the analytical procedure used to determine the minimum stair pressurization supply air and fire-floor exhaust rates. The methodology follows the algebraic equation method from ASHRAE *Handbook of Smoke Control Engineering* (HSCE), with design criteria from NFPA 92 and IBC Section 909. Equation numbers (EQ-01 through EQ-26) correspond to those in the Calculation Trace.

1. Design Objective

The goal of stair pressurization is to maintain a positive pressure differential between each pressurized stairwell and the adjacent building floor, preventing smoke migration into the stairwell during a fire. Simultaneously, the system must not over-pressurize the stairwell to the point where occupants cannot open egress doors. These competing requirements define the design window:

Min stairwell-to-corridor ΔP	12.5 Pa (0.05 in. w.g.) — IBC 909.20.5.1
Max stairwell-to-corridor ΔP	87.0 Pa (0.35 in. w.g.) — NFPA 92 §4.4.2.1
Min open-door air velocity	1 m/s — IBC 909.20.5.2
Max door-opening force	133 N (30 lbf) — IBC 1010.1.3
Fire-floor exhaust ΔP	25.0 Pa (0.10 in. w.g.) — IBC 909.20.6

2. Air Properties

All flow and pressure calculations depend on air density, which varies between the fire floor, the building interior, the outdoor environment, and the stairwell shaft. Density is determined from the ideal gas law:

EQ-01

$$\rho = \frac{P_{\mathrm{atm}}}{R_{\mathrm{air}} \cdot T}$$

where $R_{\mathrm{air}} = 287.058 \frac{\text{J}}{(\text{kg} \cdot \text{K})}$ and T is in Kelvin. Evaluated for outdoor (ρ_o), indoor (ρ_i), stairwell (ρ_s), and fire-floor (ρ_f) temperatures.

3. Building Leakage Characterization

The pressurization fan must supply enough air to overcome all leakage paths while maintaining the target pressure differential. For stairwell doors, leakage area is computed from door geometry:

EQ-02

$$A_{\mathrm{Ld}} = g_{\mathrm{d}} \cdot \bigl(2w_{\mathrm{d}} + 2h_{\mathrm{d}} - w_{\mathrm{threshold}}\bigr)$$

where g_{d} is the door gap width (m), w_{d} is door width, h_{d} is door height.

For other components, leakage areas are taken from ASHRAE HSCE tabulated values. Multiple parallel paths add directly (EQ-05: $A_{\mathrm{eff}} = A_1 + A_2 + \dots$); series paths combine as (EQ-06: $1/A_{\mathrm{eff}}^2 = 1/A_1^2 + 1/A_2^2 + \dots$).

4. Pressure Distribution Across Building Height

The net pressure differential varies with height due to **stack effect** (buoyancy) and **wind**. The stack-effect pressure at any height h relative to the neutral pressure plane (NPP) is:

EQ-07

$$\Delta P_s(h) = 3460 \left(\frac{1}{T_o} - \frac{1}{T_s} \right) (h - h_{\text{NPP}})$$

The constant 3460 Pa·K/m derives from $(g \cdot P_{\text{atm}}) / R_{\text{air}}$. The NPP height is found iteratively via mass balance (EQ-08).

Wind-induced pressure on each building face is:

EQ-09

$$\Delta P_w = \frac{1}{2} C_p \rho_o V_w^2$$

C_p : windward = +0.70, leeward = -0.45, side = -0.60.

The total pressure difference between the stairwell and each floor combines all effects:

EQ-15

$$\Delta P_{\text{net}} = \Delta P_{\text{mech}} + \Delta P_{\text{stack}} + \Delta P_{\text{wind}} - \Delta P_{\text{exhaust}}$$

$\Delta P_{\text{exhaust}}$ applies only on the fire floor. Must satisfy $\Delta P_{\text{min}} \leq \Delta P_{\text{net}} \leq \Delta P_{\text{max}}$ at every floor.

5. Airflow Through Closed and Open Doors

Flow through each leakage path is computed using the orifice equation:

EQ-03

$$Q = C_d A \sqrt{\frac{2 \Delta P}{\rho}}$$

where $C_d = 0.65$ for building leakage paths.

For open doors, a minimum air velocity prevents smoke migration:

EQ-12

$$Q_{\text{open}} = V_{\text{min}} w_d \cdot h_d$$

6. Door-Opening Force Constraint

The door-opening force includes the door closer force amplified by the pressure differential acting on the door leaf:

EQ-10b

$$F_{\text{total}} = F_{\text{closer}} + \frac{\Delta P \cdot w_d \cdot h_d^2}{2} \cdot \frac{w_d}{w_d - d}$$

Must satisfy $F_{\text{total}} \leq 133 \text{ N (30 lbf)}$. The rearranged form (EQ-10b') gives the maximum allowable ΔP .

7. Stairwell Supply Air Determination

Supply air is computed for two scenarios. All doors closed:

EQ-13

$$Q_{\mathrm{supply,closed}} = \sum Q_{\mathrm{leak,doors}} + \sum Q_{\mathrm{leak,walls}}$$

Design doors open:

EQ-14

$$Q_{\mathrm{supply,open}} = \sum Q_{\mathrm{closed\;floors}} + n_{\mathrm{open}} \cdot Q_{\mathrm{open}} + \sum Q_{\mathrm{leak,walls}}$$

The design supply air rate is the larger of the two:

DESIGN

$$Q_{\mathrm{design}} = \max(Q_{\mathrm{supply,closed}}, Q_{\mathrm{supply,open}})$$

8. Fire Floor Exhaust (Depressurization)

The exhaust system must remove all airflows entering the fire floor: stairwell leakage (EQ-19), elevator shaft leakage (EQ-20), exterior wall leakage (EQ-21), vertical leakage from adjacent floors (EQ-22), and thermal expansion (EQ-23/24).

EQ-19

$$Q_{\mathrm{stair}} = C_d \cdot A_{\mathrm{Ld}} \cdot n_d \cdot \sqrt{\frac{2 \cdot (\Delta P_{\mathrm{mech}} + \Delta P_{\mathrm{exhaust}})}{\rho_s}}$$

EQ-20

$$Q_{\mathrm{elev}} = C_d \cdot A_{\mathrm{Le}} \cdot n_{\mathrm{elev}} \cdot \sqrt{\frac{2 \cdot \Delta P_{\mathrm{elev}}}{\rho_i}}$$

EQ-21

$$Q_{\mathrm{ext}} = \sum_{\text{faces}} C_d \cdot (A_{\mathrm{Lw}} \cdot P_{\mathrm{face}} \cdot h_f) \cdot \sqrt{\frac{2 \cdot (\Delta P_{\mathrm{exhaust}} + \Delta P_{\mathrm{wind}})}{\rho_o}}$$

EQ-22

$$Q_{\mathrm{vert}} = 2 \cdot C_d \cdot (A_{\mathrm{Lf}} \cdot A_{\mathrm{floor}}) \cdot \sqrt{\frac{2 \cdot \Delta P_{\mathrm{exhaust}}}{\rho_i}}$$

Thermal expansion: air entrained by the fire (EQ-24) expands as it heats:

EQ-23/24

$$Q_{\mathrm{expansion}} = \frac{\dot{H}}{\rho_i \cdot c_p \cdot (T_f - T_i)} \cdot \left(\frac{T_f}{T_i} - 1 \right)$$

The total exhaust and its standard-conditions equivalent:

EQ-25

$$Q_{\mathrm{exhaust}} = Q_{\mathrm{stair}} + Q_{\mathrm{elev}} + Q_{\mathrm{ext}} + Q_{\mathrm{vert}} + Q_{\mathrm{expansion}}$$

EQ-26

$$Q_{\mathrm{std}} = Q_{\mathrm{exhaust}} \cdot \frac{T_{\mathrm{f}}}{T_{\mathrm{std}}}$$

$$T_{\mathrm{std}} = 293.15 \text{ K } (20^{\circ}\text{C})$$

9. Iterative Solution and Sensitivity Analysis

The equations are mutually dependent (NPP depends on flows, which depend on pressures, which depend on NPP). The solution iterates until supply and exhaust totals converge within 0.5%. A mandatory sensitivity analysis then varies wall leakage, door leakage ($\pm 50\%$), outdoor temperature, number of open doors, fire temperature, wind speed, and elevator shaft configuration to identify the governing case.

Section B — Calculation Trace (122 steps)

Equations are presented following the ASHRAE *Handbook of Smoke Control Engineering* convention: each section states the general equation, then shows one or more worked instances with numeric inputs, substitution, and result. All intermediate values are shown for plan-checker verification.

1. Air Properties

Air density at each relevant temperature is computed from the ideal gas law. Four densities are needed: outdoor (ρ_o), indoor (ρ_i), stairwell (ρ_s), and fire floor (ρ_f).

EQ-01

$$\rho = \frac{P_{atm}}{R_{air} \cdot T}$$

Outdoor air density

P_atm 101325 Pa
T 280.05 K

Substitution: rho = 101325 / (287.058 * 280.05)

⇒ rho_o = 1.2604 kg/m^3

Indoor air density

P_atm 101325 Pa
T 295.15 K

Substitution: rho = 101325 / (287.058 * 295.15)

⇒ rho_i = 1.1959 kg/m^3

Fire floor air density

P_atm 101325 Pa
T 573.15 K

Substitution: rho = 101325 / (287.058 * 573.15)

⇒ rho_f = 0.6159 kg/m^3

2. Design Pressure Targets

The mechanical ΔP target is the larger of the minimum closed-door differential and the fire-floor exhaust depressurization requirement.

SETUP

$$\Delta P_{\text{mech}} = \max(\Delta P_{\text{min,closed}}, \Delta P_{\text{exhaust}})$$

Design pressure targets

min_dp_closed 12.5 Pa (0.0502 in. w.g.)

floor_exhaust_dp 25 Pa (0.1004 in. w.g.)

Substitution: $\Delta P_{\text{mech}} = \max(12.5, 25)$

⇒ $\Delta P_{\text{mech}} = 25.00 \text{ Pa (0.1004 in. w.g.)}$

3. Wind Pressures

Wind pressure on each building face is computed using the pressure-coefficient method. Windward $C_p = +0.70$, leeward = -0.45 , side = -0.60 .

EQ-09

$$\Delta P_w = \frac{1}{2} C_p \rho_o V_w^2$$

Stair2 Floor B3 — Wind pressure contribution

P_wind_min 0.00 Pa

wall_frac 0.215

Substitution: $\Delta P_w = 0.00 \times 0.215 \times 0.2$

⇒ $\Delta P_{\text{wind}} = 0.00 \text{ Pa (0.0000 in. w.g.)}$

Stair2 Floor 17 — Wind pressure contribution

P_wind_min 0.00 Pa

wall_frac 0.215

Substitution: $\Delta P_w = 0.00 \times 0.215 \times 0.2$

⇒ $\Delta P_{\text{wind}} = 0.00 \text{ Pa (0.0000 in. w.g.)}$

Stair3 Floor B3 — Wind pressure contribution

P_wind_min 0.00 Pa

wall_frac 0.207

Substitution: $\Delta P_w = 0.00 \times 0.207 \times 0.2$

⇒ $\Delta P_{\text{wind}} = 0.00 \text{ Pa (0.0000 in. w.g.)}$

Stair3 Floor 17 — Wind pressure contribution

P_wind_min 0.00 Pa

wall_frac 0.207

Substitution: $\Delta P_w = 0.00 \times 0.207 \times 0.2$

⇒ $\Delta P_{\text{wind}} = 0.00 \text{ Pa (0.0000 in. w.g.)}$

Stair5 Floor B3 — Wind pressure contribution

P_wind_min 0.00 Pa

wall_frac 0.160

Substitution: $\Delta P_w = 0.00 \times 0.160 \times 0.2$

⇒ $\Delta P_{wind} = 0.00 \text{ Pa (0.0000 in. w.g.)}$

Stair6 Floor B3 — Wind pressure contribution

P_wind_min 0.00 Pa

wall_frac 0.157

Substitution: $\Delta P_w = 0.00 \times 0.157 \times 0.2$

⇒ $\Delta P_{wind} = 0.00 \text{ Pa (0.0000 in. w.g.)}$

Stair7 Floor B3 — Wind pressure contribution

P_wind_min 0.00 Pa

wall_frac 0.169

Substitution: $\Delta P_w = 0.00 \times 0.169 \times 0.2$

⇒ $\Delta P_{wind} = 0.00 \text{ Pa (0.0000 in. w.g.)}$

Stair8 Floor 1 — Wind pressure contribution

P_wind_min 0.00 Pa

wall_frac 0.175

Substitution: $\Delta P_w = 0.00 \times 0.175 \times 0.2$

⇒ $\Delta P_{wind} = 0.00 \text{ Pa (0.0000 in. w.g.)}$

Stair9 Floor 1 — Wind pressure contribution

P_wind_min 0.00 Pa

wall_frac 0.182

Substitution: $\Delta P_w = 0.00 \times 0.182 \times 0.2$

⇒ $\Delta P_{wind} = 0.00 \text{ Pa (0.0000 in. w.g.)}$

4. Leakage Area Characterization

Stairwell door leakage area is computed from the door crack geometry. Other component leakage areas are from ASHRAE HSCE tabulated values.

EQ-02

$$A_{Ld} = g_d \cdot (2 w_d + 2 h_d - w_{\text{threshold}})$$

OpenStair — Door leakage area

g_d 0.0030 m
w_d 1.1 m
h_d 2.1 m

Substitution: $A_{Ld} = 0.0030 \times (2 \times 1.1 + 2 \times 2.1 - 1.1)$

⇒ $A_{Ld} = 0.02000 \text{ m}^2$ (used value: 0.02000 m²)

Stair1 — Door leakage area

g_d 0.0030 m
w_d 1.1 m
h_d 2.1 m

Substitution: $A_{Ld} = 0.0030 \times (2 \times 1.1 + 2 \times 2.1 - 1.1)$

⇒ $A_{Ld} = 0.02000 \text{ m}^2$ (used value: 0.02000 m²)

Stair2 — Door leakage area

g_d 0.0030 m
w_d 1.1 m
h_d 2.1 m

Substitution: $A_{Ld} = 0.0030 \times (2 \times 1.1 + 2 \times 2.1 - 1.1)$

⇒ $A_{Ld} = 0.02000 \text{ m}^2$ (used value: 0.02000 m²)

Stair3 — Door leakage area

g_d 0.0030 m
w_d 1.1 m
h_d 2.1 m

Substitution: $A_{Ld} = 0.0030 \times (2 \times 1.1 + 2 \times 2.1 - 1.1)$

⇒ $A_{Ld} = 0.02000 \text{ m}^2$ (used value: 0.02000 m²)

Stair5 — Door leakage area

g_d 0.0030 m
w_d 1.1 m
h_d 2.1 m

Substitution: $A_{Ld} = 0.0030 \times (2 \times 1.1 + 2 \times 2.1 - 1.1)$

⇒ $A_{Ld} = 0.02000 \text{ m}^2$ (used value: 0.02000 m²)

Stair6 — Door leakage area

g_d 0.0030 m
w_d 1.1 m
h_d 2.1 m

Substitution: $A_{Ld} = 0.0030 \times (2 \times 1.1 + 2 \times 2.1 - 1.1)$

⇒ $A_{Ld} = 0.02000 \text{ m}^2$ (used value: 0.02000 m²)

Stair7 — Door leakage area

g_d 0.0030 m
w_d 1.1 m
h_d 2.1 m

Substitution: $A_{Ld} = 0.0030 \times (2 \times 1.1 + 2 \times 2.1 - 1.1)$

⇒ $A_{Ld} = 0.02000 \text{ m}^2$ (used value: 0.02000 m²)

Stair8 — Door leakage area

g_d 0.0030 m
w_d 1.1 m
h_d 2.1 m

Substitution: $A_{Ld} = 0.0030 \times (2 \times 1.1 + 2 \times 2.1 - 1.1)$

⇒ $A_{Ld} = 0.02000 \text{ m}^2$ (used value: 0.02000 m²)

Stair9 — Door leakage area

g_d 0.0030 m
w_d 1.1 m
h_d 2.1 m

Substitution: $A_{Ld} = 0.0030 \times (2 \times 1.1 + 2 \times 2.1 - 1.1)$

⇒ $A_{Ld} = 0.02000 \text{ m}^2$ (used value: 0.02000 m²)

5. Open-Door Flow Requirement

The minimum airflow through each open door is set by the code-required velocity to prevent smoke migration through the doorway.

EQ-12

$$Q_{\text{open}} = V_{\text{min}} \cdot w_d \cdot h_d$$

OpenStair — Required open-door flow

V_{min} 1 m/s
 w_d 1.1 m
 h_d 2.1 m

Substitution: $Q_{\text{open}} = 1 \times 1.1 \times 2.1$

$$\Rightarrow Q_{\text{open}} = 2.3100 \text{ m}^3/\text{s} \text{ (4895 CFM)}$$

Stair1 — Required open-door flow

V_{min} 1 m/s
 w_d 1.1 m
 h_d 2.1 m

Substitution: $Q_{\text{open}} = 1 \times 1.1 \times 2.1$

$$\Rightarrow Q_{\text{open}} = 2.3100 \text{ m}^3/\text{s} \text{ (4895 CFM)}$$

Stair2 — Required open-door flow

V_{min} 1 m/s
 w_d 1.1 m
 h_d 2.1 m

Substitution: $Q_{\text{open}} = 1 \times 1.1 \times 2.1$

$$\Rightarrow Q_{\text{open}} = 2.3100 \text{ m}^3/\text{s} \text{ (4895 CFM)}$$

Stair3 — Required open-door flow

V_{min} 1 m/s
 w_d 1.1 m
 h_d 2.1 m

Substitution: $Q_{\text{open}} = 1 \times 1.1 \times 2.1$

$$\Rightarrow Q_{\text{open}} = 2.3100 \text{ m}^3/\text{s} \text{ (4895 CFM)}$$

Stair5 — Required open-door flow

V_min 1 m/s
w_d 1.1 m
h_d 2.1 m

Substitution: $Q_{open} = 1 \times 1.1 \times 2.1$

$$\Rightarrow Q_{open} = 2.3100 \text{ m}^3/\text{s} \text{ (4895 CFM)}$$

Stair6 — Required open-door flow

V_min 1 m/s
w_d 1.1 m
h_d 2.1 m

Substitution: $Q_{open} = 1 \times 1.1 \times 2.1$

$$\Rightarrow Q_{open} = 2.3100 \text{ m}^3/\text{s} \text{ (4895 CFM)}$$

Stair7 — Required open-door flow

V_min 1 m/s
w_d 1.1 m
h_d 2.1 m

Substitution: $Q_{open} = 1 \times 1.1 \times 2.1$

$$\Rightarrow Q_{open} = 2.3100 \text{ m}^3/\text{s} \text{ (4895 CFM)}$$

Stair8 — Required open-door flow

V_min 1 m/s
w_d 1.1 m
h_d 2.1 m

Substitution: $Q_{open} = 1 \times 1.1 \times 2.1$

$$\Rightarrow Q_{open} = 2.3100 \text{ m}^3/\text{s} \text{ (4895 CFM)}$$

Stair9 — Required open-door flow

V_min 1 m/s
w_d 1.1 m
h_d 2.1 m

Substitution: $Q_{open} = 1 \times 1.1 \times 2.1$

$$\Rightarrow Q_{open} = 2.3100 \text{ m}^3/\text{s} \text{ (4895 CFM)}$$

6. Floor-by-Floor Pressure & Flow Analysis

For each floor served by each stairwell, the stack-effect pressure, net pressure differential, orifice leakage flow, and door-opening force are evaluated. Only selected representative floors are shown; all floors are computed.

EQ-07

$$\Delta P_s(h) = 3460 \left(\frac{1}{T_o} - \frac{1}{T_s} \right) (h - h_{NPP})$$

Stair1 Floor B3 — Stack effect

T_i	295.15 K
T_s	280.05 K
h	0.00 m
h_NPP	88.94 m

Substitution: $\Delta P_s = 3460 \times (1/295.15 - 1/280.05) \times (0.00 - 88.94)$

⇒ $\Delta P_{stack} = 56.22 \text{ Pa (0.2257 in. w.g.)}$

Stair1 Floor 17 — Stack effect

T _i	295.15 K
T _s	280.05 K
h	85.40 m
h _{NPP}	88.94 m

Substitution: $\Delta P_s = 3460 \times (1/295.15 - 1/280.05) \times (85.40 - 88.94)$

⇒ $\Delta P_{\text{stack}} = 2.24 \text{ Pa (0.0090 in. w.g.)}$

Stair2 Floor B3 — Stack effect

T _i	295.15 K
T _s	280.05 K
h	0.00 m
h _{NPP}	88.94 m

Substitution: $\Delta P_s = 3460 \times (1/295.15 - 1/280.05) \times (0.00 - 88.94)$

⇒ $\Delta P_{\text{stack}} = 56.22 \text{ Pa (0.2257 in. w.g.)}$

Stair2 Floor 17 — Stack effect

T _i	295.15 K
T _s	280.05 K
h	85.40 m
h _{NPP}	88.94 m

Substitution: $\Delta P_s = 3460 \times (1/295.15 - 1/280.05) \times (85.40 - 88.94)$

⇒ $\Delta P_{\text{stack}} = 2.24 \text{ Pa (0.0090 in. w.g.)}$

Stair3 Floor B3 — Stack effect

T _i	295.15 K
T _s	280.05 K
h	0.00 m
h _{NPP}	88.94 m

Substitution: $\Delta P_s = 3460 \times (1/295.15 - 1/280.05) \times (0.00 - 88.94)$

⇒ $\Delta P_{\text{stack}} = 56.22 \text{ Pa (0.2257 in. w.g.)}$

Stair3 Floor 17 — Stack effect

T _i	295.15 K
T _s	280.05 K
h	85.40 m
h _{NPP}	88.94 m

Substitution: $\Delta P_s = 3460 \times (1/295.15 - 1/280.05) \times (85.40 - 88.94)$

⇒ $\Delta P_{stack} = 2.24 \text{ Pa (0.0090 in. w.g.)}$

Stair5 Floor B3 — Stack effect

T _i	295.15 K
T _s	280.05 K
h	0.00 m
h _{NPP}	88.94 m

Substitution: $\Delta P_s = 3460 \times (1/295.15 - 1/280.05) \times (0.00 - 88.94)$

⇒ $\Delta P_{stack} = 56.22 \text{ Pa (0.2257 in. w.g.)}$

Stair6 Floor B3 — Stack effect

T _i	295.15 K
T _s	280.05 K
h	0.00 m
h _{NPP}	88.94 m

Substitution: $\Delta P_s = 3460 \times (1/295.15 - 1/280.05) \times (0.00 - 88.94)$

⇒ $\Delta P_{stack} = 56.22 \text{ Pa (0.2257 in. w.g.)}$

Stair7 Floor B3 — Stack effect

T _i	295.15 K
T _s	280.05 K
h	0.00 m
h _{NPP}	88.94 m

Substitution: $\Delta P_s = 3460 \times (1/295.15 - 1/280.05) \times (0.00 - 88.94)$

⇒ $\Delta P_{stack} = 56.22 \text{ Pa (0.2257 in. w.g.)}$

EQ-15

$$\Delta P_{\text{net}} = \Delta P_{\text{mech}} + \Delta P_{\text{stack}} + \Delta P_{\text{wind}} - \Delta P_{\text{exhaust}}$$

Stair1 Floor B3 — Net pressure differential

ΔP_{mech}	25.00 Pa
ΔP_{stack}	56.22 Pa
ΔP_{wind}	0.00 Pa
$\Delta P_{\text{exhaust}}$	0.00 Pa

Substitution: $\Delta P_{\text{net}} = 25.00 + 56.22 + 0.00 - 0.00$

$\Rightarrow \Delta P_{\text{net}} = 81.22 \text{ Pa } (0.3261 \text{ in. w.g.})$

Stair1 Floor 17 — Net pressure differential

ΔP_{mech}	25.00 Pa
ΔP_{stack}	2.24 Pa
ΔP_{wind}	0.00 Pa
$\Delta P_{\text{exhaust}}$	25.00 Pa

Substitution: $\Delta P_{\text{net}} = 25.00 + 2.24 + 0.00 - 25.00$

$\Rightarrow \Delta P_{\text{net}} = 2.24 \text{ Pa } (0.0090 \text{ in. w.g.})$

Stair2 Floor B3 — Net pressure differential

ΔP_{mech}	25.00 Pa
ΔP_{stack}	56.22 Pa
ΔP_{wind}	0.00 Pa
$\Delta P_{\text{exhaust}}$	0.00 Pa

Substitution: $\Delta P_{\text{net}} = 25.00 + 56.22 + 0.00 - 0.00$

$\Rightarrow \Delta P_{\text{net}} = 81.22 \text{ Pa } (0.3261 \text{ in. w.g.})$

Stair2 Floor 17 — Net pressure differential

ΔP_{mech}	25.00 Pa
ΔP_{stack}	2.24 Pa
ΔP_{wind}	0.00 Pa
$\Delta P_{\text{exhaust}}$	25.00 Pa

Substitution: $\Delta P_{\text{net}} = 25.00 + 2.24 + 0.00 - 25.00$

$\Rightarrow \Delta P_{\text{net}} = 2.24 \text{ Pa } (0.0090 \text{ in. w.g.})$

Stair3 Floor B3 — Net pressure differential

ΔP_{mech}	25.00 Pa
ΔP_{stack}	56.22 Pa
ΔP_{wind}	0.00 Pa
$\Delta P_{\text{exhaust}}$	0.00 Pa

Substitution: $\Delta P_{\text{net}} = 25.00 + 56.22 + 0.00 - 0.00$

⇒ $\Delta P_{\text{net}} = 81.22 \text{ Pa (0.3261 in. w.g.)}$

Stair3 Floor 17 — Net pressure differential

ΔP_{mech}	25.00 Pa
ΔP_{stack}	2.24 Pa
ΔP_{wind}	0.00 Pa
$\Delta P_{\text{exhaust}}$	25.00 Pa

Substitution: $\Delta P_{\text{net}} = 25.00 + 2.24 + 0.00 - 25.00$

⇒ $\Delta P_{\text{net}} = 2.24 \text{ Pa (0.0090 in. w.g.)}$

Stair5 Floor B3 — Net pressure differential

ΔP_{mech}	25.00 Pa
ΔP_{stack}	56.22 Pa
ΔP_{wind}	0.00 Pa
$\Delta P_{\text{exhaust}}$	0.00 Pa

Substitution: $\Delta P_{\text{net}} = 25.00 + 56.22 + 0.00 - 0.00$

⇒ $\Delta P_{\text{net}} = 81.22 \text{ Pa (0.3261 in. w.g.)}$

Stair6 Floor B3 — Net pressure differential

ΔP_{mech}	25.00 Pa
ΔP_{stack}	56.22 Pa
ΔP_{wind}	0.00 Pa
$\Delta P_{\text{exhaust}}$	0.00 Pa

Substitution: $\Delta P_{\text{net}} = 25.00 + 56.22 + 0.00 - 0.00$

⇒ $\Delta P_{\text{net}} = 81.22 \text{ Pa (0.3261 in. w.g.)}$

Stair7 Floor B3 — Net pressure differential

ΔP_{mech}	25.00 Pa
ΔP_{stack}	56.22 Pa
ΔP_{wind}	0.00 Pa
$\Delta P_{\text{exhaust}}$	0.00 Pa

Substitution: $\Delta P_{\text{net}} = 25.00 + 56.22 + 0.00 - 0.00$

⇒ $\Delta P_{\text{net}} = 81.22 \text{ Pa (0.3261 in. w.g.)}$

EQ-03

$$Q = C_d \cdot A \cdot \sqrt{\frac{2|\Delta P|}{\rho}}$$

Stair1 Floor B3 — Closed-door leakage flow

C_d 0.65
A 0.02000 m²
ΔP 81.22 Pa
ρ_s 1.2604 kg/m³

Substitution: $Q = 0.65 \times 0.02000 \times \sqrt{(2 \times 81.22 / 1.2604)}$

$$\Rightarrow Q_{\text{leak}} = 0.14758 \text{ m}^3/\text{s} \text{ (312.7 CFM)}$$

Stair2 Floor B3 — Closed-door leakage flow

C_d 0.65
A 0.02000 m²
ΔP 81.22 Pa
ρ_s 1.2604 kg/m³

Substitution: $Q = 0.65 \times 0.02000 \times \sqrt{(2 \times 81.22 / 1.2604)}$

$$\Rightarrow Q_{\text{leak}} = 0.14758 \text{ m}^3/\text{s} \text{ (312.7 CFM)}$$

Stair3 Floor B3 — Closed-door leakage flow

C_d 0.65
A 0.02000 m²
ΔP 81.22 Pa
ρ_s 1.2604 kg/m³

Substitution: $Q = 0.65 \times 0.02000 \times \sqrt{(2 \times 81.22 / 1.2604)}$

$$\Rightarrow Q_{\text{leak}} = 0.14758 \text{ m}^3/\text{s} \text{ (312.7 CFM)}$$

Stair5 Floor B3 — Closed-door leakage flow

C_d 0.65
A 0.02000 m²
ΔP 81.22 Pa
ρ_s 1.2604 kg/m³

Substitution: $Q = 0.65 \times 0.02000 \times \sqrt{(2 \times 81.22 / 1.2604)}$

$$\Rightarrow Q_{\text{leak}} = 0.14758 \text{ m}^3/\text{s} \text{ (312.7 CFM)}$$

Stair6 Floor B3 — Closed-door leakage flow

C _d	0.65
A	0.02000 m ²
ΔP	81.22 Pa
ρ _s	1.2604 kg/m ³

Substitution: $Q = 0.65 \times 0.02000 \times \sqrt{(2 \times 81.22 / 1.2604)}$

$$\Rightarrow Q_{\text{leak}} = 0.14758 \text{ m}^3/\text{s} \text{ (312.7 CFM)}$$

Stair7 Floor B3 — Closed-door leakage flow

C _d	0.65
A	0.02000 m ²
ΔP	81.22 Pa
ρ _s	1.2604 kg/m ³

Substitution: $Q = 0.65 \times 0.02000 \times \sqrt{(2 \times 81.22 / 1.2604)}$

$$\Rightarrow Q_{\text{leak}} = 0.14758 \text{ m}^3/\text{s} \text{ (312.7 CFM)}$$

EQ-10b

$$F_{\text{total}} = F_{\text{closer}} + \frac{\Delta P \cdot w_d \cdot h_d}{2} \cdot \frac{w_d}{w_d - d}$$

Stair1 Floor B3 — Door-opening force

F_closer	55.0 N (12.4 lbf)
ΔP	81.22 Pa
w_d	1.1 m
h_d	2.1 m
d	0.075 m

Substitution: $F_p = 93.81 \text{ N}$; $F_{\text{total}} = 55.0 + 93.81 \times 1.1 / (1.1 - 0.075)$

$$\Rightarrow F_{\text{total}} = 155.7 \text{ N (35.0 lbf)}$$

Stair1 Floor 17 — Door-opening force

F_closer	55.0 N (12.4 lbf)
ΔP	2.24 Pa
w_d	1.1 m
h_d	2.1 m
d	0.075 m

Substitution: $F_p = 2.59 \text{ N}$; $F_{\text{total}} = 55.0 + 2.59 \times 1.1 / (1.1 - 0.075)$

$$\Rightarrow F_{\text{total}} = 57.8 \text{ N (13.0 lbf)}$$

Stair2 Floor B3 — Door-opening force

F_closer	55.0 N (12.4 lbf)
ΔP	81.22 Pa
w_d	1.1 m
h_d	2.1 m
d	0.075 m

Substitution: $F_p = 93.81 \text{ N}$; $F_{\text{total}} = 55.0 + 93.81 \times 1.1 / (1.1 - 0.075)$

$$\Rightarrow F_{\text{total}} = 155.7 \text{ N (35.0 lbf)}$$

Stair2 Floor 17 — Door-opening force

F_closer	55.0 N (12.4 lbf)
ΔP	2.24 Pa
w_d	1.1 m
h_d	2.1 m
d	0.075 m

Substitution: $F_p = 2.59 \text{ N}$; $F_{\text{total}} = 55.0 + 2.59 \times 1.1 / (1.1 - 0.075)$

$$\Rightarrow F_{\text{total}} = 57.8 \text{ N (13.0 lbf)}$$

Stair3 Floor B3 — Door-opening force

F_closer	55.0 N (12.4 lbf)
ΔP	81.22 Pa
w_d	1.1 m
h_d	2.1 m
d	0.075 m

Substitution: $F_p = 93.81 \text{ N}$; $F_{\text{total}} = 55.0 + 93.81 \times 1.1 / (1.1 - 0.075)$

$$\Rightarrow F_{\text{total}} = 155.7 \text{ N (35.0 lbf)}$$

Stair3 Floor 17 — Door-opening force

F_closer	55.0 N (12.4 lbf)
ΔP	2.24 Pa
w_d	1.1 m
h_d	2.1 m
d	0.075 m

Substitution: $F_p = 2.59 \text{ N}$; $F_{\text{total}} = 55.0 + 2.59 \times 1.1 / (1.1 - 0.075)$

$$\Rightarrow F_{\text{total}} = 57.8 \text{ N (13.0 lbf)}$$

Stair5 Floor B3 — Door-opening force

F_closer	55.0 N (12.4 lbf)
ΔP	81.22 Pa
w_d	1.1 m
h_d	2.1 m
d	0.075 m

Substitution: $F_p = 93.81 \text{ N}$; $F_{\text{total}} = 55.0 + 93.81 \times 1.1 / (1.1 - 0.075)$

$$\Rightarrow F_{\text{total}} = 155.7 \text{ N (35.0 lbf)}$$

Stair6 Floor B3 — Door-opening force

F_closer	55.0 N (12.4 lbf)
ΔP	81.22 Pa
w_d	1.1 m
h_d	2.1 m
d	0.075 m

Substitution: $F_p = 93.81 \text{ N}$; $F_{\text{total}} = 55.0 + 93.81 \times 1.1 / (1.1 - 0.075)$

$$\Rightarrow F_{\text{total}} = 155.7 \text{ N (35.0 lbf)}$$

Stair7 Floor B3 — Door-opening force

F_closer	55.0 N (12.4 lbf)
ΔP	81.22 Pa
w_d	1.1 m
h_d	2.1 m
d	0.075 m

Substitution: $F_p = 93.81 \text{ N}$; $F_{\text{total}} = 55.0 + 93.81 \times 1.1 / (1.1 - 0.075)$

$$\Rightarrow F_{\text{total}} = 155.7 \text{ N (35.0 lbf)}$$

7. Stairwell Supply Air Determination

The supply air rate is computed for the all-doors-closed and doors-open scenarios. The design value is the larger of the two.

EQ-13

$$Q_{\text{supply,closed}} = \Sigma Q_{\text{leak,doors}} + \Sigma Q_{\text{leak,walls}}$$

Stair1 — Total supply (all doors closed)

$$\Sigma Q_{\text{leak_doors}} \quad 2.94990 \text{ m}^3/\text{s}$$

$$\Sigma Q_{\text{leak_walls}} \quad 0.00000 \text{ m}^3/\text{s}$$

$$\text{Substitution: } Q_{\text{supply_closed}} = 2.94990 + 0.00000$$

$$\Rightarrow Q_{\text{supply_closed}} = 2.94990 \text{ m}^3/\text{s} \text{ (6250 CFM)}$$

Stair2 — Total supply (all doors closed)

$$\Sigma Q_{\text{leak_doors}} \quad 2.94990 \text{ m}^3/\text{s}$$

$$\Sigma Q_{\text{leak_walls}} \quad 0.43182 \text{ m}^3/\text{s}$$

$$\text{Substitution: } Q_{\text{supply_closed}} = 2.94990 + 0.43182$$

$$\Rightarrow Q_{\text{supply_closed}} = 3.38173 \text{ m}^3/\text{s} \text{ (7165 CFM)}$$

Stair3 — Total supply (all doors closed)

$$\Sigma Q_{\text{leak_doors}} \quad 2.94990 \text{ m}^3/\text{s}$$

$$\Sigma Q_{\text{leak_walls}} \quad 0.43182 \text{ m}^3/\text{s}$$

$$\text{Substitution: } Q_{\text{supply_closed}} = 2.94990 + 0.43182$$

$$\Rightarrow Q_{\text{supply_closed}} = 3.38173 \text{ m}^3/\text{s} \text{ (7165 CFM)}$$

Stair5 — Total supply (all doors closed)

$$\Sigma Q_{\text{leak_doors}} \quad 0.97955 \text{ m}^3/\text{s}$$

$$\Sigma Q_{\text{leak_walls}} \quad 0.14221 \text{ m}^3/\text{s}$$

$$\text{Substitution: } Q_{\text{supply_closed}} = 0.97955 + 0.14221$$

$$\Rightarrow Q_{\text{supply_closed}} = 1.12176 \text{ m}^3/\text{s} \text{ (2377 CFM)}$$

Stair6 — Total supply (all doors closed)

$$\Sigma Q_{\text{leak_doors}} \quad 0.97955 \text{ m}^3/\text{s}$$

$$\Sigma Q_{\text{leak_walls}} \quad 0.14221 \text{ m}^3/\text{s}$$

$$\text{Substitution: } Q_{\text{supply_closed}} = 0.97955 + 0.14221$$

$$\Rightarrow Q_{\text{supply_closed}} = 1.12176 \text{ m}^3/\text{s} \text{ (2377 CFM)}$$

Stair7 — Total supply (all doors closed)

$$\Sigma Q_{\text{leak_doors}} \quad 0.97955 \text{ m}^3/\text{s}$$

$$\Sigma Q_{\text{leak_walls}} \quad 0.14221 \text{ m}^3/\text{s}$$

$$\text{Substitution: } Q_{\text{supply_closed}} = 0.97955 + 0.14221$$

$$\Rightarrow Q_{\text{supply_closed}} = 1.12176 \text{ m}^3/\text{s} \text{ (2377 CFM)}$$

EQ-14

$$Q_{\text{supply,open}} = \Sigma Q_{\text{closed floors}} + n_{\text{open}} \cdot Q_{\text{open}} + \Sigma Q_{\text{leak,walls}}$$

Stair1 — Total supply (doors open)

ΣQ_{closed} 2.94990 m³/s
 $Q_{\text{open_doors}}$ 2.31000 m³/s (1 doors)
 ΣQ_{walls} 0.00000 m³/s

Substitution: $Q_{\text{supply_open}} = 2.94990 + 2.31000 + 0.00000$

⇒ $Q_{\text{supply_open}} = 5.25990 \text{ m}^3/\text{s}$ (11145 CFM)

Stair2 — Total supply (doors open)

ΣQ_{closed} 2.94990 m³/s
 $Q_{\text{open_doors}}$ 2.31000 m³/s (1 doors)
 ΣQ_{walls} 0.43182 m³/s

Substitution: $Q_{\text{supply_open}} = 2.94990 + 2.31000 + 0.43182$

⇒ $Q_{\text{supply_open}} = 5.69173 \text{ m}^3/\text{s}$ (12060 CFM)

Stair3 — Total supply (doors open)

ΣQ_{closed} 2.94990 m³/s
 $Q_{\text{open_doors}}$ 2.31000 m³/s (1 doors)
 ΣQ_{walls} 0.43182 m³/s

Substitution: $Q_{\text{supply_open}} = 2.94990 + 2.31000 + 0.43182$

⇒ $Q_{\text{supply_open}} = 5.69173 \text{ m}^3/\text{s}$ (12060 CFM)

Stair5 — Total supply (doors open)

ΣQ_{closed} 0.97955 m³/s
 $Q_{\text{open_doors}}$ 2.31000 m³/s (1 doors)
 ΣQ_{walls} 0.14221 m³/s

Substitution: $Q_{\text{supply_open}} = 0.97955 + 2.31000 + 0.14221$

⇒ $Q_{\text{supply_open}} = 3.43176 \text{ m}^3/\text{s}$ (7271 CFM)

Stair6 — Total supply (doors open)

ΣQ_{closed} 0.97955 m³/s

$Q_{\text{open_doors}}$ 2.31000 m³/s (1 doors)

ΣQ_{walls} 0.14221 m³/s

Substitution: $Q_{\text{supply_open}} = 0.97955 + 2.31000 + 0.14221$

⇒ $Q_{\text{supply_open}} = 3.43176 \text{ m}^3/\text{s}$ (7271 CFM)

Stair7 — Total supply (doors open)

ΣQ_{closed} 0.97955 m³/s

$Q_{\text{open_doors}}$ 2.31000 m³/s (1 doors)

ΣQ_{walls} 0.14221 m³/s

Substitution: $Q_{\text{supply_open}} = 0.97955 + 2.31000 + 0.14221$

⇒ $Q_{\text{supply_open}} = 3.43176 \text{ m}^3/\text{s}$ (7271 CFM)

DESIGN

$$Q_{\text{design}} = \max(Q_{\text{supply,closed}}, Q_{\text{supply,open}})$$

Stair1 — Governing supply air rate

Q_closed 6250 CFM
Q_open 11145 CFM

Substitution: Governing case: doors open

$$\Rightarrow Q_{\text{design}} = 5.25990 \text{ m}^3/\text{s} \text{ (11145 CFM)}$$

Stair2 — Governing supply air rate

Q_closed 7165 CFM
Q_open 12060 CFM

Substitution: Governing case: doors open

$$\Rightarrow Q_{\text{design}} = 5.69173 \text{ m}^3/\text{s} \text{ (12060 CFM)}$$

Stair3 — Governing supply air rate

Q_closed 7165 CFM
Q_open 12060 CFM

Substitution: Governing case: doors open

$$\Rightarrow Q_{\text{design}} = 5.69173 \text{ m}^3/\text{s} \text{ (12060 CFM)}$$

Stair5 — Governing supply air rate

Q_closed 2377 CFM
Q_open 7271 CFM

Substitution: Governing case: doors open

$$\Rightarrow Q_{\text{design}} = 3.43176 \text{ m}^3/\text{s} \text{ (7271 CFM)}$$

Stair6 — Governing supply air rate

Q_{closed} 2377 CFM

Q_{open} 7271 CFM

Substitution: Governing case: doors open

$$\Rightarrow Q_{\text{design}} = 3.43176 \text{ m}^3/\text{s} \text{ (7271 CFM)}$$

Stair7 — Governing supply air rate

Q_{closed} 2377 CFM

Q_{open} 7271 CFM

Substitution: Governing case: doors open

$$\Rightarrow Q_{\text{design}} = 3.43176 \text{ m}^3/\text{s} \text{ (7271 CFM)}$$

Section C — Stair1 Floor-by-Floor

Floor	Height (m)	ΔP_{stack} (Pa / in. w.g.)	ΔP_{wind} (Pa / in. w.g.)	ΔP_{net} (Pa / in. w.g.)	Q_{leak} (CFM)	Q_{open} (CFM)	F_{total} (N / lbf)	Status
B3	0.0	56.22 0.2257	0.00 0.0000	81.22 0.3261	313	—	155.7 35.0	FAIL
B2	4.3	53.52 0.2149	0.00 0.0000	78.52 0.3152	307	—	152.3 34.2	FAIL
B1	8.5	50.82 0.2040	0.00 0.0000	75.82 0.3044	302	—	149.0 33.5	FAIL
G	12.8	48.12 0.1932	0.00 0.0000	73.12 0.2936	297	—	145.6 32.7	FAIL
1	17.1	45.42 0.1824	0.00 0.0000	70.42 0.2827	291	—	142.3 32.0	FAIL
2	21.3	42.72 0.1715	0.00 0.0000	67.72 0.2719	286	—	138.9 31.2	FAIL
3	25.6	40.02 0.1607	0.00 0.0000	65.02 0.2610	280	—	135.6 30.5	FAIL
4	29.9	37.33 0.1498	0.00 0.0000	62.33 0.2502	274	—	132.3 29.7	PASS
5	34.2	34.63 0.1390	0.00 0.0000	59.63 0.2394	268	—	128.9 29.0	PASS
6	38.4	31.93 0.1282	0.00 0.0000	56.93 0.2285	262	—	125.6 28.2	PASS
7	42.7	29.23 0.1173	0.00 0.0000	54.23 0.2177	256	—	122.2 27.5	PASS
8	47.0	26.53 0.1065	0.00 0.0000	51.53 0.2069	249	—	118.9 26.7	PASS
9	51.2	23.83 0.0957	0.00 0.0000	48.83 0.1960	242	—	115.5 26.0	PASS
10	55.5	21.13 0.0848	0.00 0.0000	46.13 0.1852	236	—	112.2 25.2	PASS
11	59.8	18.43 0.0740	0.00 0.0000	43.43 0.1744	229	—	108.8 24.5	PASS
12	64.0	15.73 0.0632	0.00 0.0000	40.73 0.1635	221	—	105.5 23.7	PASS
13	68.3	13.03 0.0523	0.00 0.0000	38.03 0.1527	214	—	102.1 23.0	PASS
14	72.6	10.34 0.0415	0.00 0.0000	35.34 0.1419	206	—	98.8 22.2	PASS

Floor	Height (m)	ΔP_{stack} (Pa / in. w.g.)	ΔP_{wind} (Pa / in. w.g.)	ΔP_{net} (Pa / in. w.g.)	Q_leak (CFM)	Q_open (CFM)	F_total (N / lbf)	Status
15	76.9	7.64 0.0307	0.00 0.0000	32.64 0.1310	198	—	95.5 21.5	PASS
16	81.1	4.94 0.0198	0.00 0.0000	29.94 0.1202	190	—	92.1 20.7	PASS
17	85.4	2.24 0.0090	0.00 0.0000	2.24 0.0090	0	4895	57.8 13.0	PASS
18	89.7	-0.46 -0.0018	0.00 0.0000	24.54 0.0985	172	—	85.4 19.2	PASS
19	93.9	-3.16 -0.0127	0.00 0.0000	21.84 0.0877	162	—	82.1 18.5	PASS
20	98.2	-5.86 -0.0235	0.00 0.0000	19.14 0.0768	152	—	78.7 17.7	PASS
21	102.5	-8.56 -0.0344	0.00 0.0000	16.44 0.0660	141	—	75.4 16.9	PASS
22	106.7	-11.26 -0.0452	0.00 0.0000	13.74 0.0552	129	—	72.0 16.2	PASS
23	111.0	-13.96 -0.0560	0.00 0.0000	11.04 0.0443	115	—	68.7 15.4	FAIL
24	115.3	-16.65 -0.0669	0.00 0.0000	8.35 0.0335	100	—	65.3 14.7	FAIL
25	119.6	-19.35 -0.0777	0.00 0.0000	5.65 0.0227	82	—	62.0 13.9	FAIL
26	123.8	-22.05 -0.0885	0.00 0.0000	2.95 0.0118	60	—	58.7 13.2	FAIL
27	128.1	-24.75 -0.0994	0.00 0.0000	0.25 0.0010	17	—	55.3 12.4	FAIL
28	132.4	-27.45 -0.1102	0.00 0.0000	-2.45 -0.0098	0	—	52.0 11.7	FAIL
29	136.6	-30.15 -0.1210	0.00 0.0000	-5.15 -0.0207	0	—	48.6 10.9	FAIL
30	140.9	-32.85 -0.1319	0.00 0.0000	-7.85 -0.0315	0	—	45.3 10.2	FAIL
31	145.2	-35.55 -0.1427	0.00 0.0000	-10.55 -0.0423	0	—	41.9 9.4	FAIL
32	149.4	-38.25 -0.1535	0.00 0.0000	-13.25 -0.0532	0	—	38.6 8.7	FAIL
33	153.7	-40.95 -0.1644	0.00 0.0000	-15.95 -0.0640	0	—	35.2 7.9	FAIL

Floor	Height (m)	ΔP_{stack} (Pa / in. w.g.)	ΔP_{wind} (Pa / in. w.g.)	ΔP_{net} (Pa / in. w.g.)	Q_{leak} (CFM)	Q_{open} (CFM)	F_{total} (N / lbf)	Status
34	158.0	-43.64 -0.1752	0.00 0.0000	-18.64 -0.0749	0	—	31.9 7.2	FAIL
35	162.3	-46.34 -0.1861	0.00 0.0000	-21.34 -0.0857	0	—	28.5 6.4	FAIL
36	166.5	-49.04 -0.1969	0.00 0.0000	-24.04 -0.0965	0	—	25.2 5.7	FAIL
37	170.8	-51.74 -0.2077	0.00 0.0000	-26.74 -0.1074	0	—	21.9 4.9	FAIL
38	175.1	-54.44 -0.2186	0.00 0.0000	-29.44 -0.1182	0	—	18.5 4.2	FAIL

Critical floor (min ΔP): **38** | Critical floor (max force): **B3**

Section C — Stair2 Floor-by-Floor

Floor	Height (m)	ΔP_{stack} (Pa / in. w.g.)	ΔP_{wind} (Pa / in. w.g.)	ΔP_{net} (Pa / in. w.g.)	Q_{leak} (CFM)	Q_{open} (CFM)	F_{total} (N / lbf)	Status
B3	0.0	56.22 0.2257	0.00 0.0000	81.22 0.3261	313	—	155.7 35.0	FAIL
B2	4.3	53.52 0.2149	0.00 0.0000	78.52 0.3152	307	—	152.3 34.2	FAIL
B1	8.5	50.82 0.2040	0.00 0.0000	75.82 0.3044	302	—	149.0 33.5	FAIL
G	12.8	48.12 0.1932	0.00 0.0000	73.12 0.2936	297	—	145.6 32.7	FAIL
1	17.1	45.42 0.1824	0.00 0.0000	70.42 0.2827	291	—	142.3 32.0	FAIL
2	21.3	42.72 0.1715	0.00 0.0000	67.72 0.2719	286	—	138.9 31.2	FAIL
3	25.6	40.02 0.1607	0.00 0.0000	65.02 0.2610	280	—	135.6 30.5	FAIL
4	29.9	37.33 0.1498	0.00 0.0000	62.33 0.2502	274	—	132.3 29.7	PASS
5	34.2	34.63 0.1390	0.00 0.0000	59.63 0.2394	268	—	128.9 29.0	PASS
6	38.4	31.93 0.1282	0.00 0.0000	56.93 0.2285	262	—	125.6 28.2	PASS
7	42.7	29.23 0.1173	0.00 0.0000	54.23 0.2177	256	—	122.2 27.5	PASS
8	47.0	26.53 0.1065	0.00 0.0000	51.53 0.2069	249	—	118.9 26.7	PASS
9	51.2	23.83 0.0957	0.00 0.0000	48.83 0.1960	242	—	115.5 26.0	PASS
10	55.5	21.13 0.0848	0.00 0.0000	46.13 0.1852	236	—	112.2 25.2	PASS
11	59.8	18.43 0.0740	0.00 0.0000	43.43 0.1744	229	—	108.8 24.5	PASS
12	64.0	15.73 0.0632	0.00 0.0000	40.73 0.1635	221	—	105.5 23.7	PASS
13	68.3	13.03 0.0523	0.00 0.0000	38.03 0.1527	214	—	102.1 23.0	PASS
14	72.6	10.34 0.0415	0.00 0.0000	35.34 0.1419	206	—	98.8 22.2	PASS

Floor	Height (m)	ΔP_{stack} (Pa / in. w.g.)	ΔP_{wind} (Pa / in. w.g.)	ΔP_{net} (Pa / in. w.g.)	Q_leak (CFM)	Q_open (CFM)	F_total (N / lbf)	Status
15	76.9	7.64 0.0307	0.00 0.0000	32.64 0.1310	198	—	95.5 21.5	PASS
16	81.1	4.94 0.0198	0.00 0.0000	29.94 0.1202	190	—	92.1 20.7	PASS
17	85.4	2.24 0.0090	0.00 0.0000	2.24 0.0090	0	4895	57.8 13.0	PASS
18	89.7	-0.46 -0.0018	0.00 0.0000	24.54 0.0985	172	—	85.4 19.2	PASS
19	93.9	-3.16 -0.0127	0.00 0.0000	21.84 0.0877	162	—	82.1 18.5	PASS
20	98.2	-5.86 -0.0235	0.00 0.0000	19.14 0.0768	152	—	78.7 17.7	PASS
21	102.5	-8.56 -0.0344	0.00 0.0000	16.44 0.0660	141	—	75.4 16.9	PASS
22	106.7	-11.26 -0.0452	0.00 0.0000	13.74 0.0552	129	—	72.0 16.2	PASS
23	111.0	-13.96 -0.0560	0.00 0.0000	11.04 0.0443	115	—	68.7 15.4	FAIL
24	115.3	-16.65 -0.0669	0.00 0.0000	8.35 0.0335	100	—	65.3 14.7	FAIL
25	119.6	-19.35 -0.0777	0.00 0.0000	5.65 0.0227	82	—	62.0 13.9	FAIL
26	123.8	-22.05 -0.0885	0.00 0.0000	2.95 0.0118	60	—	58.7 13.2	FAIL
27	128.1	-24.75 -0.0994	0.00 0.0000	0.25 0.0010	17	—	55.3 12.4	FAIL
28	132.4	-27.45 -0.1102	0.00 0.0000	-2.45 -0.0098	0	—	52.0 11.7	FAIL
29	136.6	-30.15 -0.1210	0.00 0.0000	-5.15 -0.0207	0	—	48.6 10.9	FAIL
30	140.9	-32.85 -0.1319	0.00 0.0000	-7.85 -0.0315	0	—	45.3 10.2	FAIL
31	145.2	-35.55 -0.1427	0.00 0.0000	-10.55 -0.0423	0	—	41.9 9.4	FAIL
32	149.4	-38.25 -0.1535	0.00 0.0000	-13.25 -0.0532	0	—	38.6 8.7	FAIL
33	153.7	-40.95 -0.1644	0.00 0.0000	-15.95 -0.0640	0	—	35.2 7.9	FAIL

Floor	Height (m)	ΔP_{stack} (Pa / in. w.g.)	ΔP_{wind} (Pa / in. w.g.)	ΔP_{net} (Pa / in. w.g.)	Q_{leak} (CFM)	Q_{open} (CFM)	F_{total} (N / lbf)	Status
34	158.0	-43.64 -0.1752	0.00 0.0000	-18.64 -0.0749	0	—	31.9 7.2	FAIL
35	162.3	-46.34 -0.1861	0.00 0.0000	-21.34 -0.0857	0	—	28.5 6.4	FAIL
36	166.5	-49.04 -0.1969	0.00 0.0000	-24.04 -0.0965	0	—	25.2 5.7	FAIL
37	170.8	-51.74 -0.2077	0.00 0.0000	-26.74 -0.1074	0	—	21.9 4.9	FAIL
38	175.1	-54.44 -0.2186	0.00 0.0000	-29.44 -0.1182	0	—	18.5 4.2	FAIL

Critical floor (min ΔP): **38** | Critical floor (max force): **B3**

Section C — Stair3 Floor-by-Floor

Floor	Height (m)	ΔP_{stack} (Pa / in. w.g.)	ΔP_{wind} (Pa / in. w.g.)	ΔP_{net} (Pa / in. w.g.)	Q_{leak} (CFM)	Q_{open} (CFM)	F_{total} (N / lbf)	Status
B3	0.0	56.22 0.2257	0.00 0.0000	81.22 0.3261	313	—	155.7 35.0	FAIL
B2	4.3	53.52 0.2149	0.00 0.0000	78.52 0.3152	307	—	152.3 34.2	FAIL
B1	8.5	50.82 0.2040	0.00 0.0000	75.82 0.3044	302	—	149.0 33.5	FAIL
G	12.8	48.12 0.1932	0.00 0.0000	73.12 0.2936	297	—	145.6 32.7	FAIL
1	17.1	45.42 0.1824	0.00 0.0000	70.42 0.2827	291	—	142.3 32.0	FAIL
2	21.3	42.72 0.1715	0.00 0.0000	67.72 0.2719	286	—	138.9 31.2	FAIL
3	25.6	40.02 0.1607	0.00 0.0000	65.02 0.2610	280	—	135.6 30.5	FAIL
4	29.9	37.33 0.1498	0.00 0.0000	62.33 0.2502	274	—	132.3 29.7	PASS
5	34.2	34.63 0.1390	0.00 0.0000	59.63 0.2394	268	—	128.9 29.0	PASS
6	38.4	31.93 0.1282	0.00 0.0000	56.93 0.2285	262	—	125.6 28.2	PASS
7	42.7	29.23 0.1173	0.00 0.0000	54.23 0.2177	256	—	122.2 27.5	PASS
8	47.0	26.53 0.1065	0.00 0.0000	51.53 0.2069	249	—	118.9 26.7	PASS
9	51.2	23.83 0.0957	0.00 0.0000	48.83 0.1960	242	—	115.5 26.0	PASS
10	55.5	21.13 0.0848	0.00 0.0000	46.13 0.1852	236	—	112.2 25.2	PASS
11	59.8	18.43 0.0740	0.00 0.0000	43.43 0.1744	229	—	108.8 24.5	PASS
12	64.0	15.73 0.0632	0.00 0.0000	40.73 0.1635	221	—	105.5 23.7	PASS
13	68.3	13.03 0.0523	0.00 0.0000	38.03 0.1527	214	—	102.1 23.0	PASS
14	72.6	10.34 0.0415	0.00 0.0000	35.34 0.1419	206	—	98.8 22.2	PASS

Floor	Height (m)	ΔP_{stack} (Pa / in. w.g.)	ΔP_{wind} (Pa / in. w.g.)	ΔP_{net} (Pa / in. w.g.)	Q_leak (CFM)	Q_open (CFM)	F_total (N / lbf)	Status
15	76.9	7.64 0.0307	0.00 0.0000	32.64 0.1310	198	—	95.5 21.5	PASS
16	81.1	4.94 0.0198	0.00 0.0000	29.94 0.1202	190	—	92.1 20.7	PASS
17	85.4	2.24 0.0090	0.00 0.0000	2.24 0.0090	0	4895	57.8 13.0	PASS
18	89.7	-0.46 -0.0018	0.00 0.0000	24.54 0.0985	172	—	85.4 19.2	PASS
19	93.9	-3.16 -0.0127	0.00 0.0000	21.84 0.0877	162	—	82.1 18.5	PASS
20	98.2	-5.86 -0.0235	0.00 0.0000	19.14 0.0768	152	—	78.7 17.7	PASS
21	102.5	-8.56 -0.0344	0.00 0.0000	16.44 0.0660	141	—	75.4 16.9	PASS
22	106.7	-11.26 -0.0452	0.00 0.0000	13.74 0.0552	129	—	72.0 16.2	PASS
23	111.0	-13.96 -0.0560	0.00 0.0000	11.04 0.0443	115	—	68.7 15.4	FAIL
24	115.3	-16.65 -0.0669	0.00 0.0000	8.35 0.0335	100	—	65.3 14.7	FAIL
25	119.6	-19.35 -0.0777	0.00 0.0000	5.65 0.0227	82	—	62.0 13.9	FAIL
26	123.8	-22.05 -0.0885	0.00 0.0000	2.95 0.0118	60	—	58.7 13.2	FAIL
27	128.1	-24.75 -0.0994	0.00 0.0000	0.25 0.0010	17	—	55.3 12.4	FAIL
28	132.4	-27.45 -0.1102	0.00 0.0000	-2.45 -0.0098	0	—	52.0 11.7	FAIL
29	136.6	-30.15 -0.1210	0.00 0.0000	-5.15 -0.0207	0	—	48.6 10.9	FAIL
30	140.9	-32.85 -0.1319	0.00 0.0000	-7.85 -0.0315	0	—	45.3 10.2	FAIL
31	145.2	-35.55 -0.1427	0.00 0.0000	-10.55 -0.0423	0	—	41.9 9.4	FAIL
32	149.4	-38.25 -0.1535	0.00 0.0000	-13.25 -0.0532	0	—	38.6 8.7	FAIL
33	153.7	-40.95 -0.1644	0.00 0.0000	-15.95 -0.0640	0	—	35.2 7.9	FAIL

Floor	Height (m)	ΔP_{stack} (Pa / in. w.g.)	ΔP_{wind} (Pa / in. w.g.)	ΔP_{net} (Pa / in. w.g.)	Q_leak (CFM)	Q_open (CFM)	F_total (N / lbf)	Status
34	158.0	-43.64 -0.1752	0.00 0.0000	-18.64 -0.0749	0	—	31.9 7.2	FAIL
35	162.3	-46.34 -0.1861	0.00 0.0000	-21.34 -0.0857	0	—	28.5 6.4	FAIL
36	166.5	-49.04 -0.1969	0.00 0.0000	-24.04 -0.0965	0	—	25.2 5.7	FAIL
37	170.8	-51.74 -0.2077	0.00 0.0000	-26.74 -0.1074	0	—	21.9 4.9	FAIL

Critical floor (min ΔP): **37** | Critical floor (max force): **B3**

Section C — Stair5 Floor-by-Floor

Floor	Height (m)	ΔP_{stack} (Pa / in. w.g.)	ΔP_{wind} (Pa / in. w.g.)	ΔP_{net} (Pa / in. w.g.)	Q_{leak} (CFM)	Q_{open} (CFM)	F_{total} (N / lbf)	Status
B3	0.0	56.22 0.2257	0.00 0.0000	81.22 0.3261	313	—	155.7 35.0	FAIL
B2	4.3	53.52 0.2149	0.00 0.0000	78.52 0.3152	307	—	152.3 34.2	FAIL
B1	8.5	50.82 0.2040	0.00 0.0000	75.82 0.3044	302	—	149.0 33.5	FAIL
G	12.8	48.12 0.1932	0.00 0.0000	73.12 0.2936	297	—	145.6 32.7	FAIL
1	17.1	45.42 0.1824	0.00 0.0000	70.42 0.2827	291	—	142.3 32.0	FAIL
2	21.3	42.72 0.1715	0.00 0.0000	67.72 0.2719	286	—	138.9 31.2	FAIL
3	25.6	40.02 0.1607	0.00 0.0000	65.02 0.2610	280	—	135.6 30.5	FAIL

Critical floor (min ΔP): 3 | Critical floor (max force): B3

Section C — Stair6 Floor-by-Floor

Floor	Height (m)	ΔP_{stack} (Pa / in. w.g.)	ΔP_{wind} (Pa / in. w.g.)	ΔP_{net} (Pa / in. w.g.)	Q_{leak} (CFM)	Q_{open} (CFM)	F_{total} (N / lbf)	Status
B3	0.0	56.22 0.2257	0.00 0.0000	81.22 0.3261	313	—	155.7 35.0	FAIL
B2	4.3	53.52 0.2149	0.00 0.0000	78.52 0.3152	307	—	152.3 34.2	FAIL
B1	8.5	50.82 0.2040	0.00 0.0000	75.82 0.3044	302	—	149.0 33.5	FAIL
G	12.8	48.12 0.1932	0.00 0.0000	73.12 0.2936	297	—	145.6 32.7	FAIL
1	17.1	45.42 0.1824	0.00 0.0000	70.42 0.2827	291	—	142.3 32.0	FAIL
2	21.3	42.72 0.1715	0.00 0.0000	67.72 0.2719	286	—	138.9 31.2	FAIL
3	25.6	40.02 0.1607	0.00 0.0000	65.02 0.2610	280	—	135.6 30.5	FAIL

Critical floor (min ΔP): 3 | Critical floor (max force): B3

Section C — Stair7 Floor-by-Floor

Floor	Height (m)	ΔP_{stack} (Pa / in. w.g.)	ΔP_{wind} (Pa / in. w.g.)	ΔP_{net} (Pa / in. w.g.)	Q_{leak} (CFM)	Q_{open} (CFM)	F_{total} (N / lbf)	Status
B3	0.0	56.22 0.2257	0.00 0.0000	81.22 0.3261	313	—	155.7 35.0	FAIL
B2	4.3	53.52 0.2149	0.00 0.0000	78.52 0.3152	307	—	152.3 34.2	FAIL
B1	8.5	50.82 0.2040	0.00 0.0000	75.82 0.3044	302	—	149.0 33.5	FAIL
G	12.8	48.12 0.1932	0.00 0.0000	73.12 0.2936	297	—	145.6 32.7	FAIL
1	17.1	45.42 0.1824	0.00 0.0000	70.42 0.2827	291	—	142.3 32.0	FAIL
2	21.3	42.72 0.1715	0.00 0.0000	67.72 0.2719	286	—	138.9 31.2	FAIL
3	25.6	40.02 0.1607	0.00 0.0000	65.02 0.2610	280	—	135.6 30.5	FAIL

Critical floor (min ΔP): 3 | Critical floor (max force): B3

Section D — System Summary

Stair1 Supply (all closed)	2.9499 m ³ /s (6250 CFM)
Stair1 Supply (doors open)	5.2599 m ³ /s (11145 CFM)
Stair1 Design Supply	5.2599 m³/s (11145 CFM)
Stair2 Supply (all closed)	3.3817 m ³ /s (7165 CFM)
Stair2 Supply (doors open)	5.6917 m ³ /s (12060 CFM)
Stair2 Design Supply	5.6917 m³/s (12060 CFM)
Stair3 Supply (all closed)	3.3817 m ³ /s (7165 CFM)
Stair3 Supply (doors open)	5.6917 m ³ /s (12060 CFM)
Stair3 Design Supply	5.6917 m³/s (12060 CFM)
Stair5 Supply (all closed)	1.1218 m ³ /s (2377 CFM)
Stair5 Supply (doors open)	3.4318 m ³ /s (7271 CFM)
Stair5 Design Supply	3.4318 m³/s (7271 CFM)
Stair6 Supply (all closed)	1.1218 m ³ /s (2377 CFM)
Stair6 Supply (doors open)	3.4318 m ³ /s (7271 CFM)
Stair6 Design Supply	3.4318 m³/s (7271 CFM)
Stair7 Supply (all closed)	1.1218 m ³ /s (2377 CFM)
Stair7 Supply (doors open)	3.4318 m ³ /s (7271 CFM)
Stair7 Design Supply	3.4318 m³/s (7271 CFM)

Appendix C

Smoke Zones



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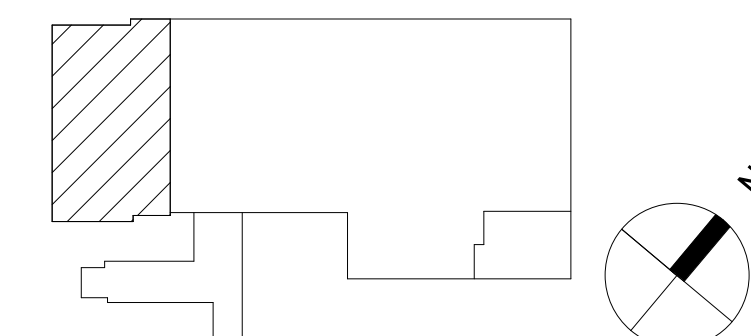
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SMOKE CONTROL LEGEND

- ACTIVE SMOKE CONTROL ZONE
- NON-PRESSURIZED STAIR/ELEVATOR
- PRESSURIZED STAIRS
- ENCLOSED ELEVATOR LOBBY
- PASSIVE SMOKE CONTROL ZONE
- OPEN TO BELOW
- 1 HOUR SMOKE BARRIER
- 2 HOUR FIRE BARRIER
- 1 HOUR FIRE PARTITION
- STAIR IDENTIFICATION NUMBER

DAMPER LEGEND

- NORMALLY CLOSED DAMPER; OPENS DURING SMOKE MODE WHEN INDICATED BY SEQUENCE OF OPERATIONS. DAMPER SERVES SMOKE EXHAUST FAN (SEF)
- NORMALLY OPEN DAMPER; CLOSSES DURING SMOKE MODE WHEN INDICATED BY SEQUENCE OF OPERATIONS. DAMPER SERVES TENANT EXHAUST FAN (TEF)
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2	04/09/2025	CONSTRUCTION BULLETIN 2
1	03/14/2025	CONSTRUCTION BULLETIN 1
0	02/14/2025	ISSUE FOR CONSTRUCTION
G		ADDENDUM 4
B	11/18/2024	ADDENDUM 2 PLAN UPDATES
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PARTITION PLAN - LEVEL 05

IA2-05

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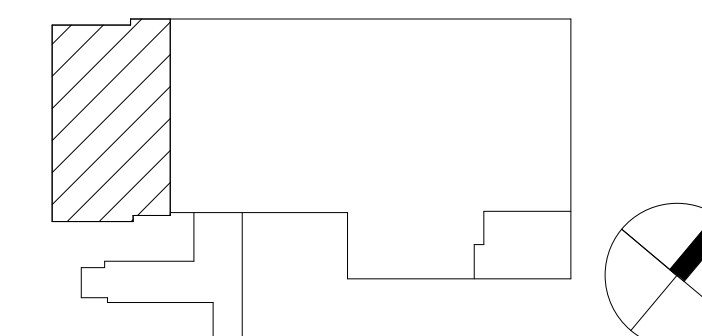
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G		
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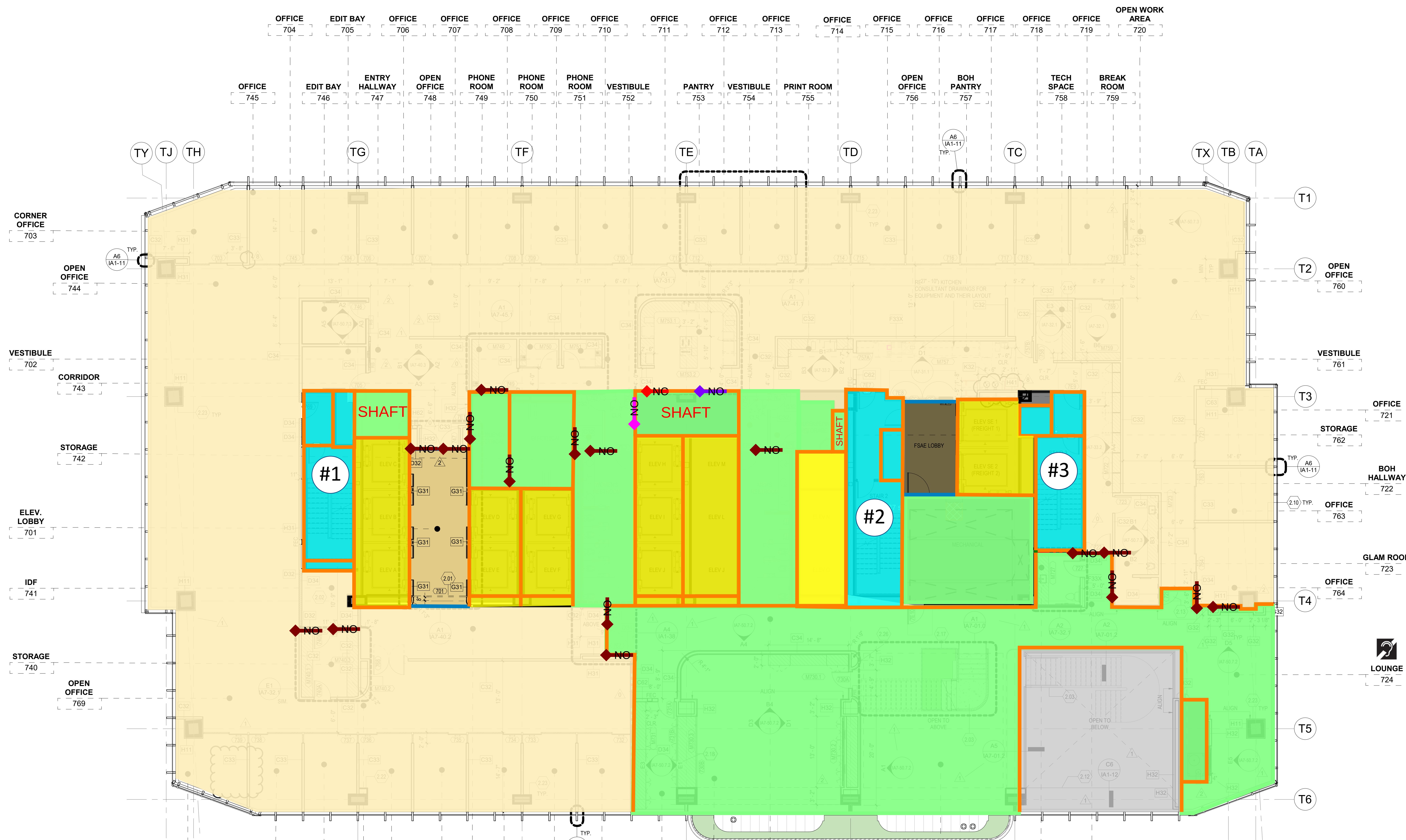
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SMOKE CONTROL LEGEND

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	STAIR IDENTIFICATION NUMBER

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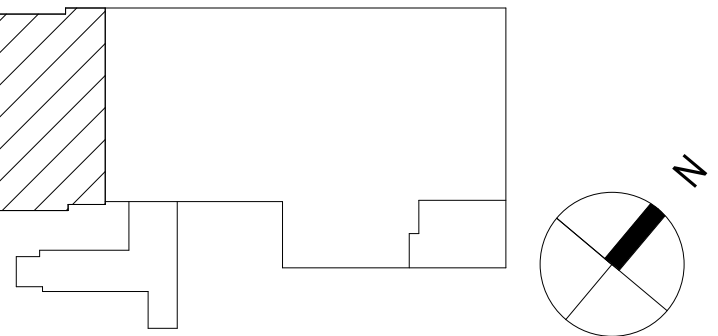
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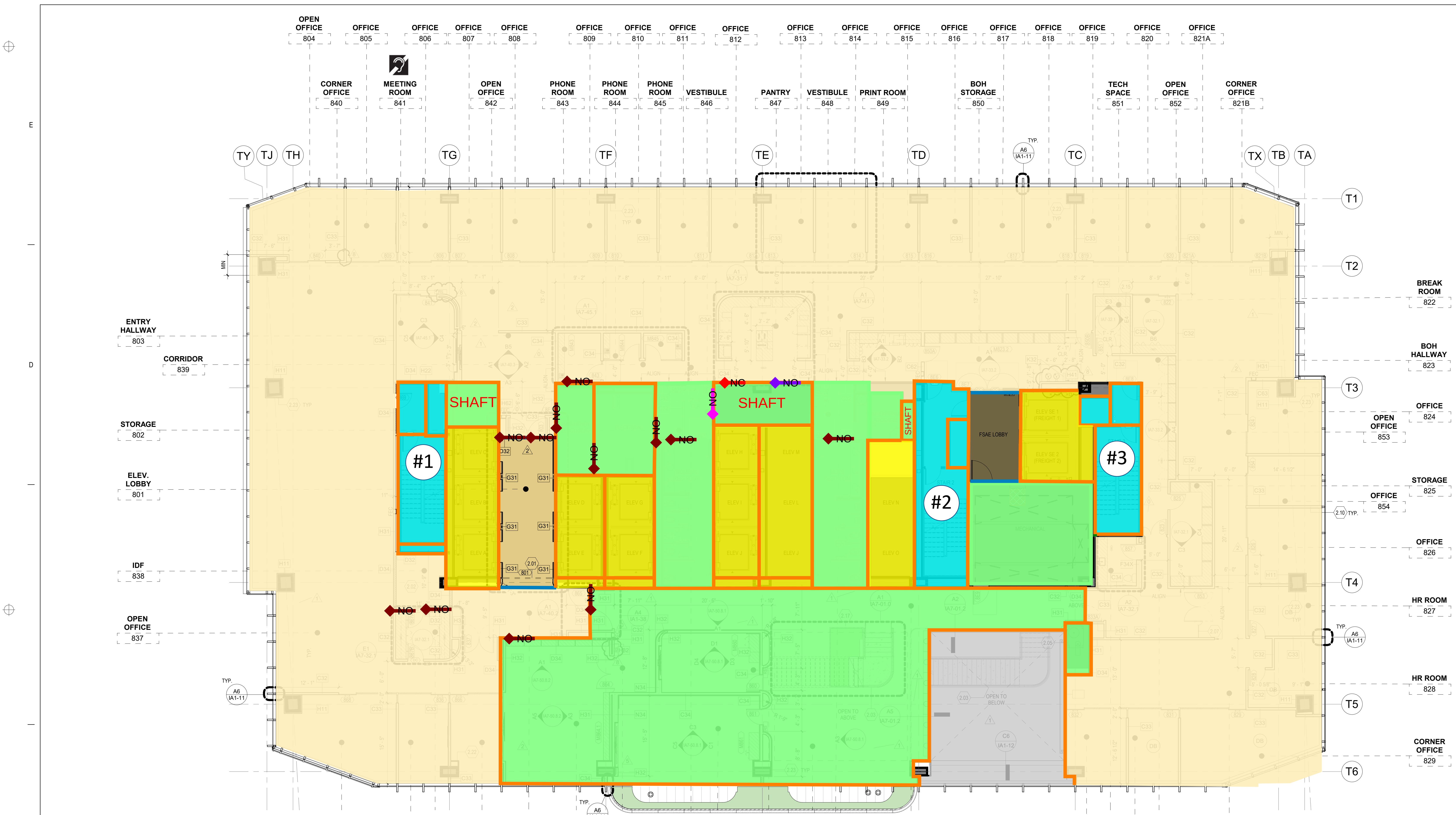


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- PRESSURIZED STAIRS
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- OPEN TO BELOW
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- 2 HOUR FIRE BARRIER
- 1 HOUR FIRE PARTITION
- STAIR IDENTIFICATION NUMBER

DAMPER LEGEND

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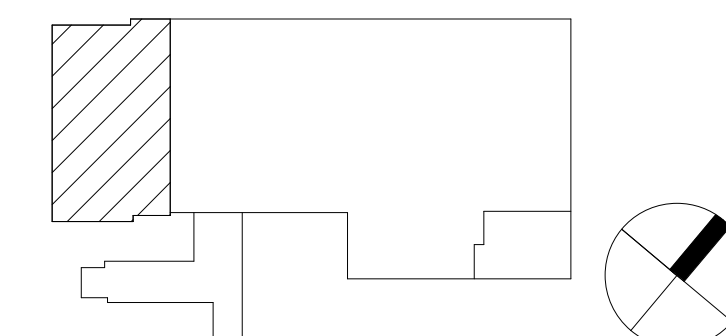
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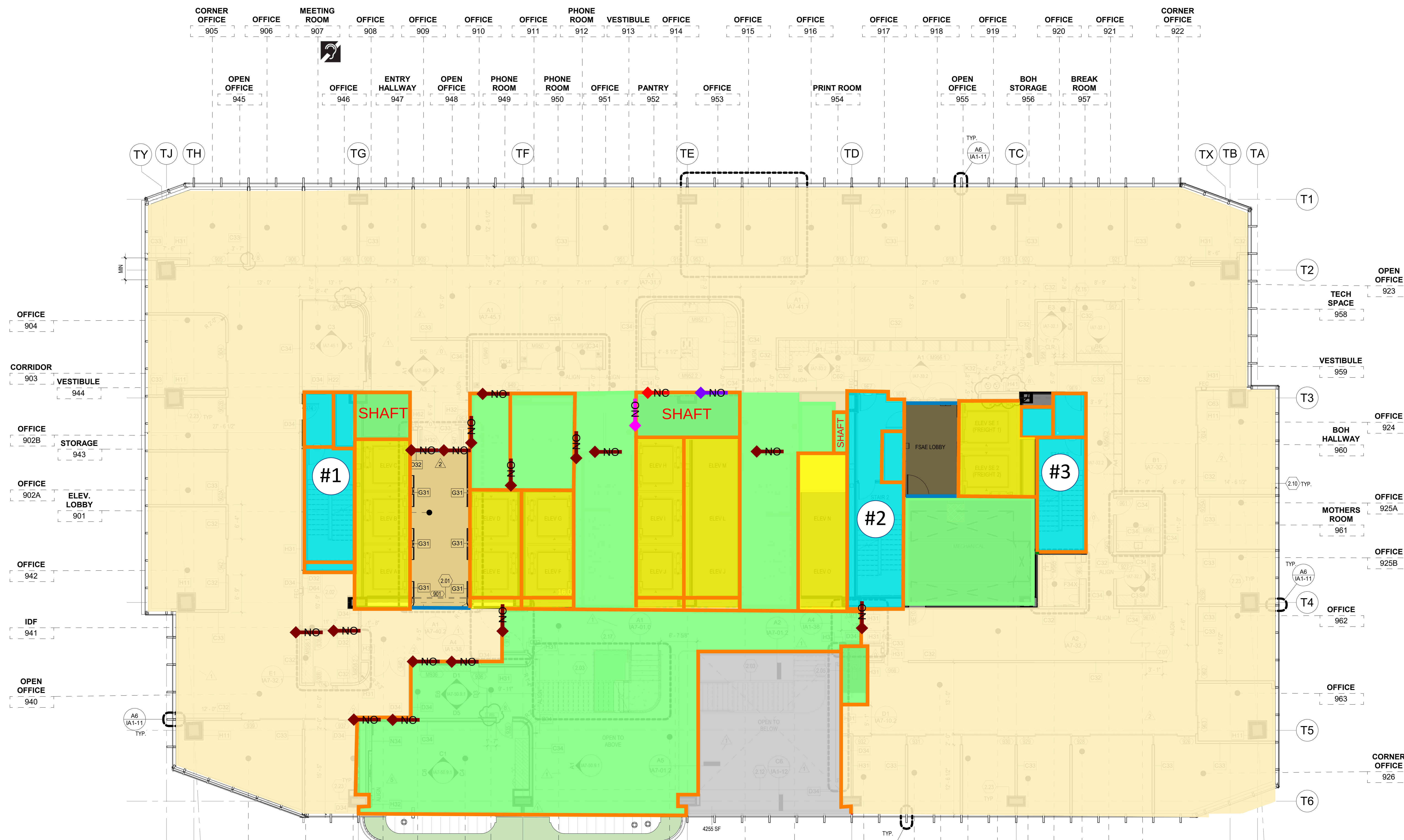
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PARTITION PLAN - LEVEL
09

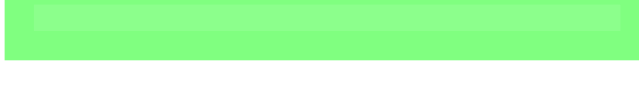
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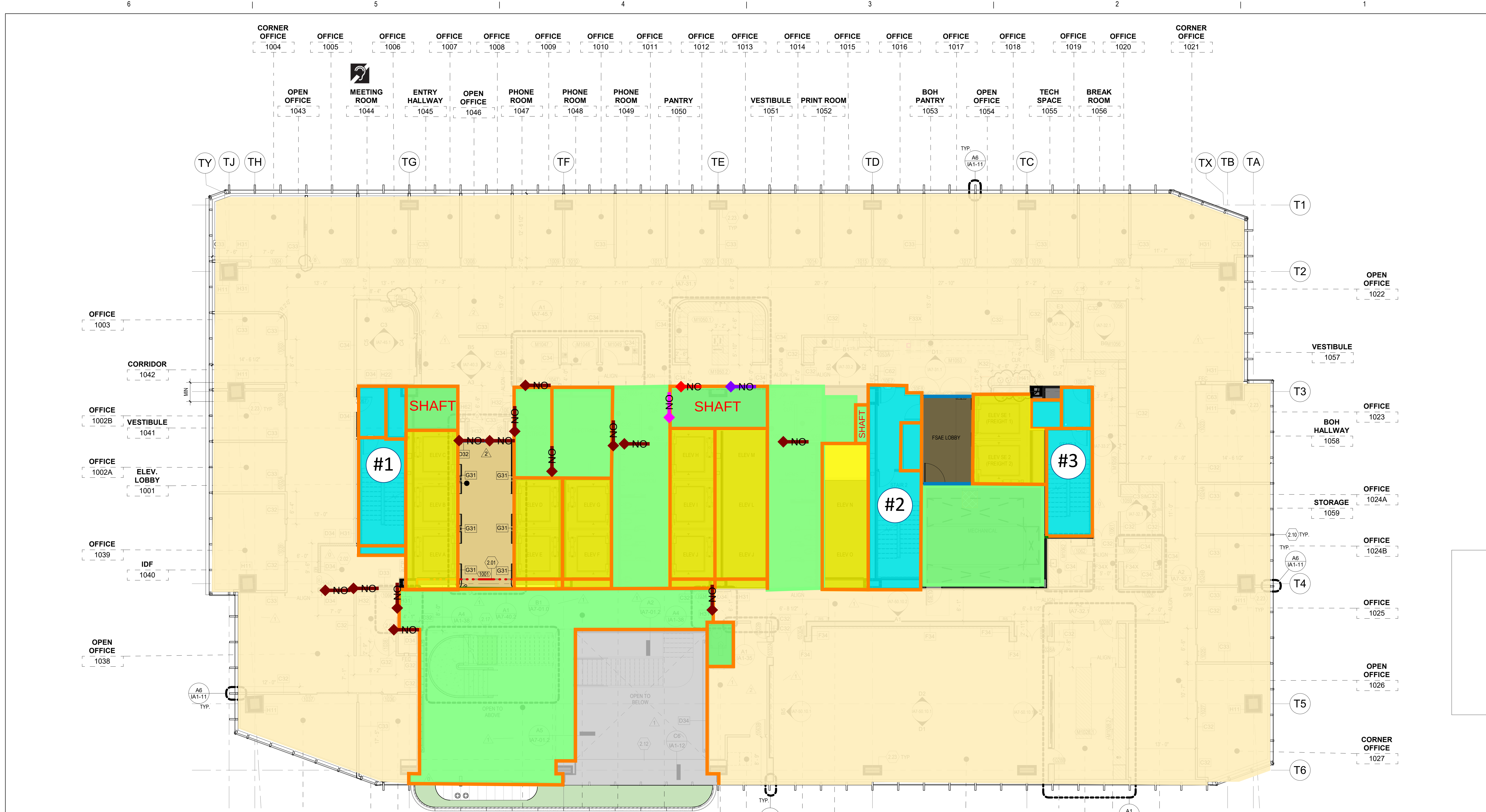
SMOKE CONTROL LEGEND

- | | |
|---|--------------------------------|
|  | ACTIVE SMOKE CONTROL ZONE |
|  | NON-PRESSURIZED STAIR/ELEVATOR |
|  | PRESSURIZED STAIRS |
|  | ENCLOSED ELEVATOR LOBBY |
|  | PASSIVE SMOKE CONTROL ZONE |
|  | OPEN TO BELOW |
|  | 1 HOUR SMOKE BARRIER |
|  | 2 HOUR FIRE BARRIER |
|  | 1 HOUR FIRE PARTITION |
|  | STAIR IDENTIFICATION NUMBER |

DAMPER LEGEND

- | | |
|---|--|
|  | <p>NORMALLY CLOSED DAMPER; OPENS DURING SMOKE MODE WHEN INDICATED BY SEQUENCE OF OPERATIONS. DAMPER SERVES SMOKE EXHAUST FAN (SEF)</p> |
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SMOKE CONTROL LEGEND

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##	STAIR IDENTIFICATION NUMBER

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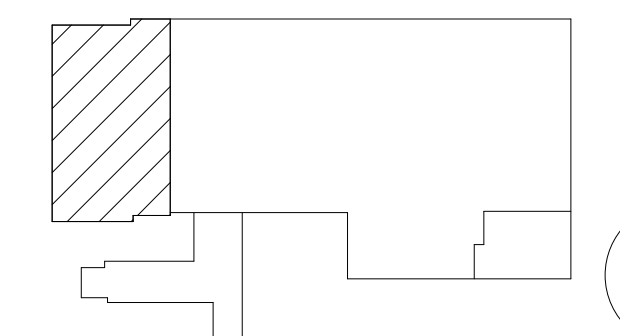
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Cini Little

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Tel: 626.463.7673 D
747.228.4224 M

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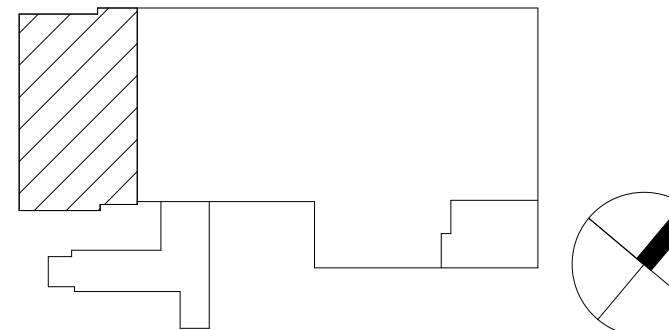
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2	04/09/2025	CONSTRUCTION BULLETIN 2
1	03/14/2025	CONSTRUCTION BULLETIN 1
0	02/14/2025	ISSUE FOR CONSTRUCTION
G		ADDENDUM 4
B	11/18/2024	ADDENDUM 2 PLAN UPDATES
	10/25/2024	BID
A	08/30/2024	PLAN CHECK
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PARTITION PLAN - LEVEL 12

IA2-12

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SMOKE CONTROL LEGEND

- ACTIVE SMOKE CONTROL ZONE
- NON-PRESSURIZED STAIR/ELEVATOR
- PRESSURIZED STAIRS
- ENCLOSED ELEVATOR LOBBY
- PASSIVE SMOKE CONTROL ZONE
- OPEN TO BELOW
- 1 HOUR SMOKE BARRIER
- 2 HOUR FIRE BARRIER
- 1 HOUR FIRE PARTITION
- STAIR IDENTIFICATION NUMBER

DAMPER LEGEND

- NORMALLY CLOSED DAMPER; OPENS DURING SMOKE MODE WHEN INDICATED BY SEQUENCE OF OPERATIONS. DAMPER SERVES SMOKE EXHAUST FAN (SEF)
- NORMALLY OPEN DAMPER; CLOSSES DURING SMOKE MODE WHEN INDICATED BY SEQUENCE OF OPERATIONS. DAMPER SERVES TENANT EXHAUST FAN (TEF)
- NORMALLY OPEN DAMPER; CLOSSES DURING SMOKE MODE WHEN INDICATED BY SEQUENCE OF OPERATIONS. DAMPER SERVES TOILET EXHAUST FAN (TLEF)
- NORMALLY OPEN DAMPER; CLOSSES DURING SMOKE MODE WHEN INDICATED BY SEQUENCE OF OPERATIONS. DAMPER SERVES VARIOUS COMFORT HVAC SYSTEMS.

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Cini Little

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Bjarke Ingels Group

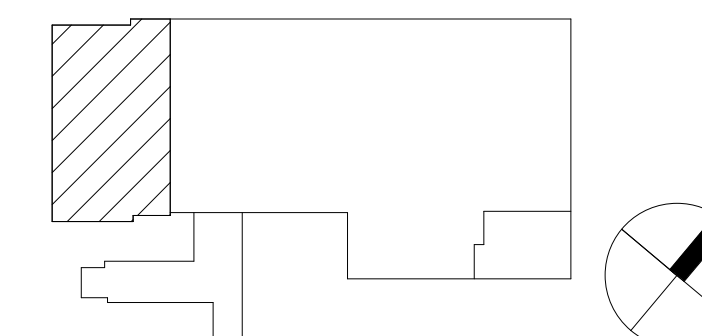
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T	03/14/2025	CONSTRUCTION BULLETIN 1
O	02/14/2025	ISSUE FOR CONSTRUCTION
G		ADDENDUM 4
B	11/18/2024	ADDENDUM 2 PLAN UPDATES
	19/25/2024	BID
A		ADDENDUM A
	08/30/2024	PLAN CHECK
	06/14/2024	PRELIMINARY PRICING
	02/26/2024	ANALOGY REVIEW

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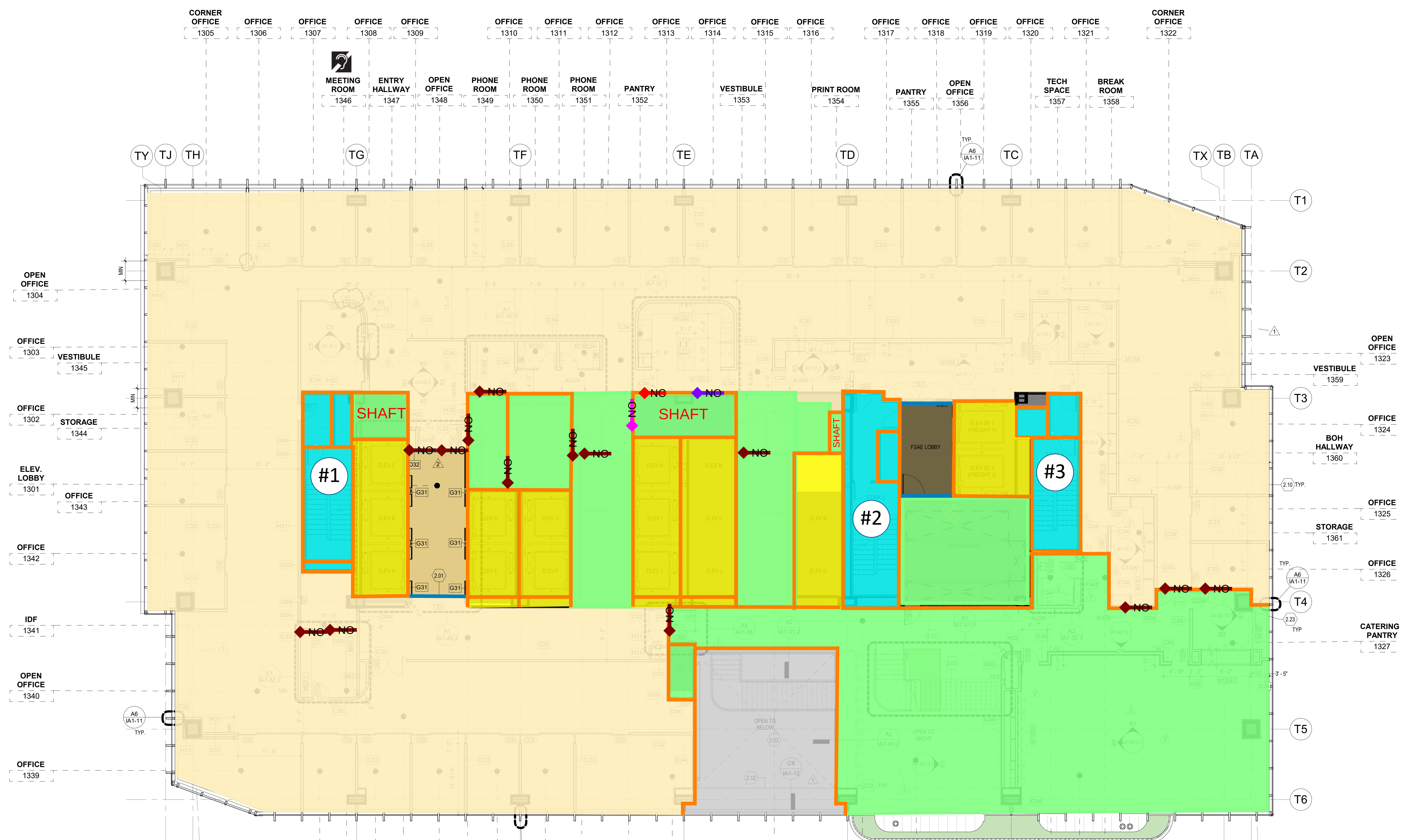
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PARTITION PLAN - LEVEL
13

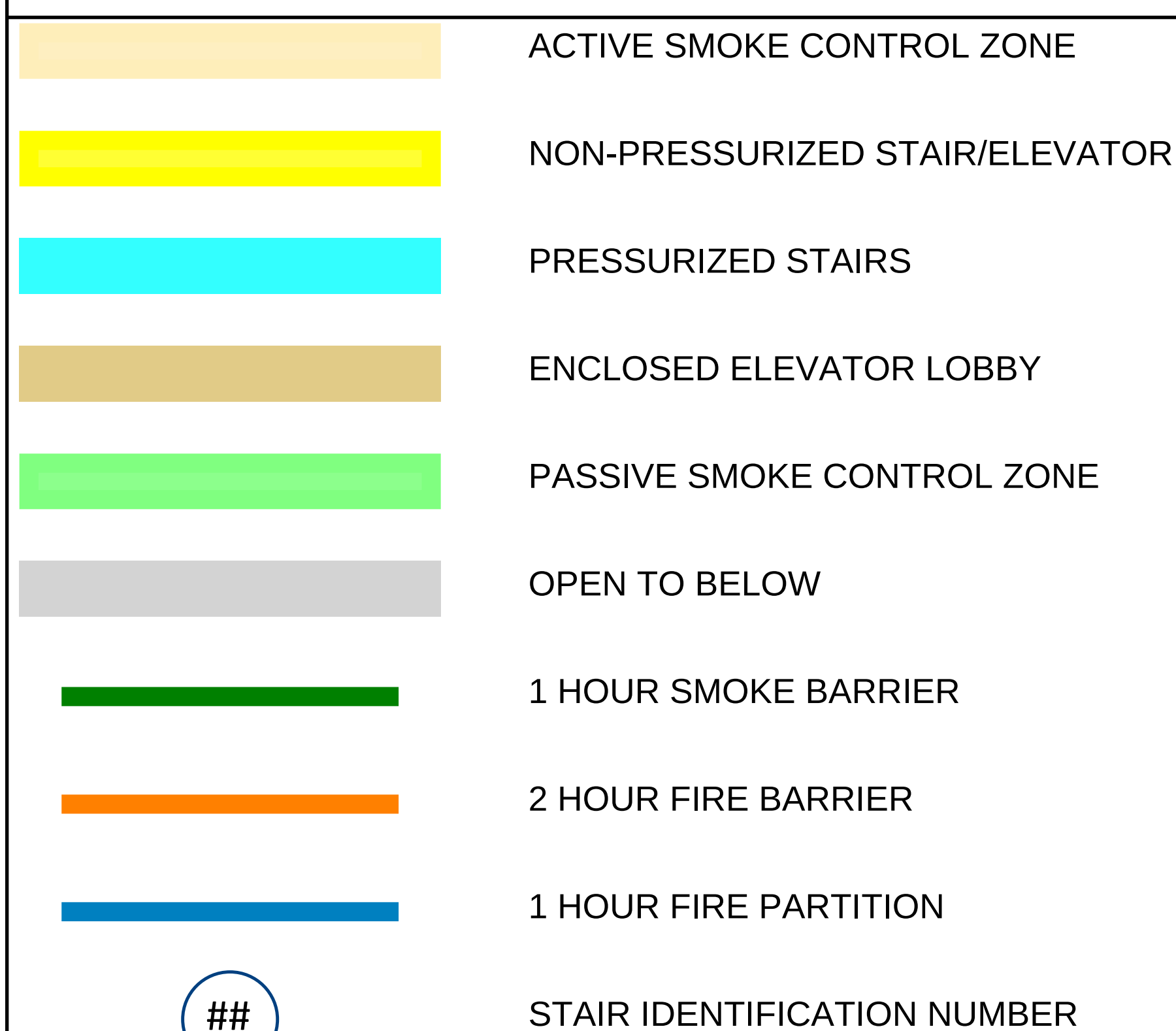
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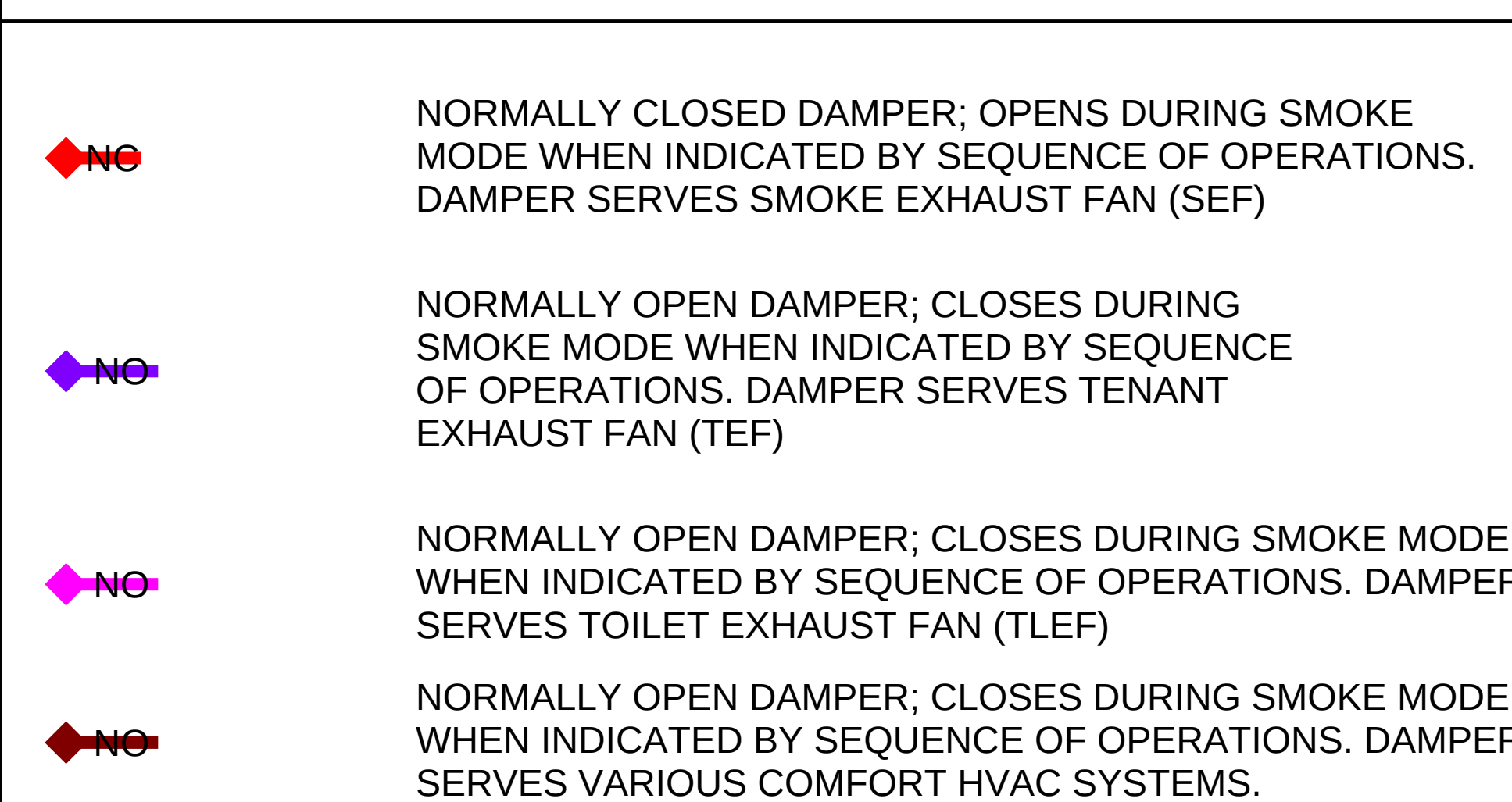
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SMOKE CONTROL LEGEND



DAMPER LEGEND





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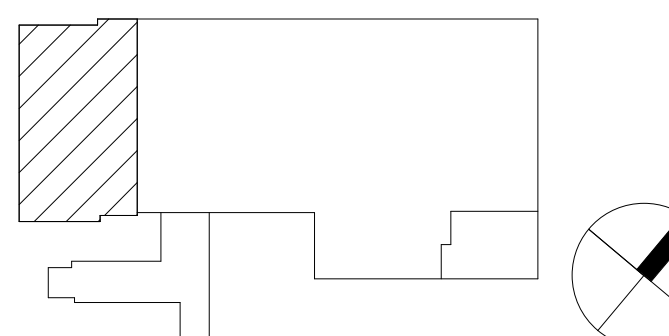
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	02/28/2024	LANDLORD REVIEW

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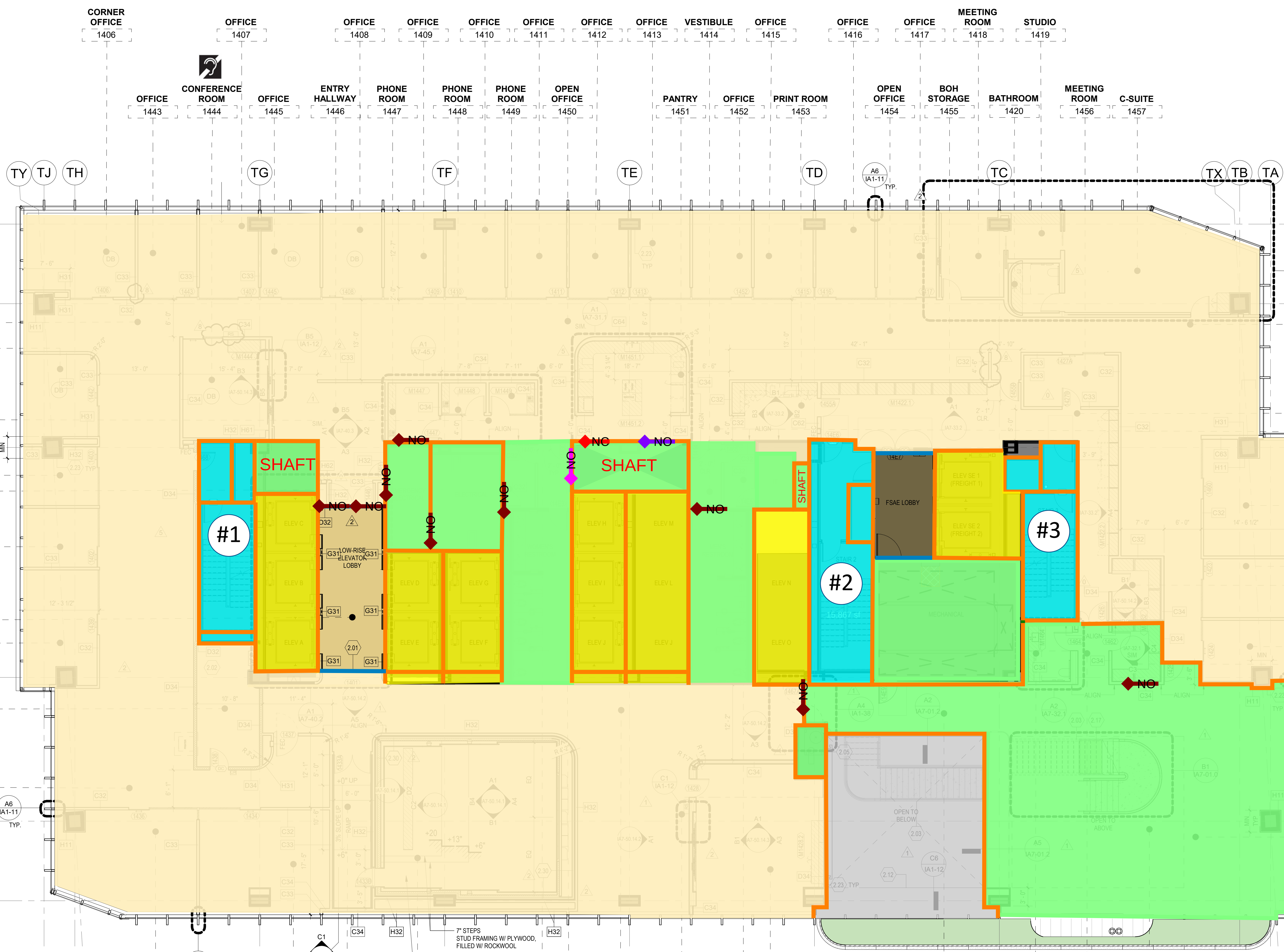


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PARTITION PLAN - LEVEL 14

IA2-14

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SMOKE CONTROL LEGEND

- ACTIVE SMOKE CONTROL ZONE
- NON-PRESSURIZED STAIR/ELEVATOR
- PRESSURIZED STAIRS
- ENCLOSED ELEVATOR LOBBY
- PASSIVE SMOKE CONTROL ZONE
- OPEN TO BELOW
- 1 HOUR SMOKE BARRIER
- 2 HOUR FIRE BARRIER
- 1 HOUR FIRE PARTITION
- STAIR IDENTIFICATION NUMBER

DAMPER LEGEND

- NORMALLY CLOSED DAMPER; OPENS DURING SMOKE MODE WHEN INDICATED BY SEQUENCE OF OPERATIONS. DAMPER SERVES SMOKE EXHAUST FAN (SEF)
- NORMALLY OPEN DAMPER; CLOSSES DURING SMOKE MODE WHEN INDICATED BY SEQUENCE OF OPERATIONS. DAMPER SERVES TENANT EXHAUST FAN (TEF)
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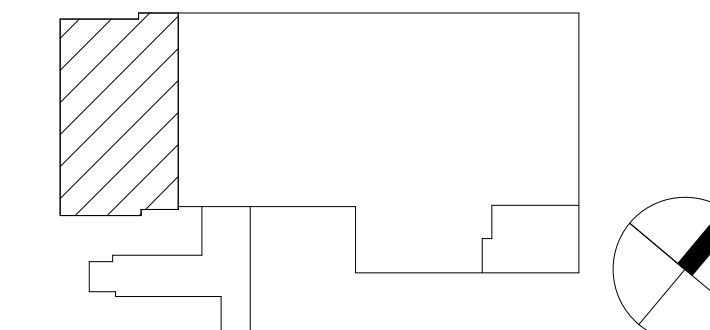
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1	03/14/2025	CONSTRUCTION BULLETIN 1
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G		ADDENDUM 4
B	11/18/2024	ADDENDUM 2 PLAN UPDATES
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PARTITION PLAN - LEVEL 15

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SMOKE CONTROL LEGEND

- ACTIVE SMOKE CONTROL ZONE
- NON-PRESSURIZED STAIR/ELEVATOR
- PRESSURIZED STAIRS
- ENCLOSED ELEVATOR LOBBY
- PASSIVE SMOKE CONTROL ZONE
- OPEN TO BELOW
- 1 HOUR SMOKE BARRIER
- 2 HOUR FIRE BARRIER
- 1 HOUR FIRE PARTITION
- STAIR IDENTIFICATION NUMBER

DAMPER LEGEND

- NORMALLY CLOSED DAMPER; OPENS DURING SMOKE MODE WHEN INDICATED BY SEQUENCE OF OPERATIONS. DAMPER SERVES SMOKE EXHAUST FAN (SEF)
- NORMALLY OPEN DAMPER; CLOSSES DURING SMOKE MODE WHEN INDICATED BY SEQUENCE OF OPERATIONS. DAMPER SERVES TENANT EXHAUST FAN (TEF)
- NORMALLY OPEN DAMPER; CLOSSES DURING SMOKE MODE WHEN INDICATED BY SEQUENCE OF OPERATIONS. DAMPER SERVES TOILET EXHAUST FAN (TLEF)
- NORMALLY OPEN DAMPER; CLOSSES DURING SMOKE MODE WHEN INDICATED BY SEQUENCE OF OPERATIONS. DAMPER SERVES VARIOUS COMFORT HVAC SYSTEMS.



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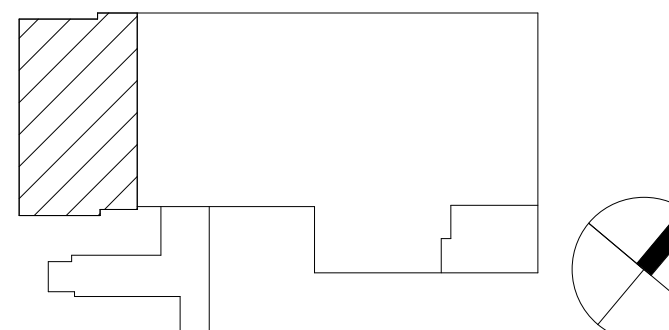
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B	11/18/2024	ADDENDUM 2 PLAN UPDATES
	10/25/2024	BID
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PARTITION PLAN - LEVEL 16

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SMOKE CONTROL LEGEND

- ACTIVE SMOKE CONTROL ZONE
- NON-PRESSURIZED STAIR/ELEVATOR
- PRESSURIZED STAIRS
- ENCLOSED ELEVATOR LOBBY
- PASSIVE SMOKE CONTROL ZONE
- OPEN TO BELOW
- 1 HOUR SMOKE BARRIER
- 2 HOUR FIRE BARRIER
- 1 HOUR FIRE PARTITION
- STAIR IDENTIFICATION NUMBER

DAMPER LEGEND

- NORMALLY CLOSED DAMPER; OPENS DURING SMOKE MODE WHEN INDICATED BY SEQUENCE OF OPERATIONS. DAMPER SERVES SMOKE EXHAUST FAN (SEF)
- NORMALLY OPEN DAMPER; CLOSSES DURING SMOKE MODE WHEN INDICATED BY SEQUENCE OF OPERATIONS. DAMPER SERVES TENANT EXHAUST FAN (TEF)
- NORMALLY OPEN DAMPER; CLOSSES DURING SMOKE MODE WHEN INDICATED BY SEQUENCE OF OPERATIONS. DAMPER SERVES TOILET EXHAUST FAN (TLEF)
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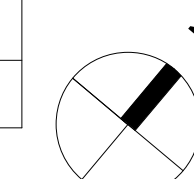
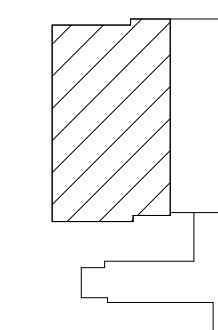
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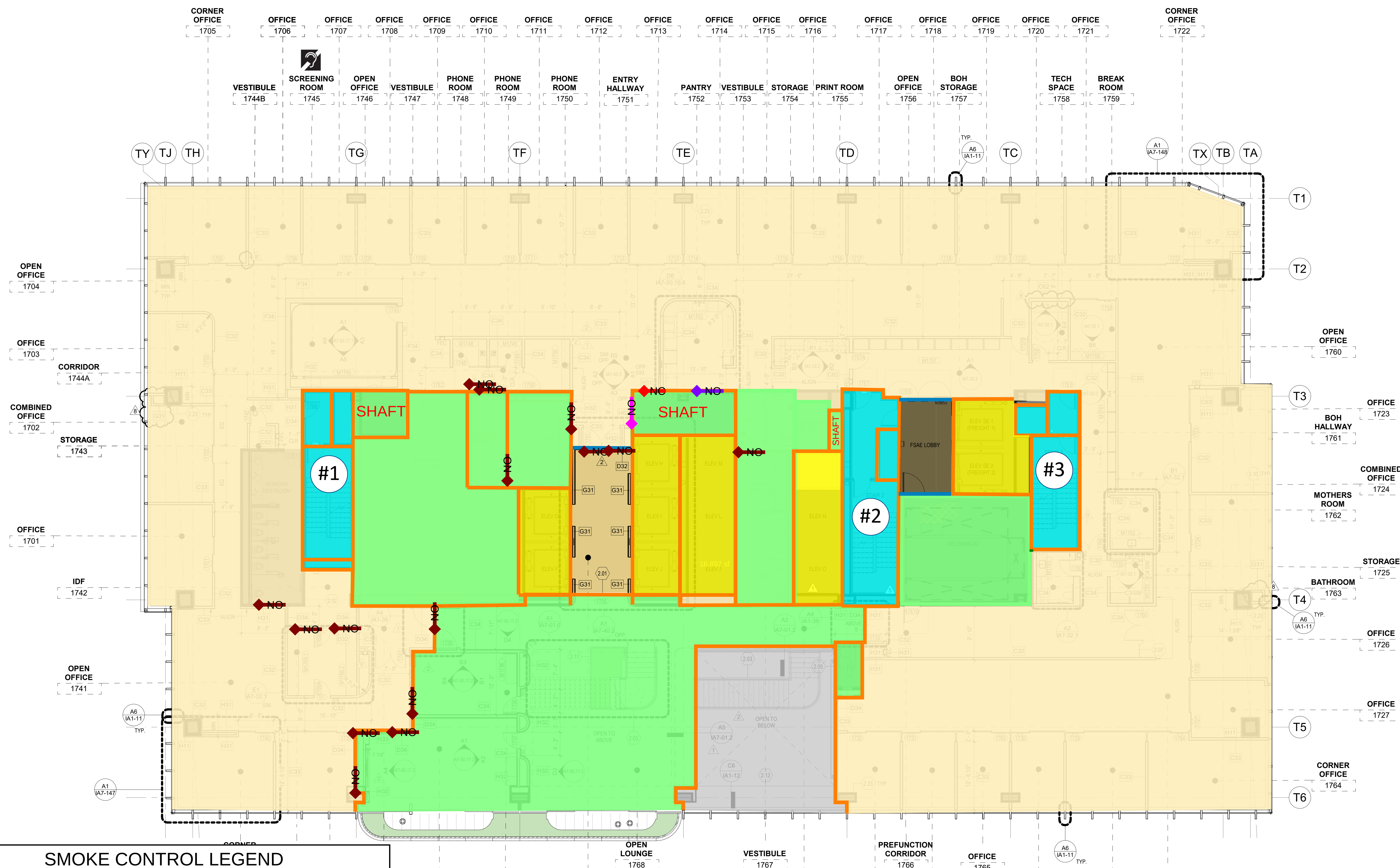
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PARTITION PLAN - LEVEL
17

IA2-17

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SMOKE CONTROL LEGEND

ACTIVE SMOKE CONTROL ZONE

NON-PRESSURIZED STAIR/ELEVATOR

PRESSURIZED STAIRS

ENCLOSED ELEVATOR LOBBY

PASSIVE SMOKE CONTROL ZONE

OPEN TO BELOW

1 HOUR SMOKE BARRIER

2 HOUR FIRE BARRIER

1 HOUR FIRE PARTITION

STAIR IDENTIFICATION NUMBER

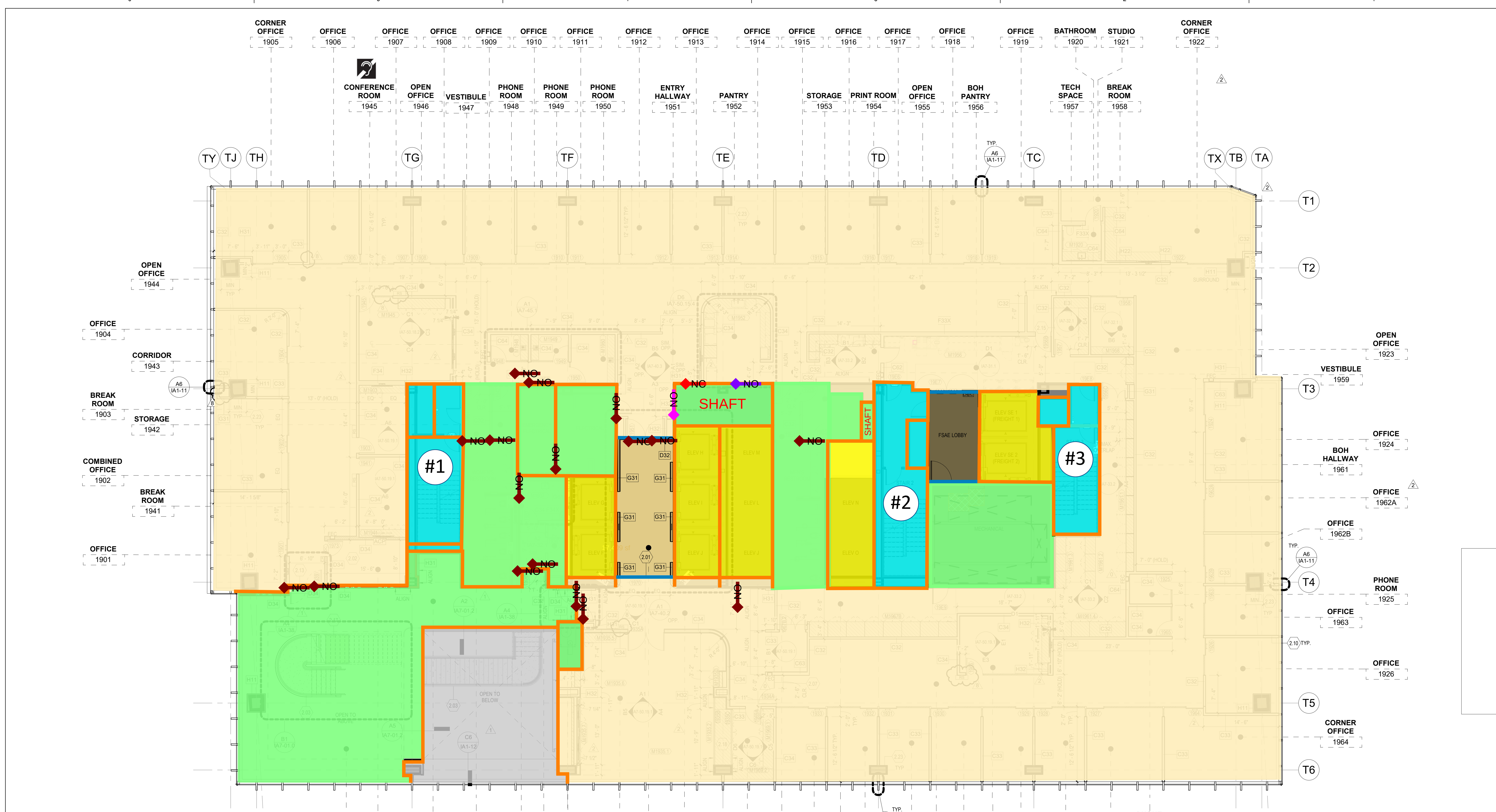
DAMPER LEGEND

NORMALLY CLOSED DAMPER; OPENS DURING SMOKE
MODE WHEN INDICATED BY SEQUENCE OF OPERATIONS.
DAMPER SERVES SMOKE EXHAUST FAN (SEE)

NORMALLY OPEN DAMPER; CLOSSES DURING SMOKE MODE WHEN INDICATED BY SEQUENCE OF OPERATIONS. DAMPER SERVES TENANT EXHAUST FAN (TEF)

NORMALLY OPEN DAMPER; CLOSSES DURING SMOKE MODE
WHEN INDICATED BY SEQUENCE OF OPERATIONS. DAMPER
SERVES TOILET EXHAUST FAN (TLEF)

NORMALLY OPEN DAMPER; CLOSSES DURING SMOKE MODE WHEN INDICATED BY SEQUENCE OF OPERATIONS. DAMPER SERVES VARIOUS COMFORT HVAC SYSTEMS.



SMOKE CONTROL LEGEND

	ACTIVE SMOKE CONTROL ZONE
	NON-PRESSURIZED STAIR/ELEVATOR
	PRESSURIZED STAIRS
	ENCLOSED ELEVATOR LOBBY
	PASSIVE SMOKE CONTROL ZONE
	OPEN TO BELOW
	1 HOUR SMOKE BARRIER
	2 HOUR FIRE BARRIER
	1 HOUR FIRE PARTITION
##	STAIR IDENTIFICATION NUMBER

DAMPER LEGEND

NO	NORMALLY CLOSED DAMPER; OPENS DURING SMOKE MODE WHEN INDICATED BY SEQUENCE OF OPERATIONS. DAMPER SERVES SMOKE EXHAUST FAN (SEF)
NO	NORMALLY OPEN DAMPER; CLOSSES DURING SMOKE MODE WHEN INDICATED BY SEQUENCE OF OPERATIONS. DAMPER SERVES TENANT EXHAUST FAN (TEF)
NO	NORMALLY OPEN DAMPER; CLOSSES DURING SMOKE MODE WHEN INDICATED BY SEQUENCE OF OPERATIONS. DAMPER SERVES TOILET EXHAUST FAN (TLEF)
NO	NORMALLY OPEN DAMPER; CLOSSES DURING SMOKE MODE WHEN INDICATED BY SEQUENCE OF OPERATIONS. DAMPER SERVES VARIOUS COMFORT HVAC SYSTEMS.

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NO.	DATE	REVISION/ISSUE
8	09/12/2025	CONSTRUCTION BULLETIN 8
2	04/09/2025	CONSTRUCTION BULLETIN 2
1	03/14/2025	CONSTRUCTION BULLETIN 1
0	02/14/2025	ISSUE FOR CONSTRUCTION
G		ADDENDUM 4
B	11/18/2024	ADDENDUM 2 PLAN UPDATES
	10/25/2024	BID
	08/30/2024	PLAN CHECK
	06/14/2024	PRELIMINARY PRICING
	02/28/2024	LANDLORD REVIEW

PATRICIA
McCAUL
C-268315
01-31-2025
RENEWAL
DATE

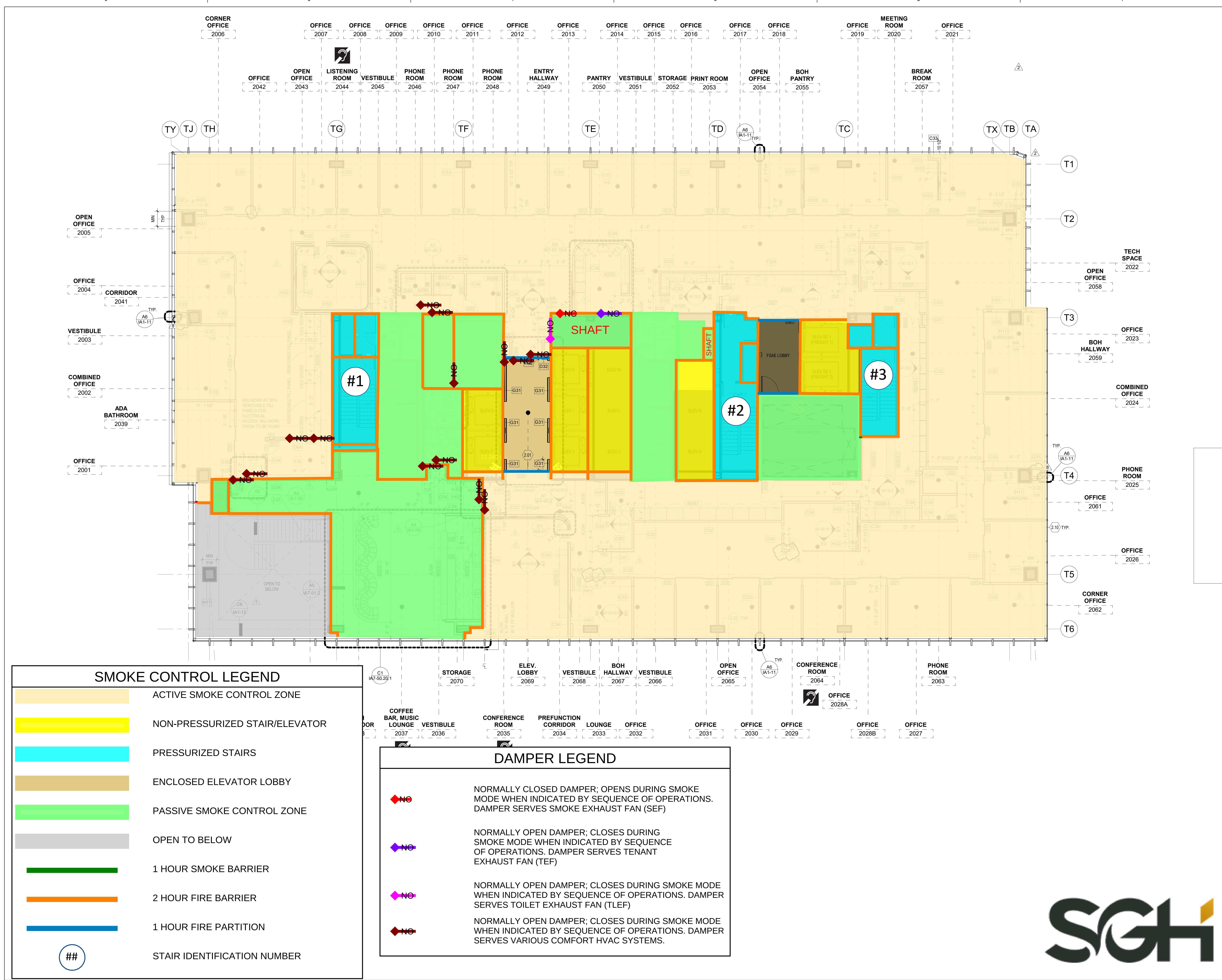
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PARTITION PLAN - LEVEL
19

IA2-19

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		LANDLORD REVIEW

NO. DATE REVISION/ISSUE

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PARTITION PLAN - LEVEL 20

IA2-20

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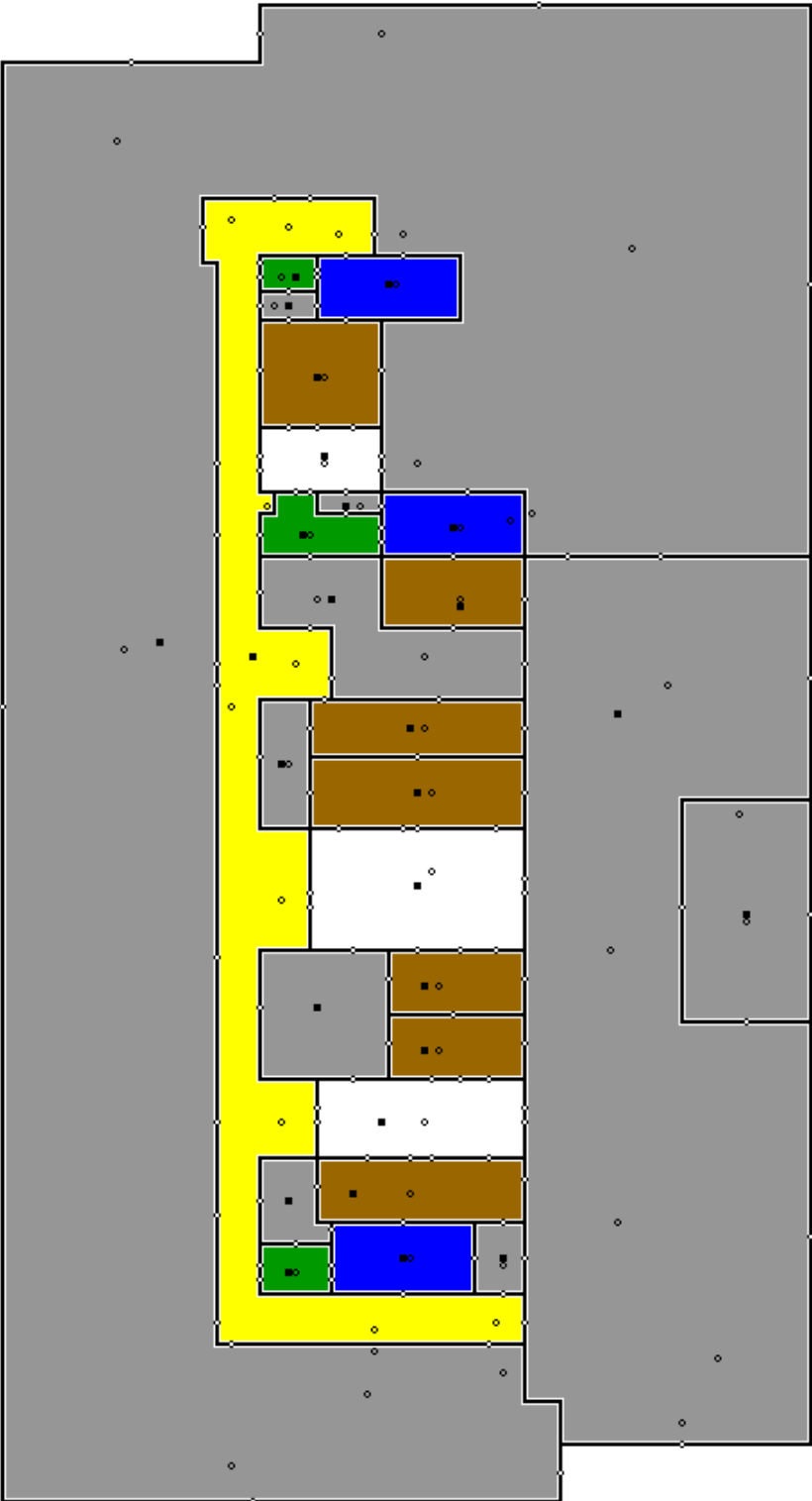
Appendix D
CONTAM Results

Winter Wind

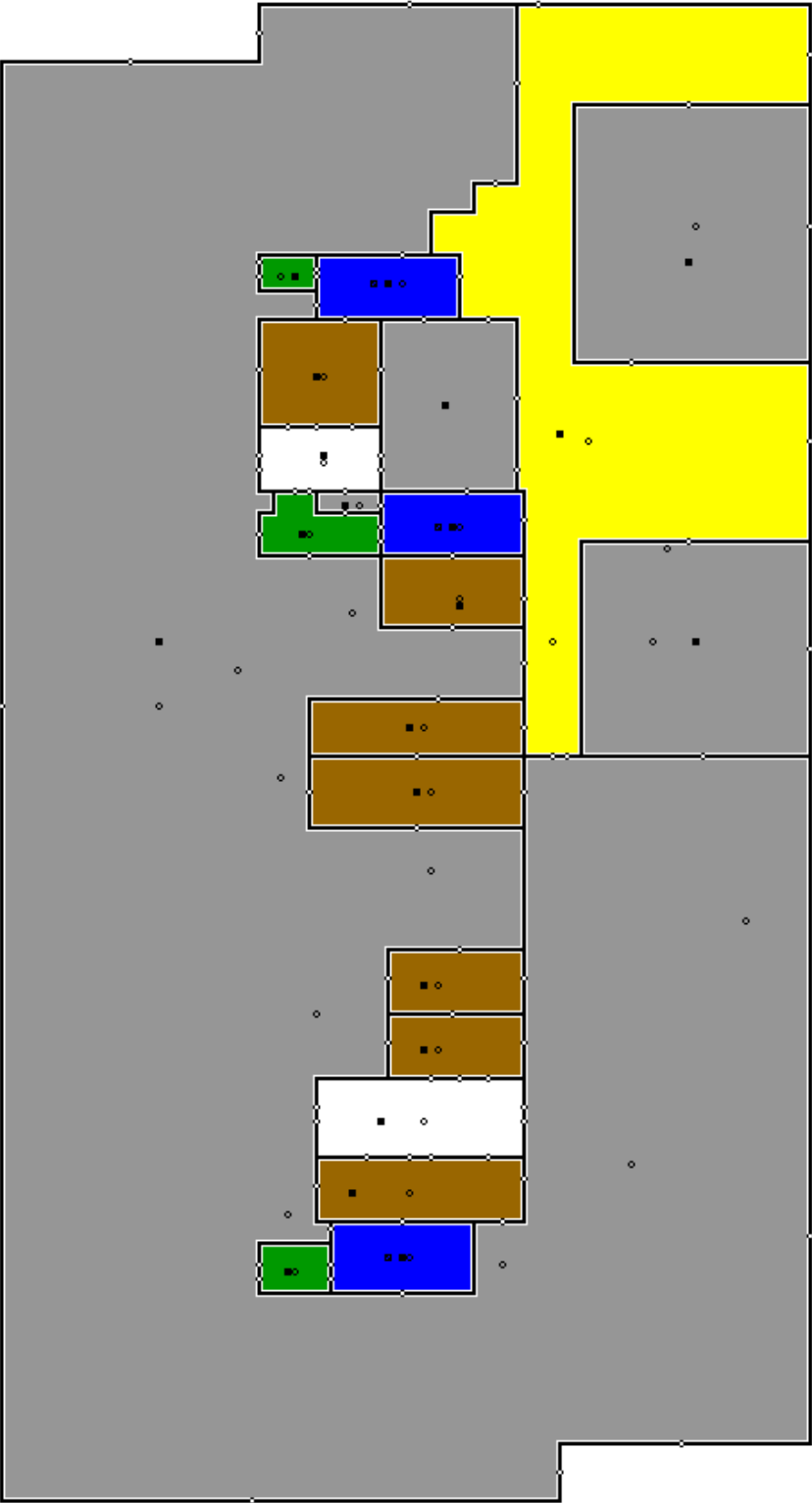
Fire Floor	Floor Exhaust Rate [cfm]	ΔP from Active Floor to Adjacent Floors [in. H ₂ O]		ΔP from Stairwell to Stair Vestibule (or Directly From Stair to Floor) [in. H ₂ O]			ΔP from Stair Vestibule to Floor/Lobby (or Directly From Stair to Exterior) [in. H ₂ O]		
		Above	Below	Stair 1	Stair 2	Stair 3	Stair 1	Stair 2	Stair 3
4	8000	0.055	0.072	0.120	0.136	0.126	0.105	0.056	0.110
5	8000	0.107	0.106	0.149	0.145	0.155	0.130	0.102	0.136
6	8000	0.098	0.102	0.145	0.149	0.151	0.127	0.091	0.133
11	8000	0.104	0.105	0.150	0.160	0.155	0.131	0.088	0.136
12	8000	0.103	0.107	0.151	0.161	0.156	0.131	0.089	0.137
13	8000	0.116	0.115	0.157	0.149	0.163	0.137	0.113	0.143
19	8000	0.061	0.061	0.129	0.144	0.136	0.113	0.067	0.119
20	6000	0.103	0.110	0.153	0.162	0.159	0.133	0.093	0.139
21	6000	0.105	0.107	0.154	0.163	0.161	0.135	0.095	0.141

Appendix E
CONTAM Screenshots

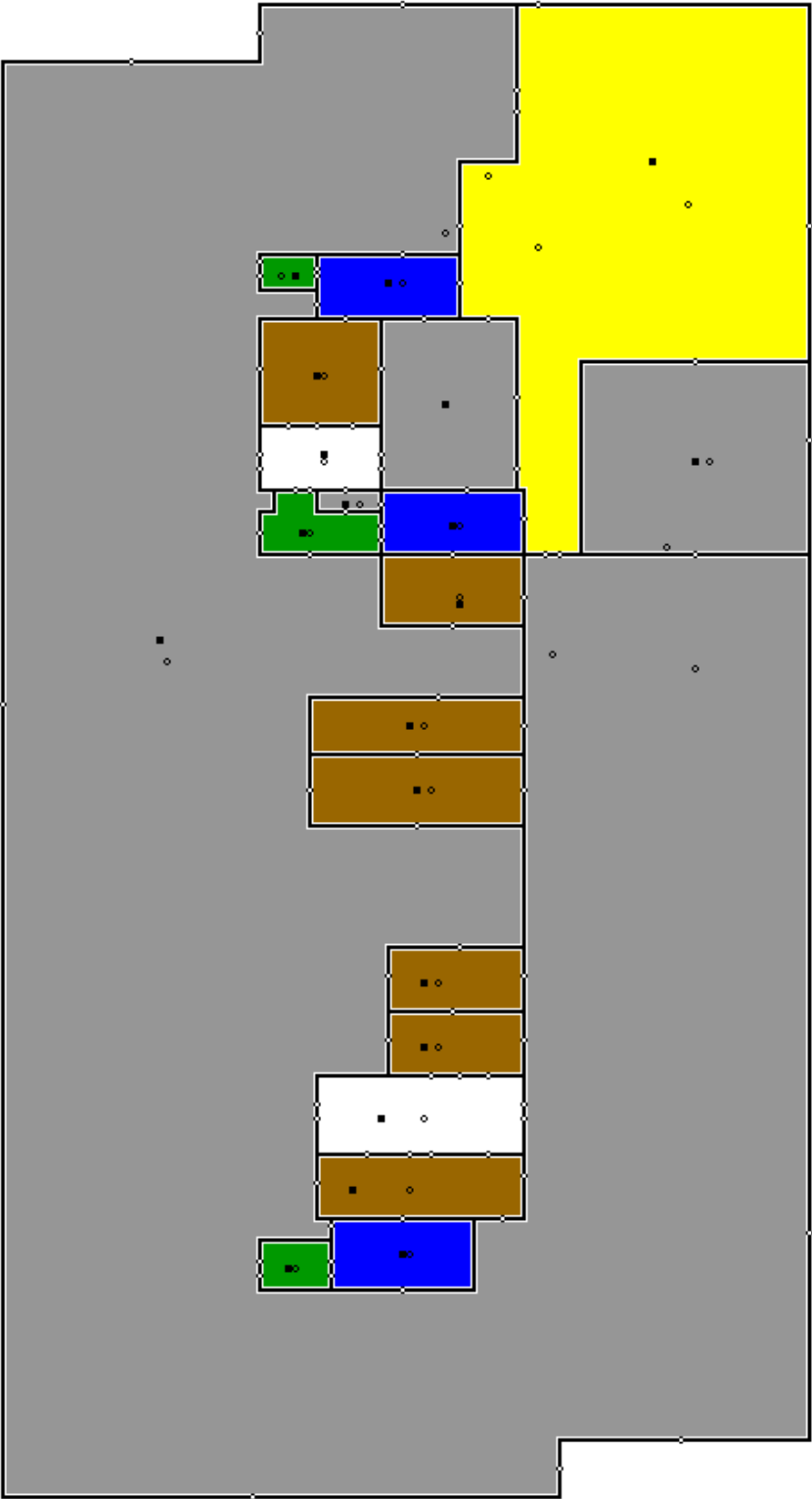
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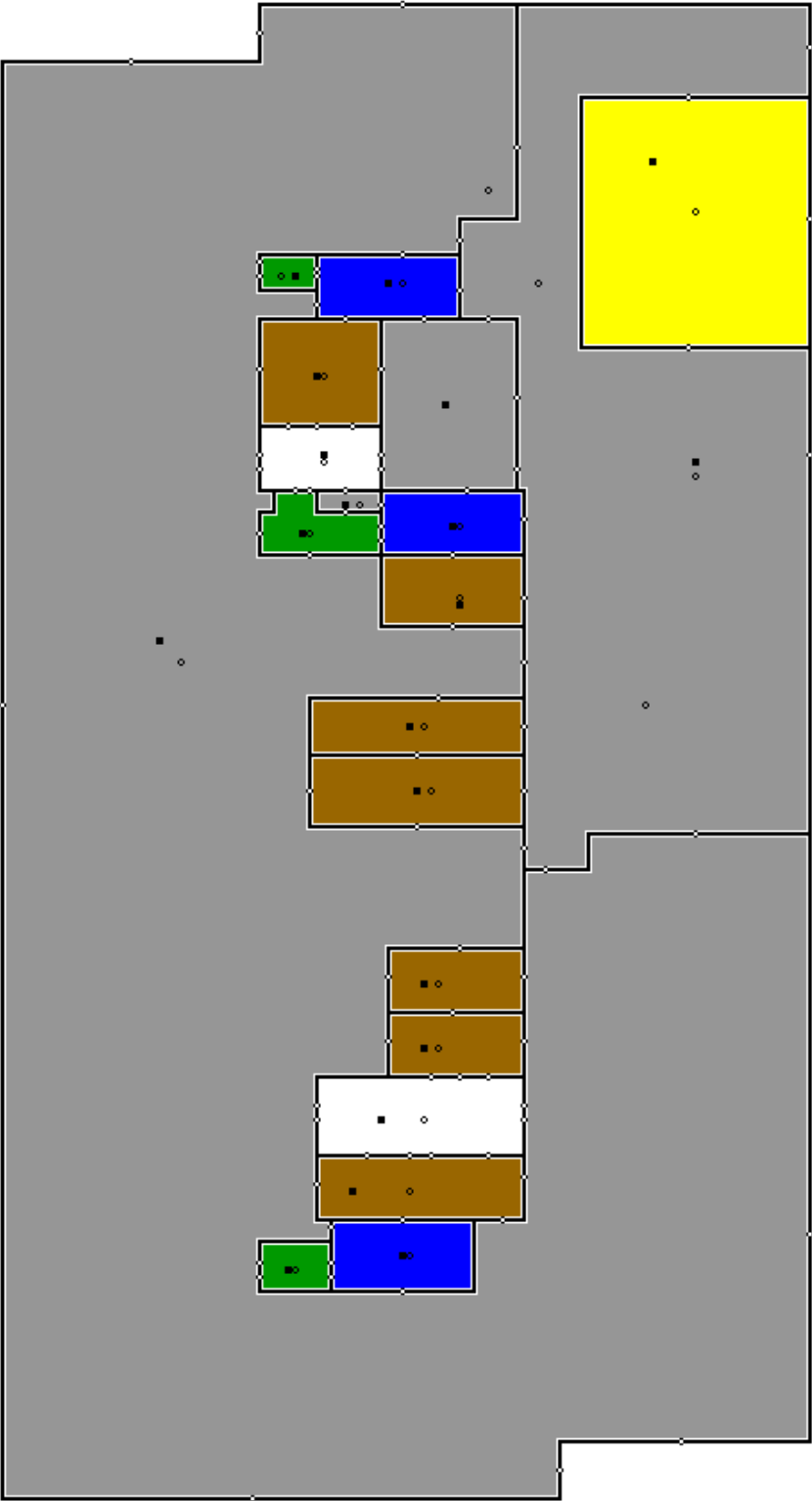
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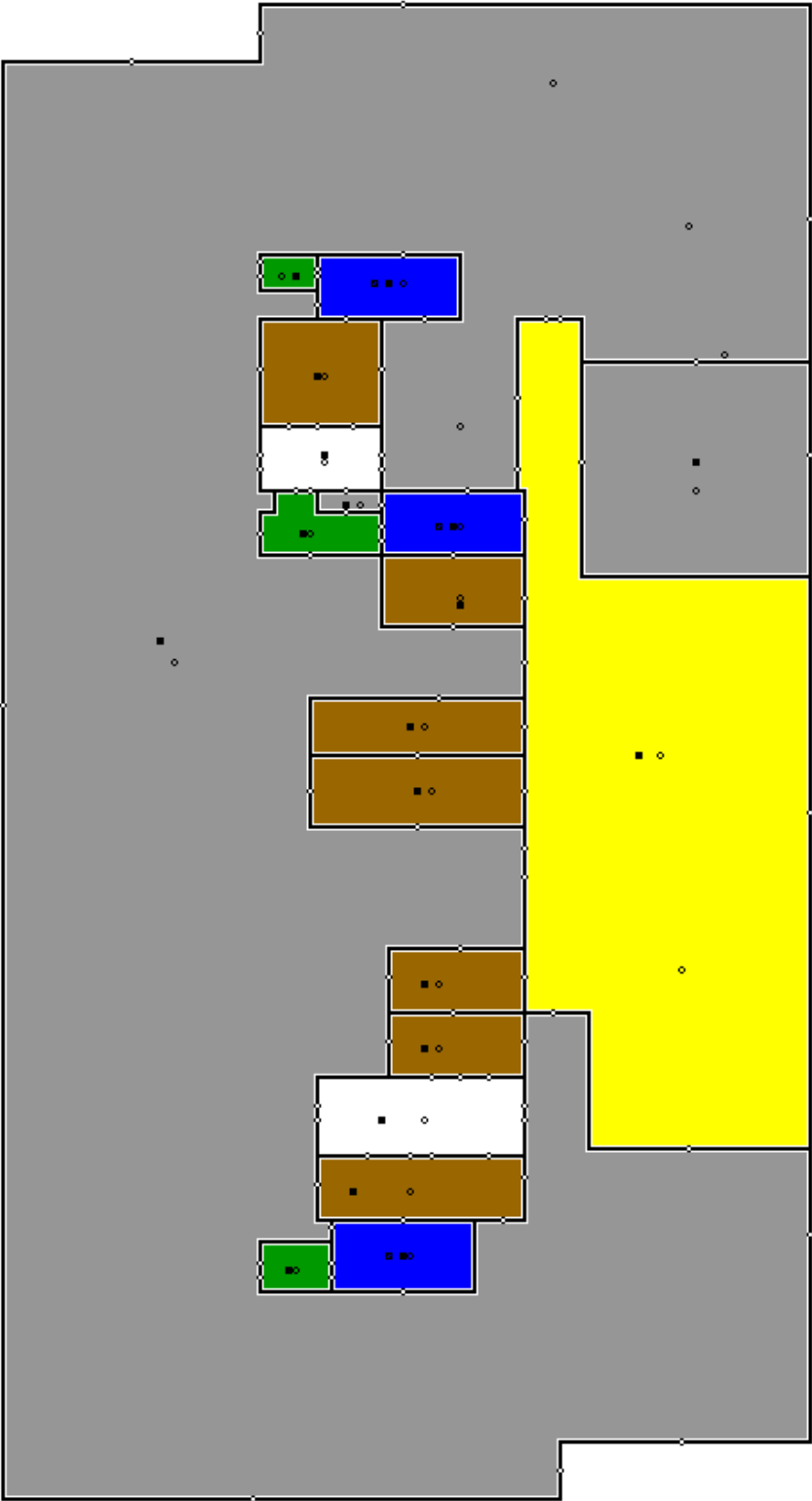
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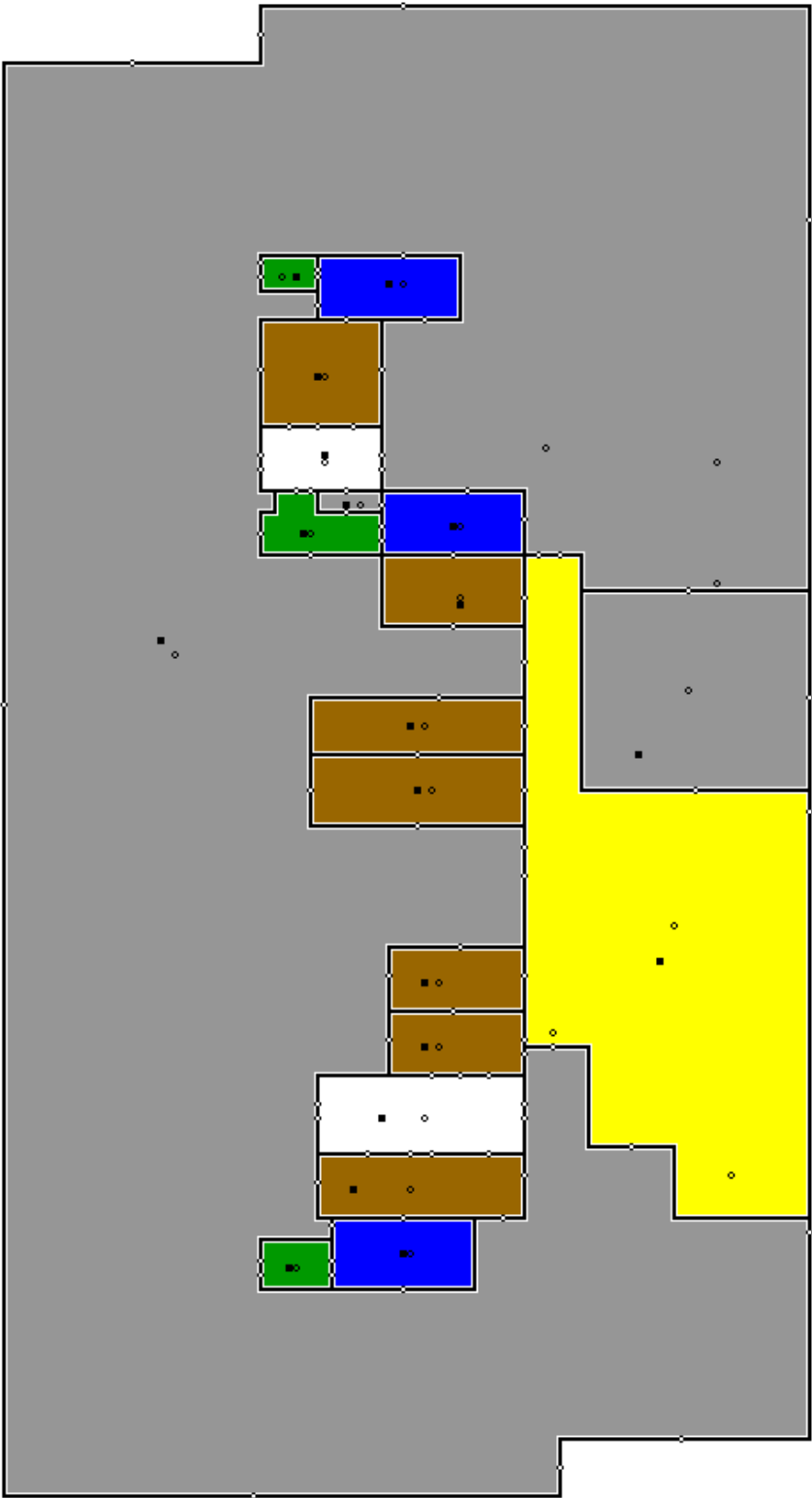
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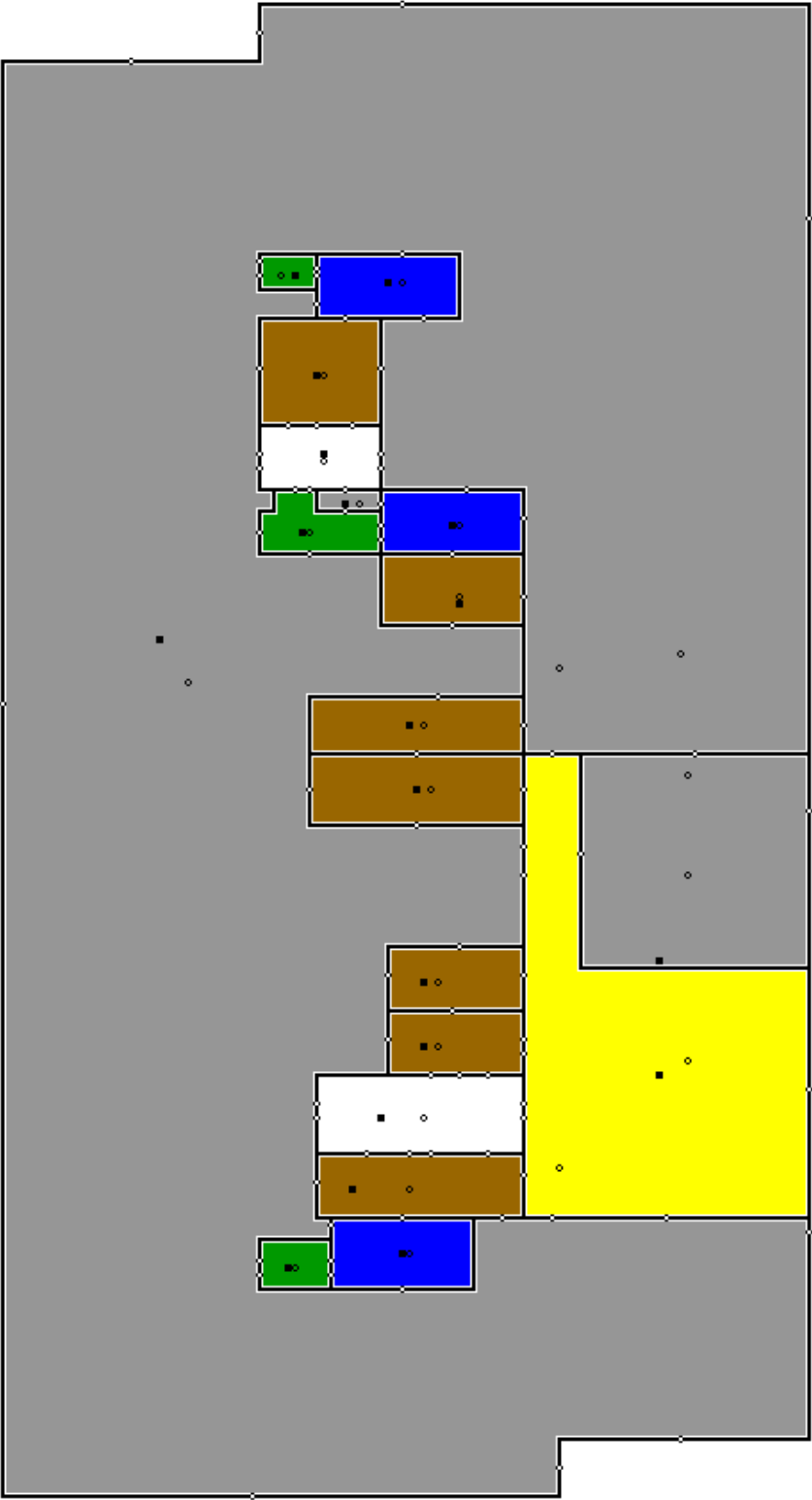
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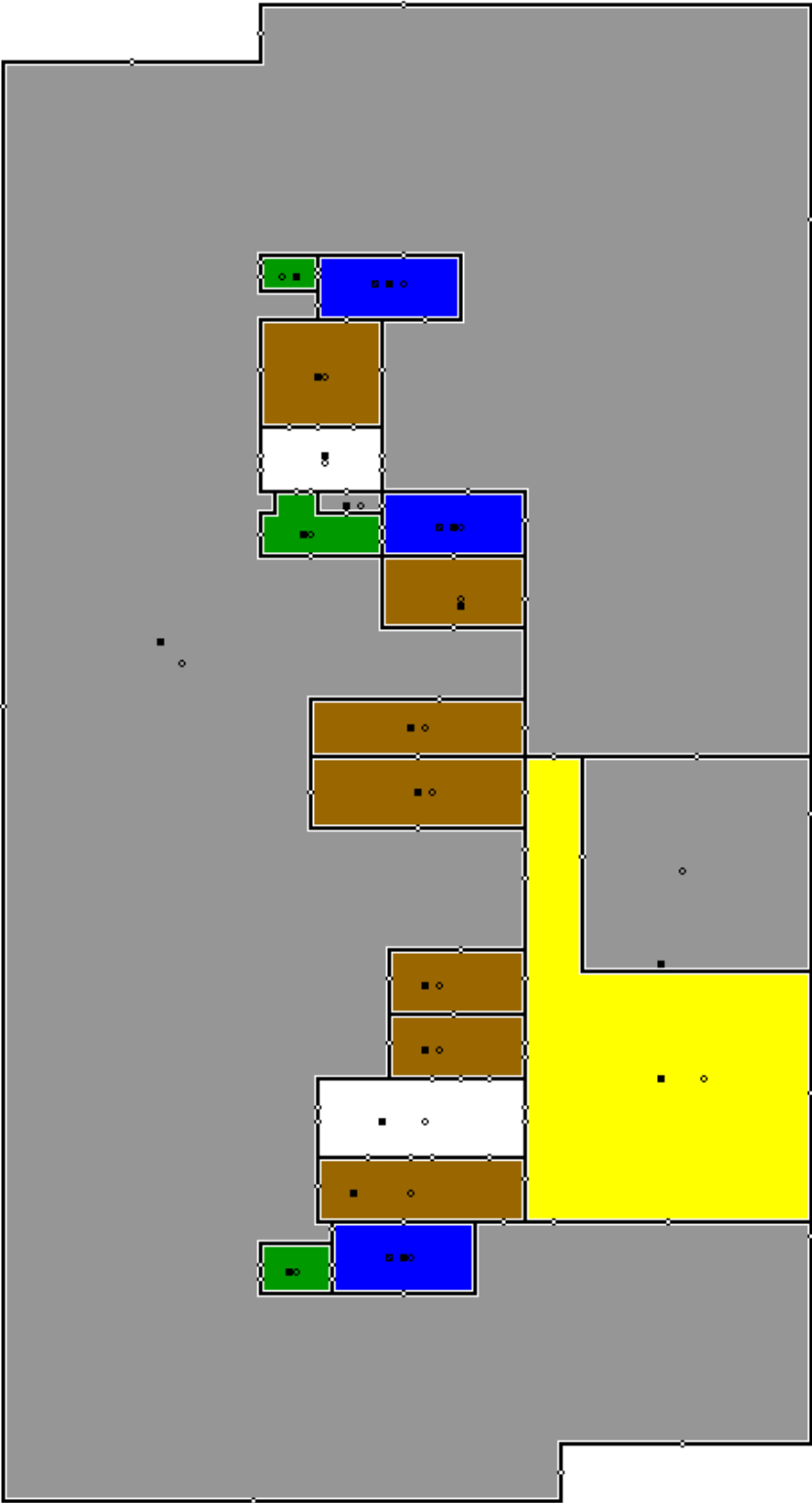
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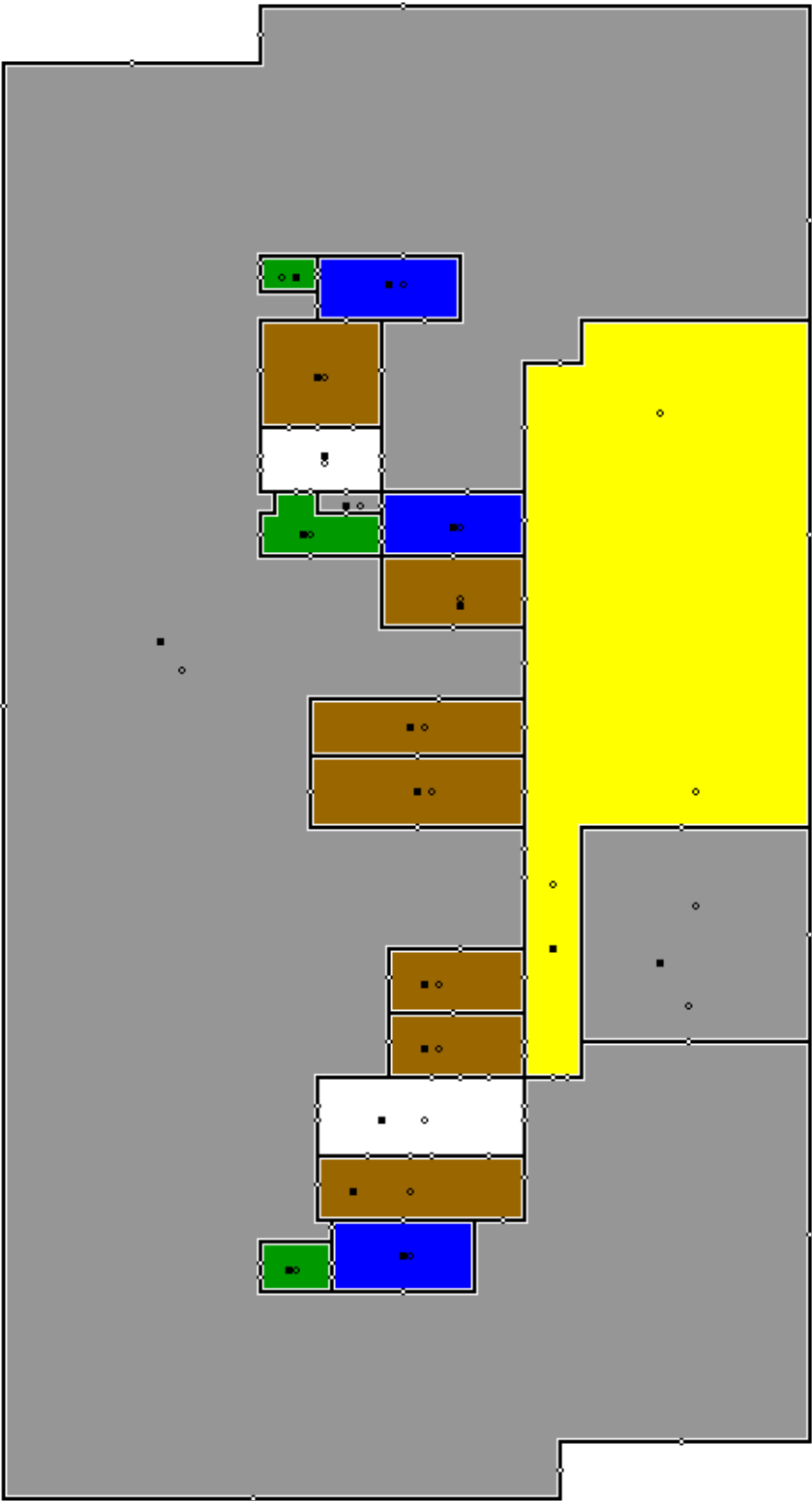
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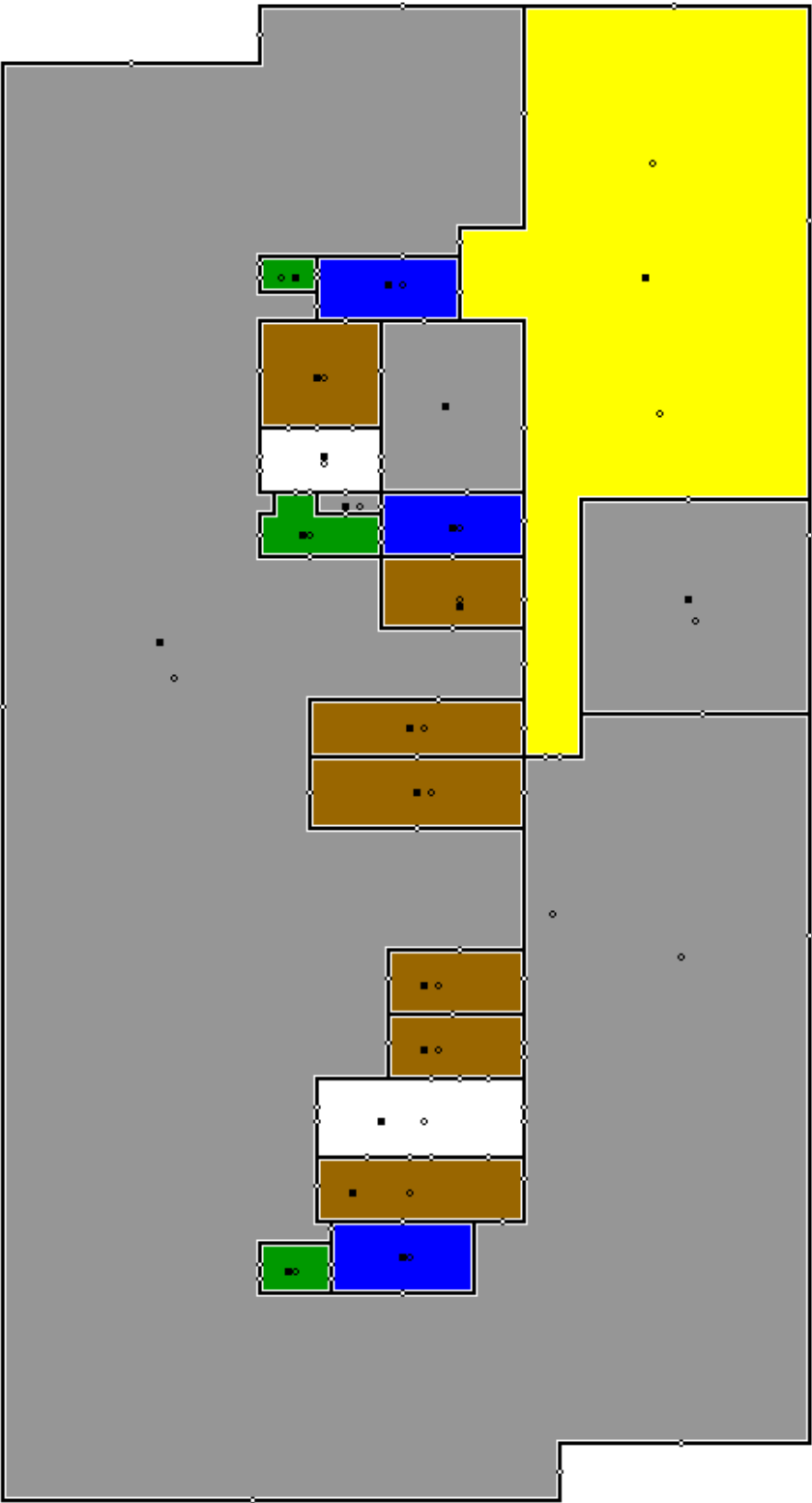
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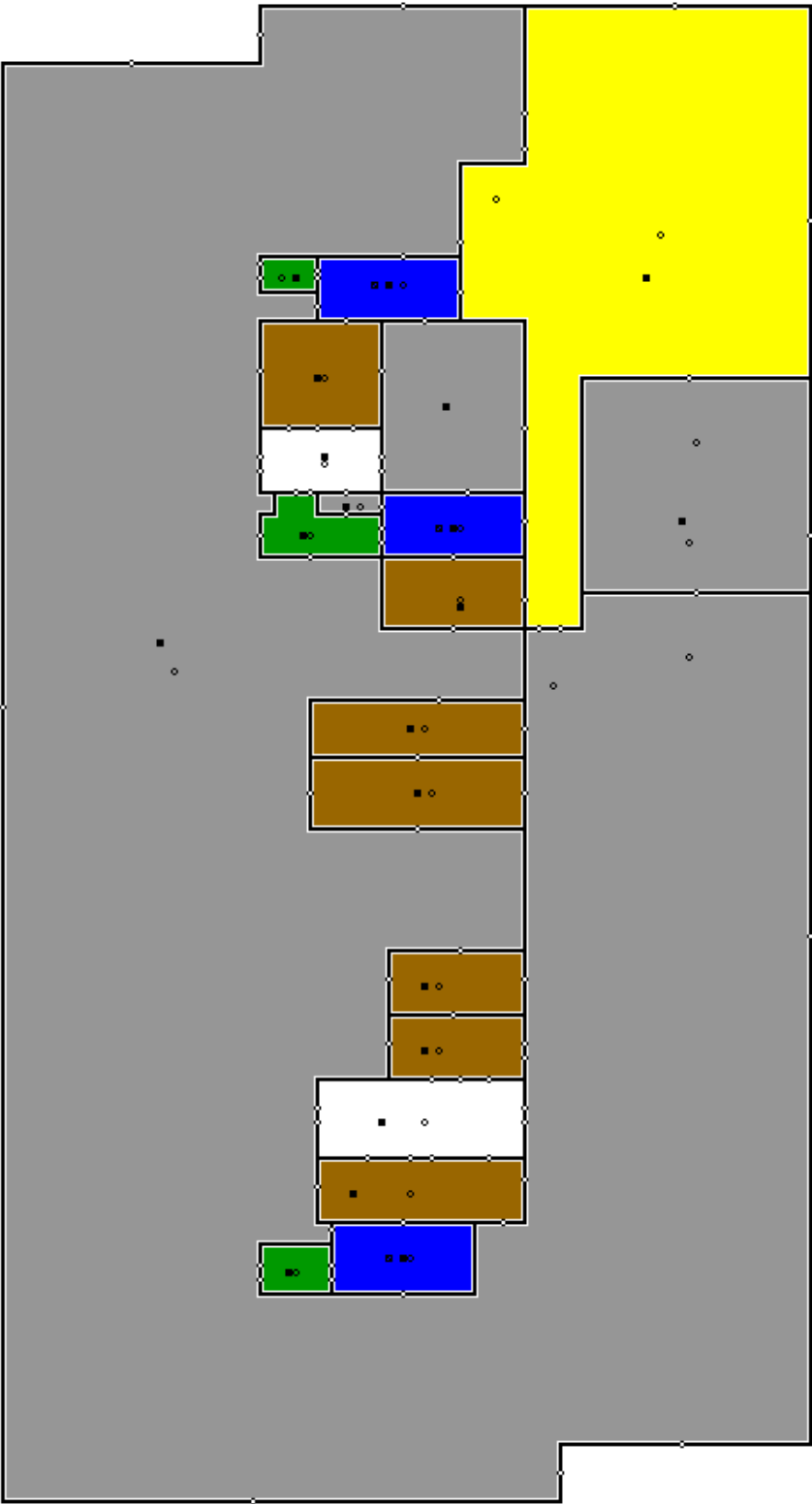
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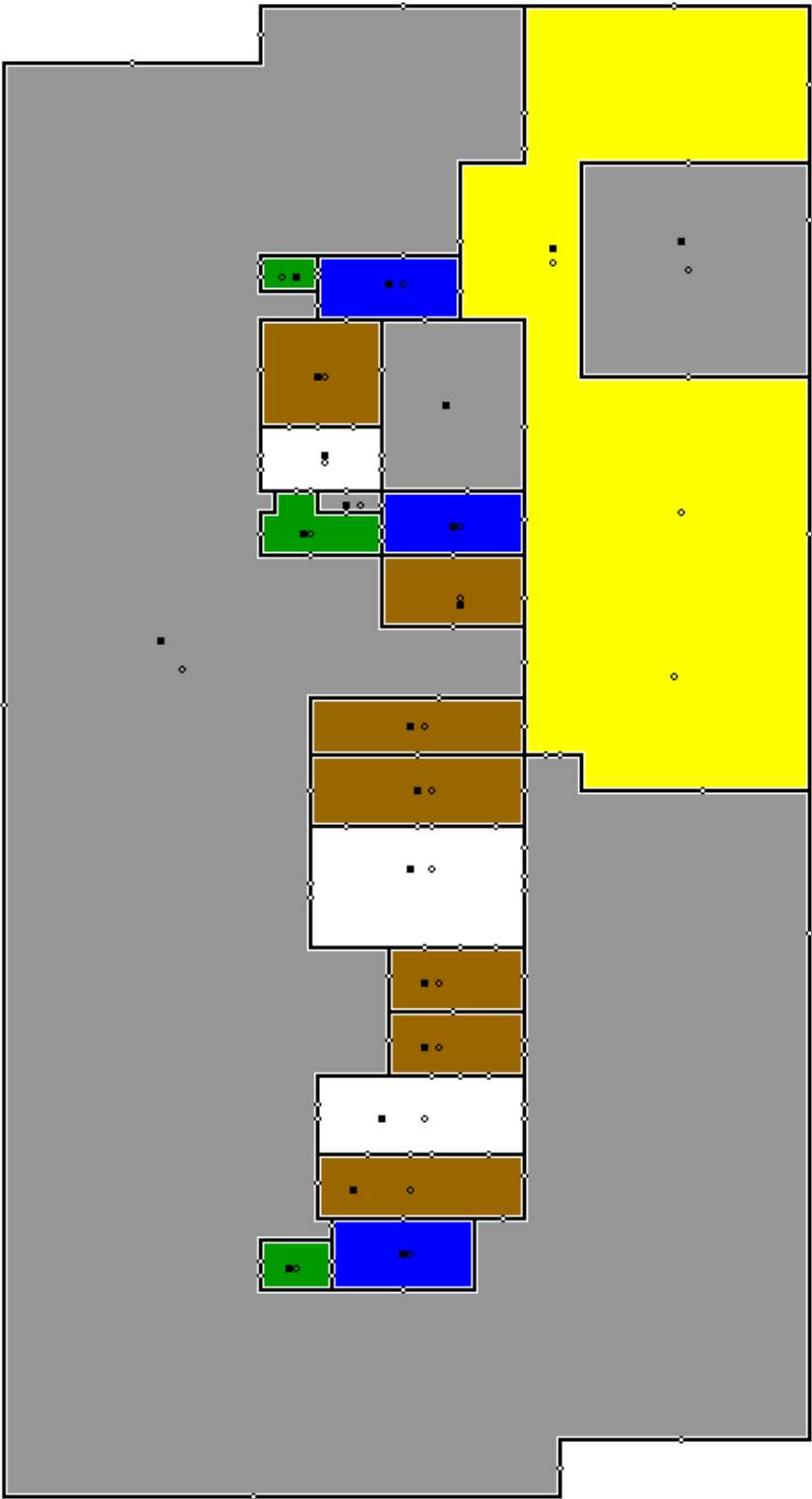
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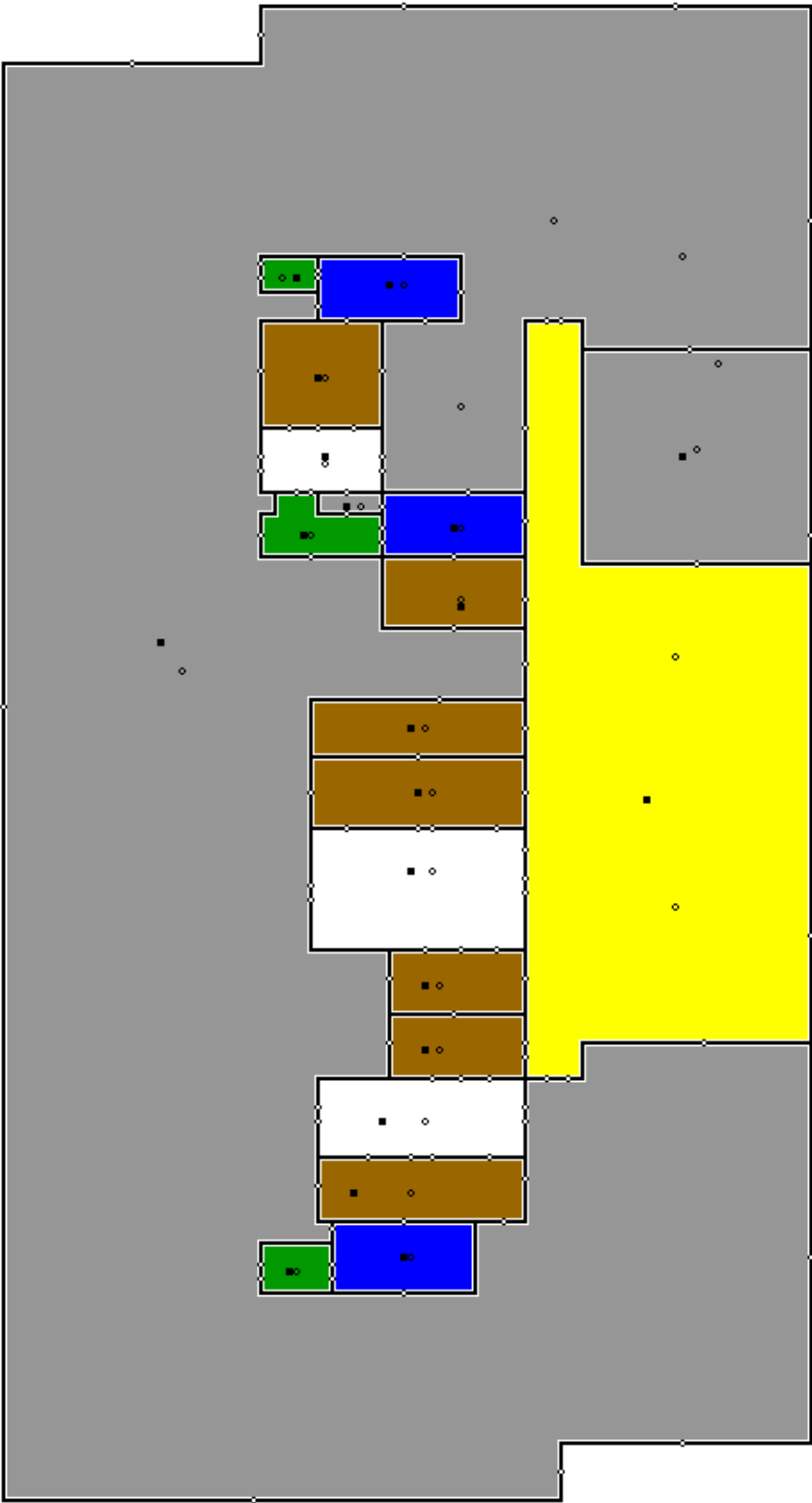
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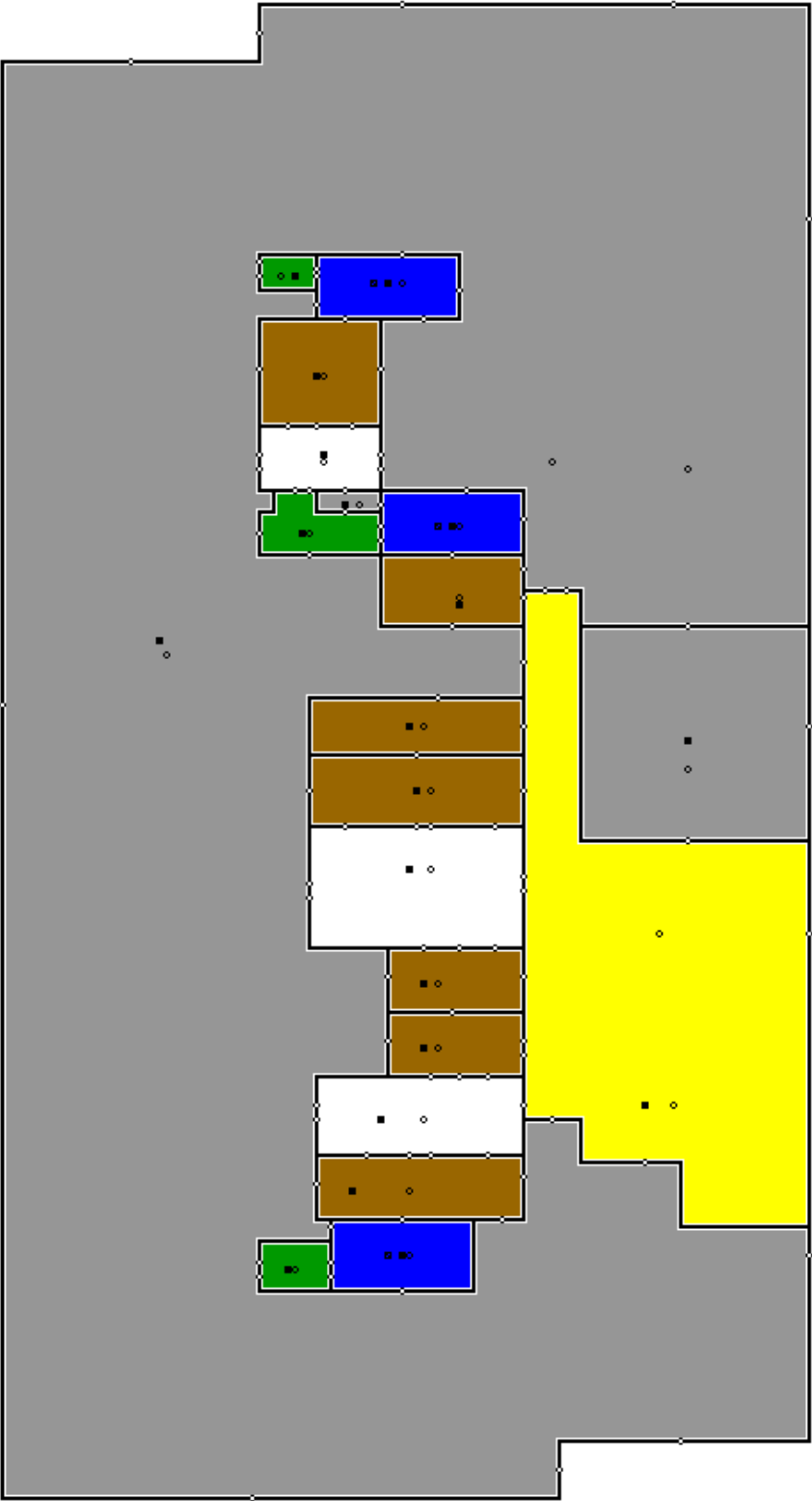
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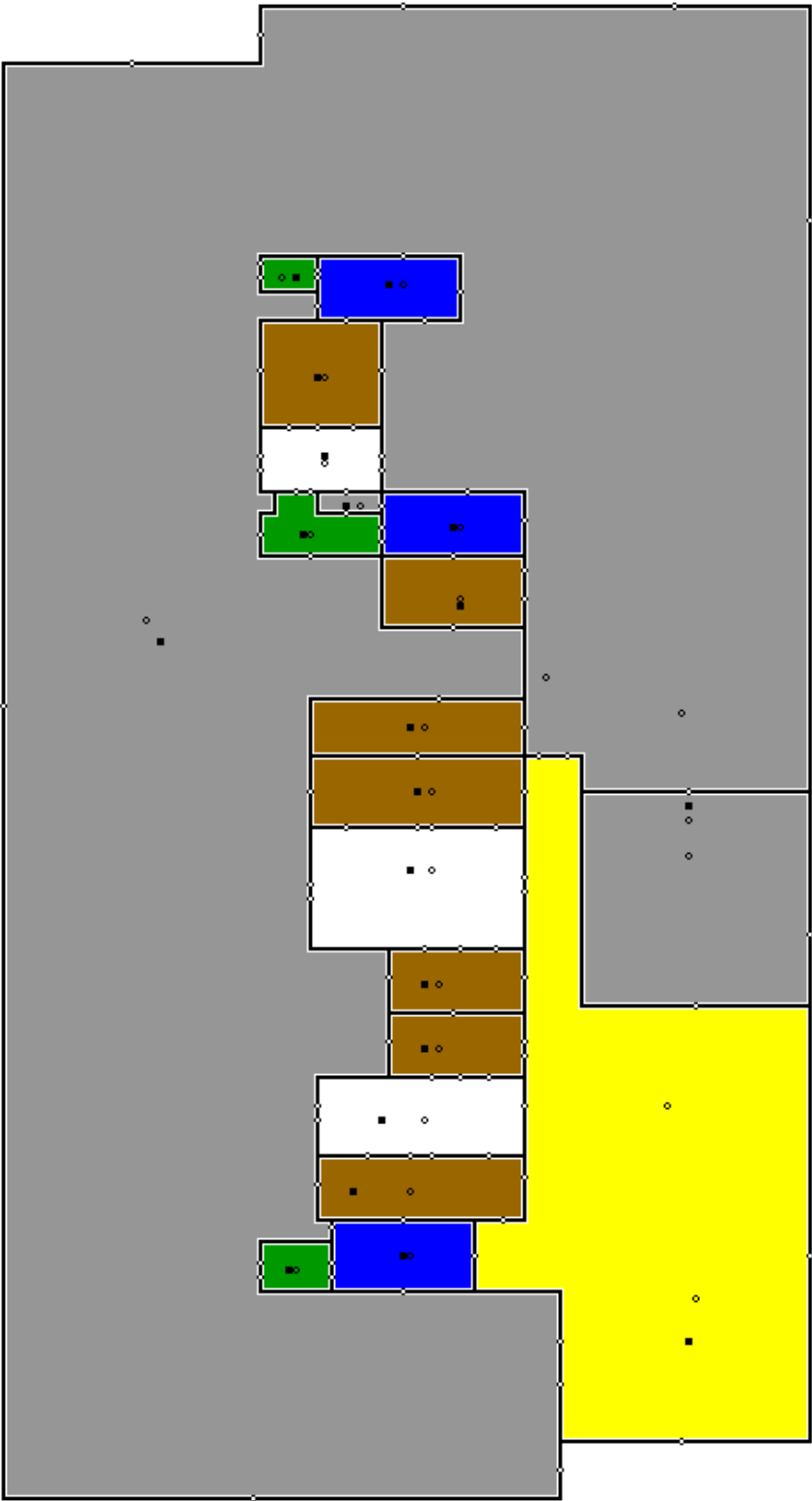
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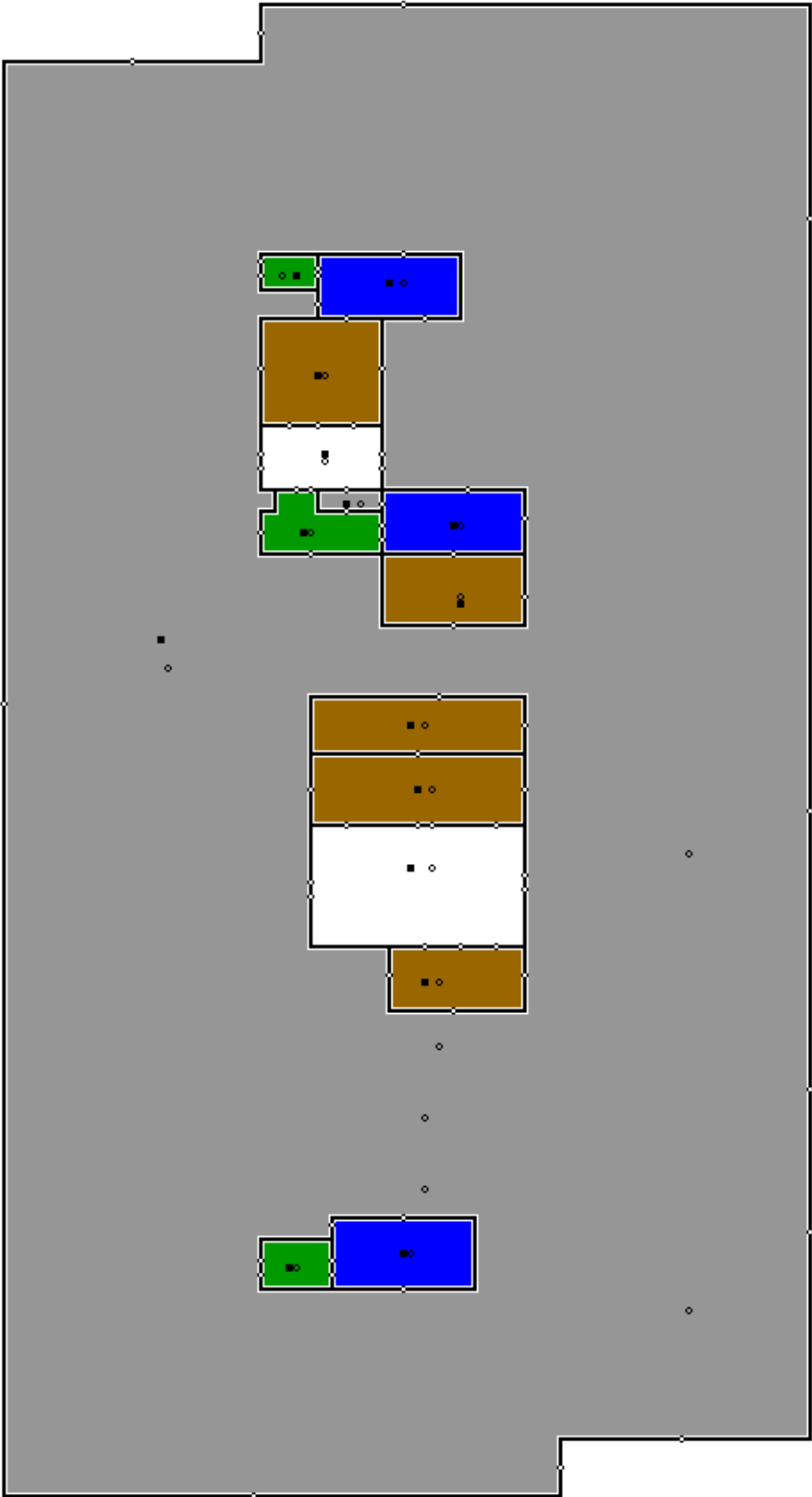
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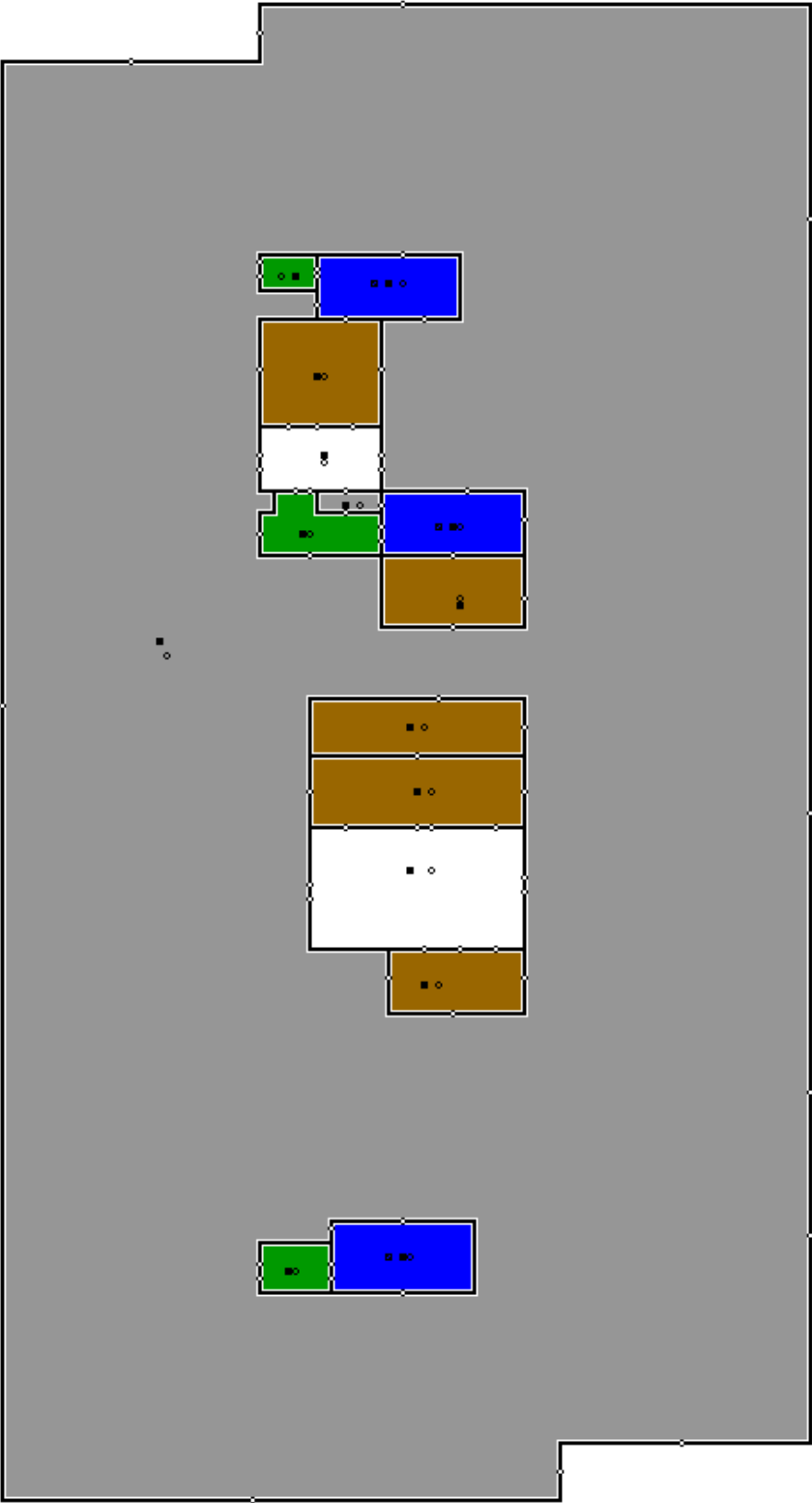
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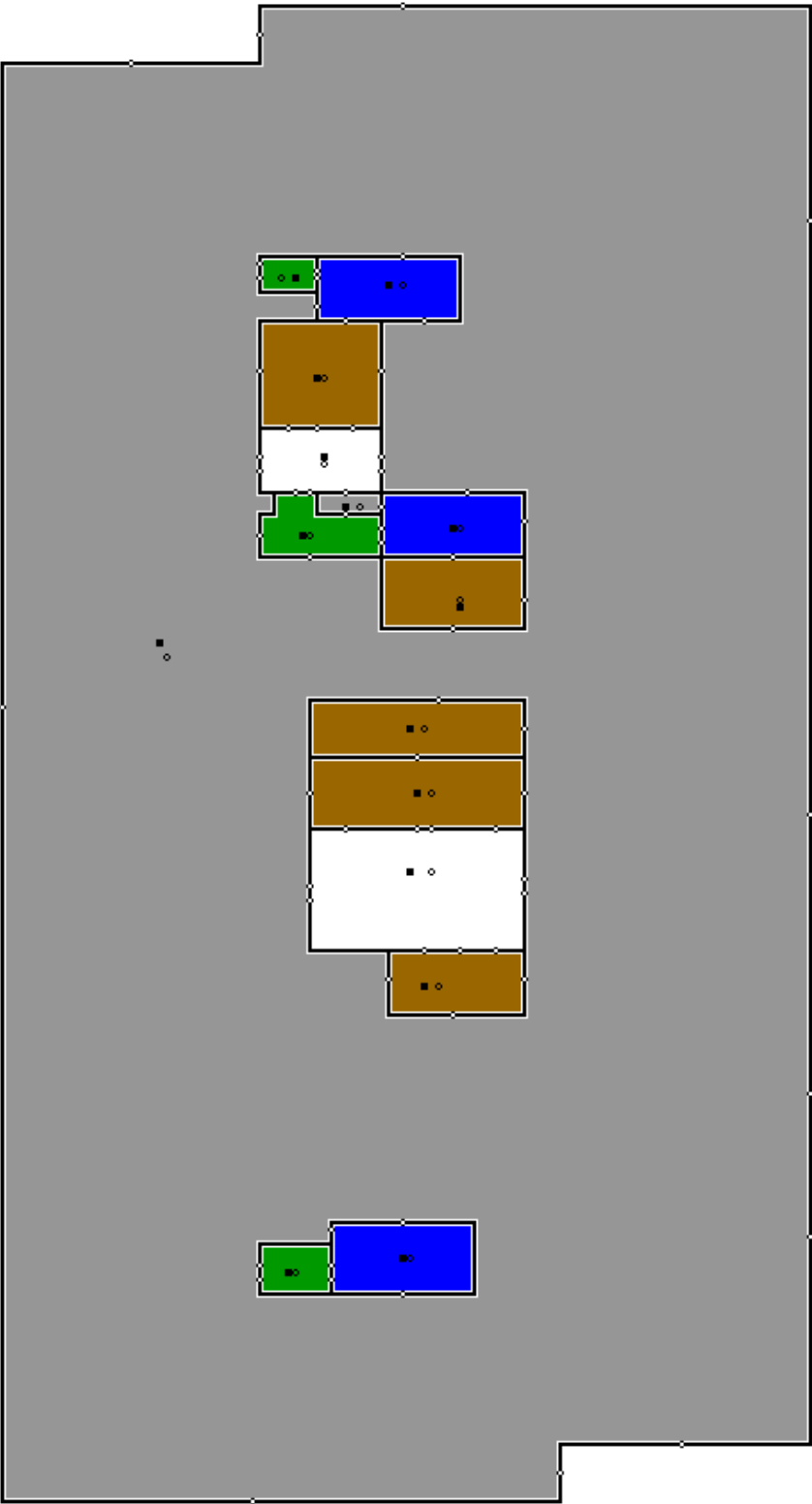
LEVEL 19



LEVEL 20



LEVEL 21



Appendix F
Smoke Control Commissioning

Prior to the issuance of a Certificate of Occupancy, smoke control systems are required to be tested as required by the City Los Angeles to show compliance with the minimum requirements of the rational analysis and the Los Angeles Building Code (LABC). The tests will be conducted by a qualified deputy inspector approved by Los Angeles Fire Department (LAFD) and the Los Angeles Department of Building and Safety (LADBS). A report of the results will be submitted to the inspection groups for LAFD and LADBS.

Typical Floors

The typical floors are passive and will be isolated via rated floor construction, rated elevator lobbies, and combination fire/smoke dampers (CFSDs) at common shafts.

1. Smoke/Fire Dampers

- Inspect and label all dampers, fault test for power and position, witness proper reporting to smoke control panel (airflow and power), and confirm response time. Perform functional test from smoke control panel to verify fail safe function and operation based on activation from detector in zone. Verify listing, record model number and manufacturer, and confirm that location matches approved plans and intended smoke zone. Record and verify damper number/address.

2. Smoke Control Ducts, Fans, and Control Equipment (stair pressurization)

- Verify duct detector locations at ducts and air handling units, as well as proper reporting of status to smoke control panel.
- Inspect supply and exhaust fans to confirm monitoring downstream of the last disconnect, motor service factor, presence of non-combustible fan restraints, quantity of belts (1.5 times minimum), belt tension, RPMs, voltage and amperage. Record smoke zone and unit IDs.
- Review inlets and outlets for location relative to potential fire exposures and to minimize potential for introducing smoke or flame into the buildings.
- Witness duct leakage testing to 1.5 times the maximum design pressure for all duct sections (except for flexible connections).

3. Waterflows, Detectors, and Manual Pull Stations

- Verify that stair pressurization system activates, and that fans turn on/off per design sequence of operations. Verify proper reporting to Fire Alarm Control Panel (matches physical location); confirm that fire alarm evacuation system activates, elevator recall sequence is initiated, and that magnetic door holders

release, and doors latch in elevator lobbies. Confirm proper location of device in smoke zone and record device address.

4. Functional Testing of Smoke Control Panel

- Verify status for all fans and dampers and control priority over test and inspect panel. Confirm response time.

5. Smoke Curtains and Automatic Closing Doors

- Verify functional sequence for smoke curtains/automatic closing door assemblies in designated smoke zones.

6. Pressurized Stairwells and Firefighter's Smoke Control Panel

- Verify pressure difference from building to vestibule and vestibule to stair shaft, door opening force, and confirm presence of door labels. Verify status for all fans and dampers on smoke control panel and control priority over the test-and-inspect panel and other life-safety system panels (e.g., carbon monoxide monitoring, if provided).

7. Door Fan Testing

- Witness door fan testing to confirm maximum leakage from areas identified by LAFD. Review of up to 10% of typical spaces on each floor is assumed.

8. Deficiencies

- Re-inspect all deficiencies to prepare for final inspection.

9. Final Inspection with Authority Having Jurisdiction (AHJ)

- Re-inspect building with AHJ.

Testing of Smokeproof Enclosures

Smokeproof enclosures will be tested to confirm that the minimum pressure differentials described in Section 2.2.1 are achieved (0.05 in. water gage for vestibule-to-stair and vestibule-to-fire-floor door sets; or 0.10 in. water gage for stair-to-floor where no vestibule is provided). Pressure differentials between pressurized stairs (e.g., Stairs 1, 2, 3, 5, 6, and 7, as applicable) and the associated exit passageways, and between exit passageways and the associated floors, will be verified to meet the applicable criteria. Maximum allowable door opening forces will be verified at each stair and exit passageway door, as required by the Code and as described in this narrative. Where the mechanical ventilation method is used for a smokeproof enclosure,

openings at the top of each smokeproof enclosure will be traversed to confirm the minimum required flow across the openings relative to the minimum code-required flow of 2,500 cfm (3,000 cfm targeted).

Periodic Retesting

After the building is occupied, all operating parts of the smoke control systems must be retested every six months in accordance with the retest requirements established by the LAFD and LADBS. The retest will be conducted by an approved inspection agency, and a report of the test results will be submitted to LAFD and LADBS.

To facilitate periodic retesting, we recommend installing pressure testing tubing in various locations for verification of pressure gradients across stairwell doors and from floor to floor, as applicable. Such tubing should be installed in electrical and phone rooms or similar spaces.